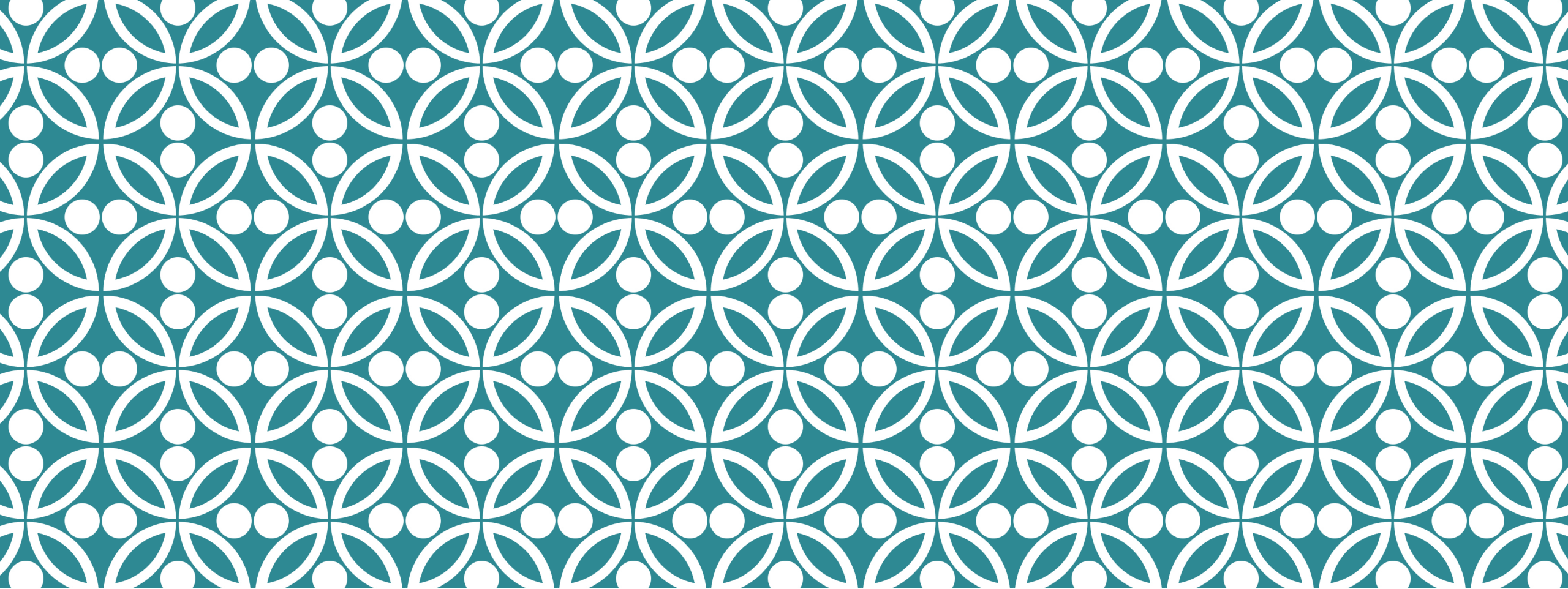




SESSION 3: VU TOOLS FOR TEACHING & RESEARCH

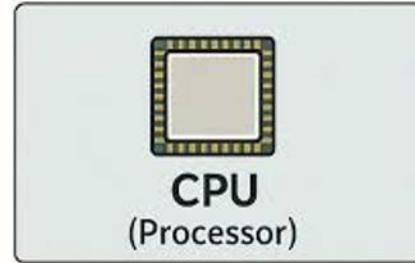
DR. SOFIA GIL-CLAVEL

- ❖ Introduction to servers.
- ❖ Introduction to JupyterHub.
- ❖ Usage and data storage.
- ❖ Introduction to AmCAT.
- ❖ Use AmCAT to share, access, and analyze data.

1. INTRODUCTION TO SERVERS

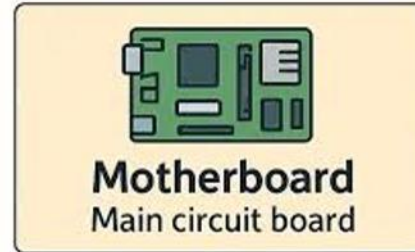
HOW DOES YOUR COMPUTER WORK?

The brain of the computer. It performs all processing calculations.



CPU
(Processor)

The main circuit board in a computer that acts as the central hub connecting the CPU, RAM, and storage.



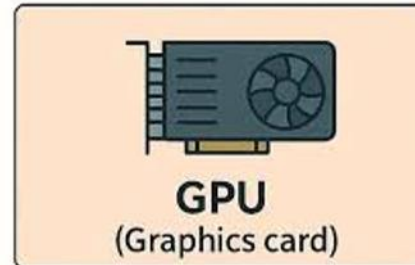
Motherboard
Main circuit board

Keyboard and mouse are examples of hardware used to send data and control a computer.

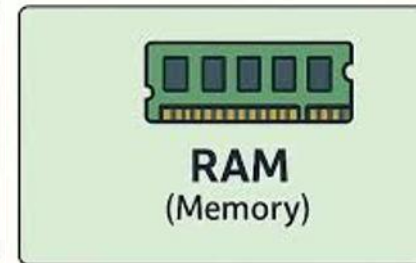


Keyboard
Input device

A specialized electronic circuit designed to rapidly process and render images, video, and complex math through parallel processing.



GPU
(Graphics card)



RAM
(Memory)

Random Access Memory stores data temporarily while a computer is running.



Storage drive
Data storage device

It is the hardware component used to save, keep, and get back digital data.



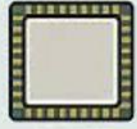
Monitor
Displays output

A monitor is an example of a hardware that takes processed data from a computer and turns it into a form people can understand.

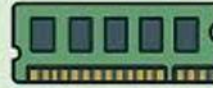


Cooling Fan
Keeps components cool

Prevents overheating by drawing in cool outside air, pushing it over hot parts like the CPU and GPU, and blowing the warm air out of the case.



CPU
(Processor)



RAM
(Memory)



Storage drive
Data storage device



Motherboard
Main circuit board

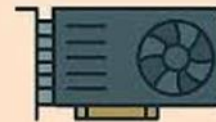
To understand how a server works, it helps to look at it as a high-powered, industrial version of a regular computer.



Monitor
Displays output



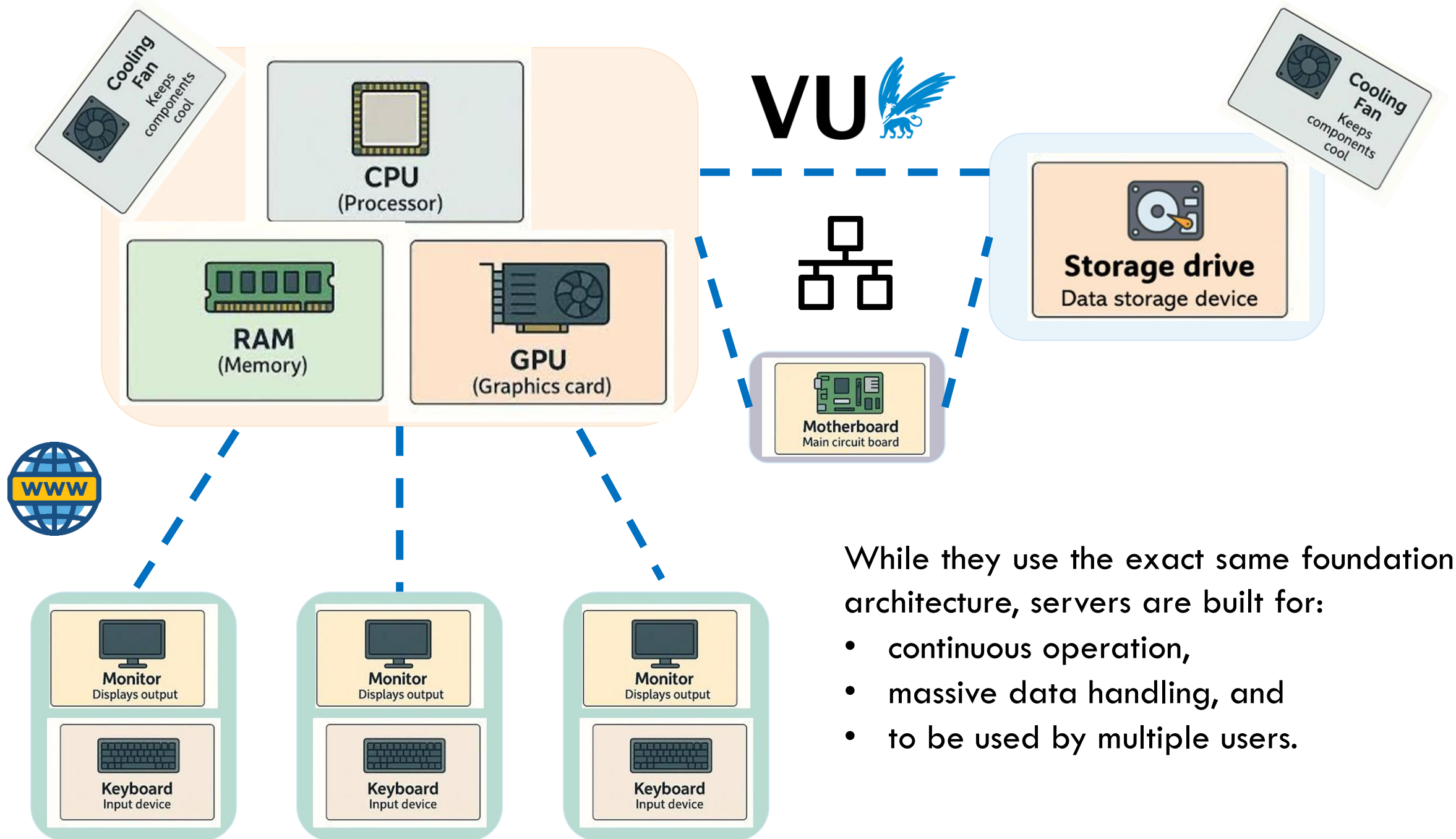
Keyboard
Input device

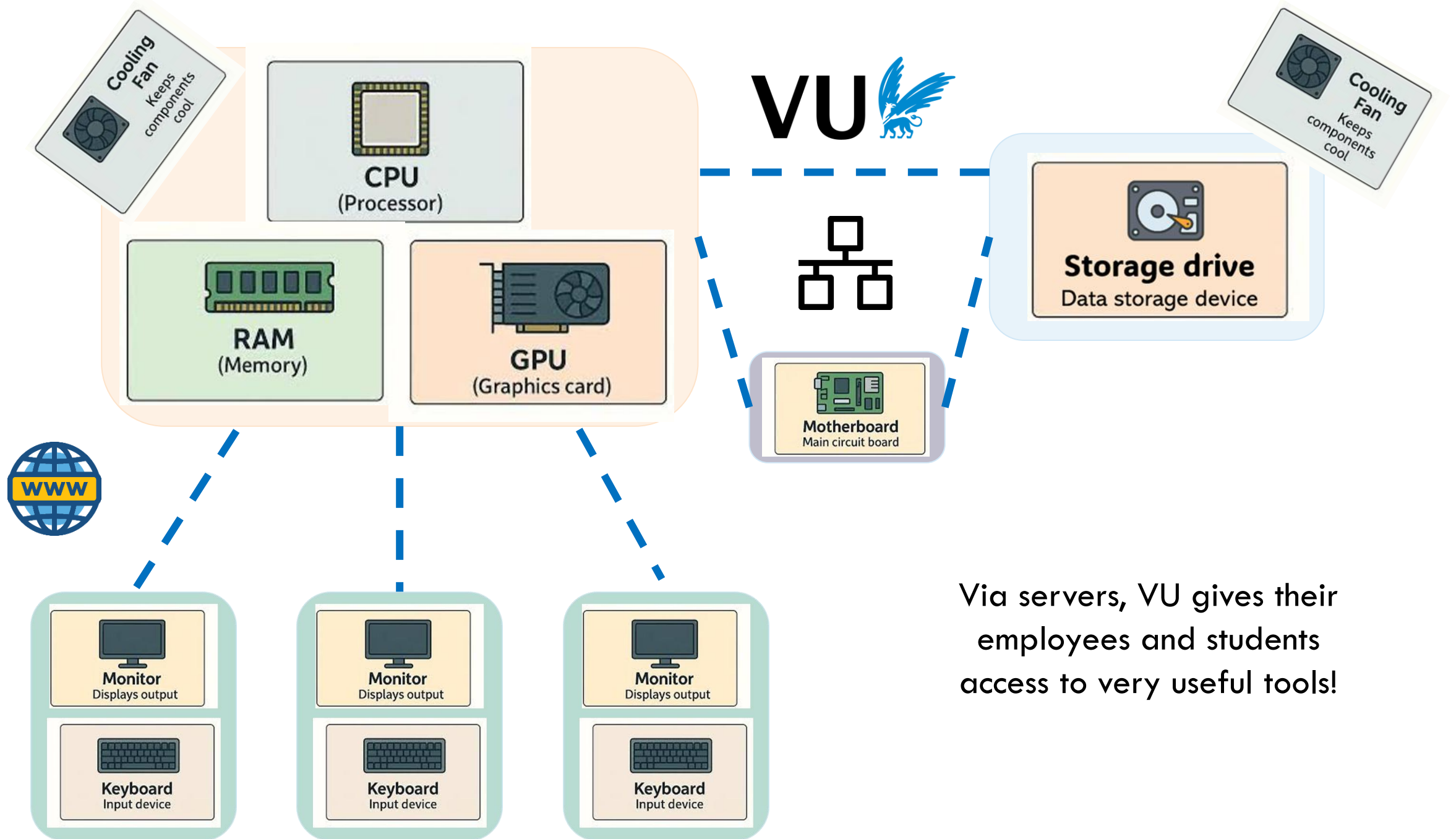


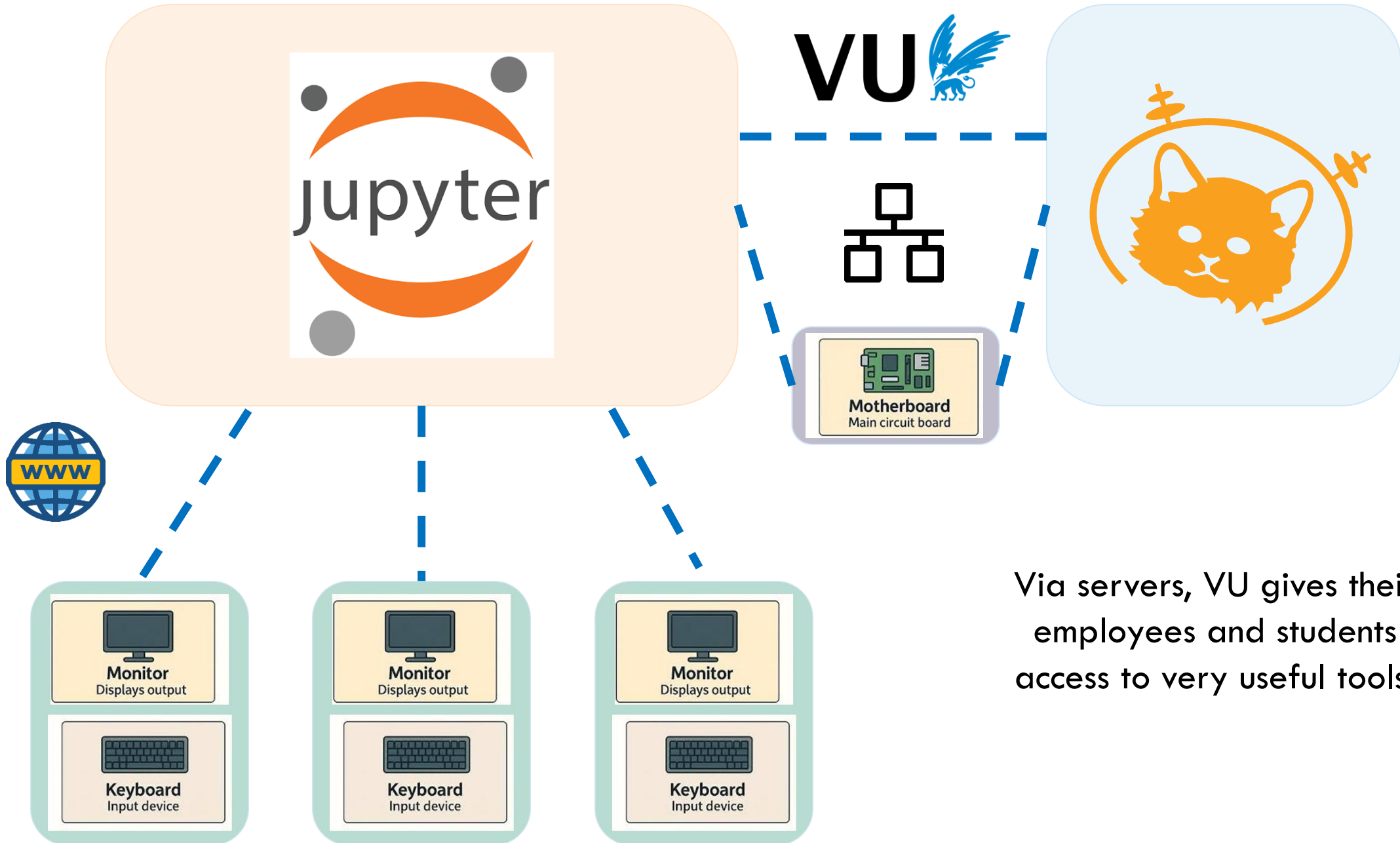
GPU
(Graphics card)



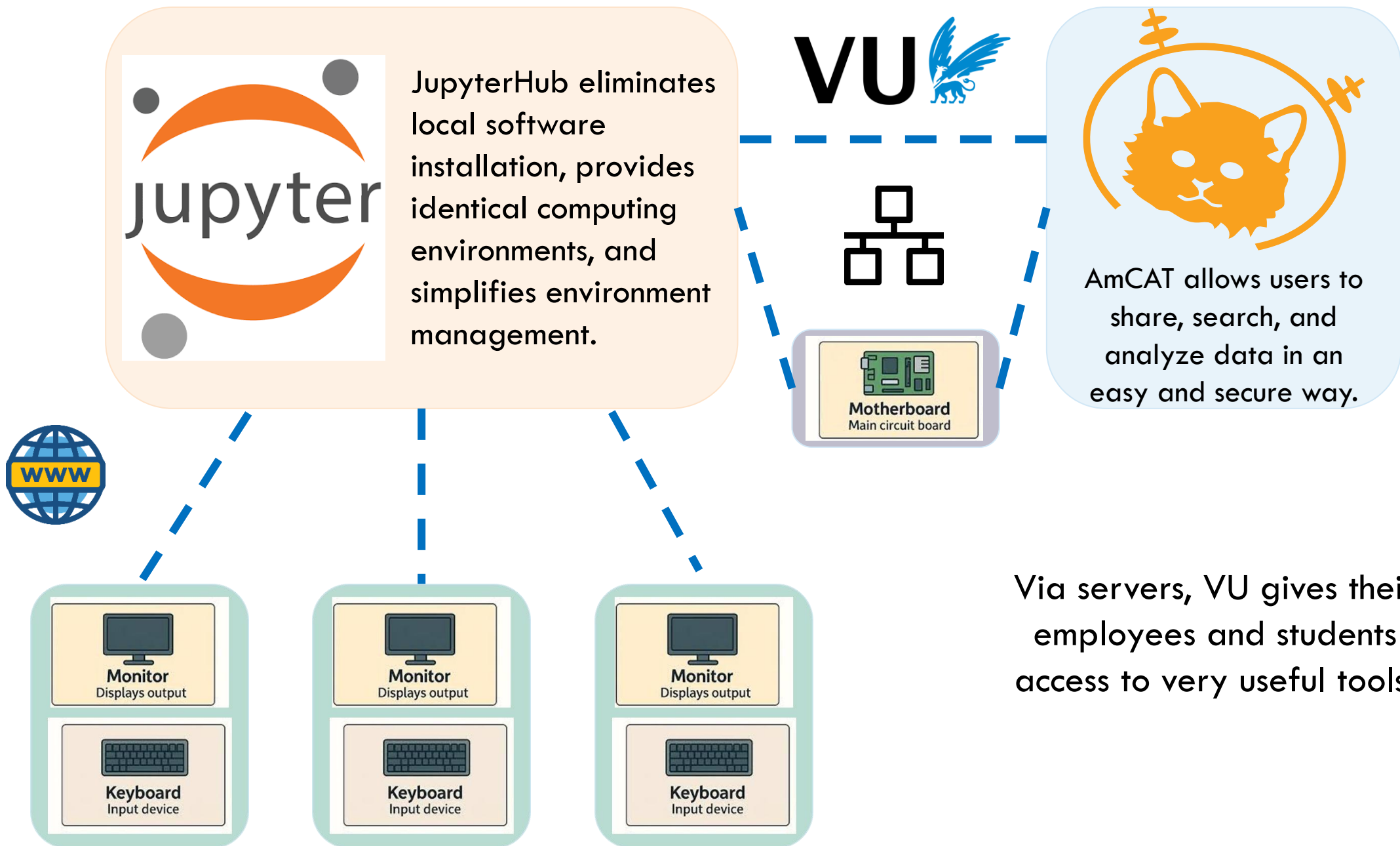
Cooling Fan
Keeps components cool







Via servers, VU gives their employees and students access to very useful tools!



Via servers, VU gives their employees and students access to very useful tools!

2.1 Introduction

2.2 Access

2.3 Environment usage

2.4 Scripts, folders, and storage
of small data

2.5 Installing & loading R-
packages.

2.6 Storage of bigger data

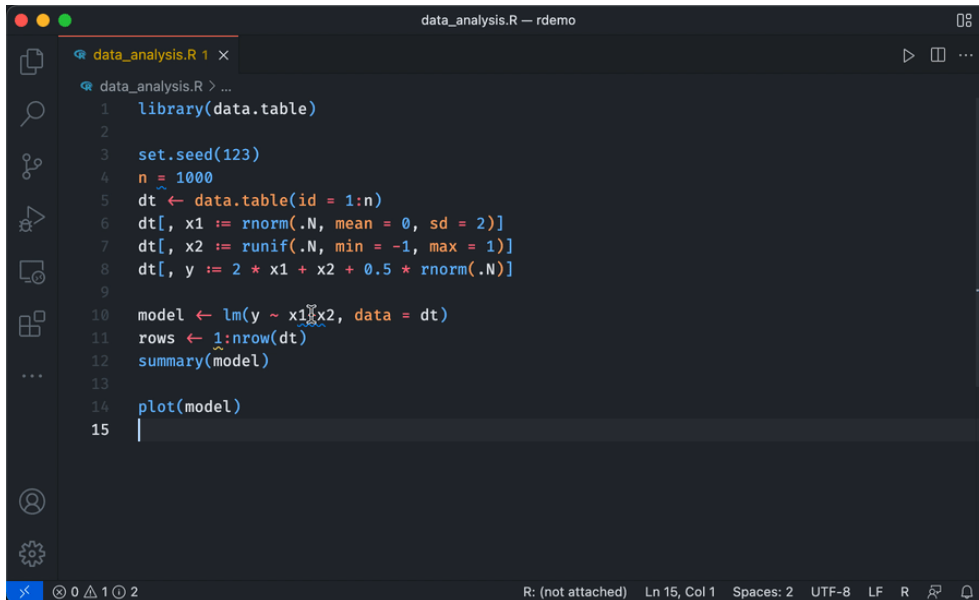
2.7 Downloading files

2. JUPYTERHUB



2.1 INTRODUCTION

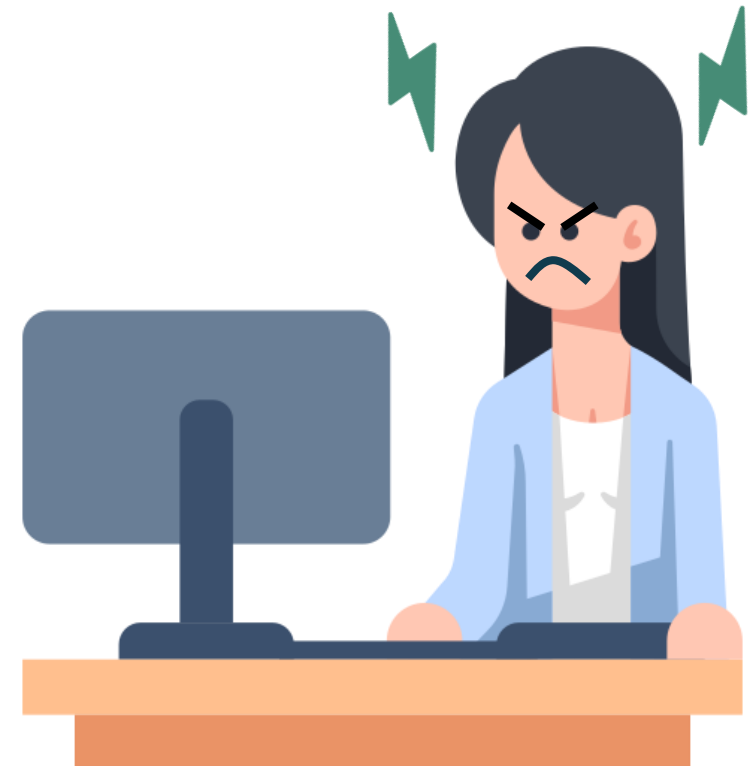
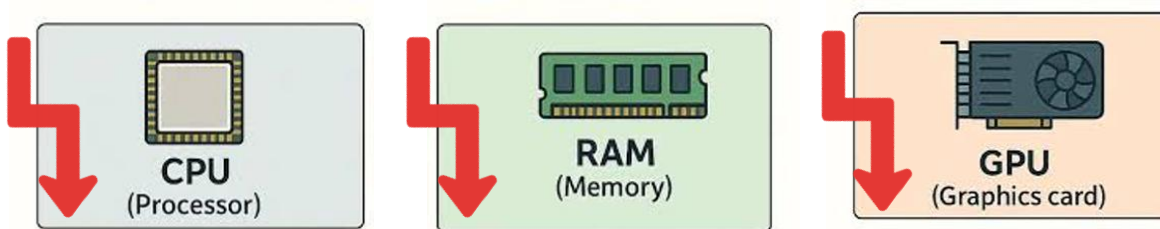
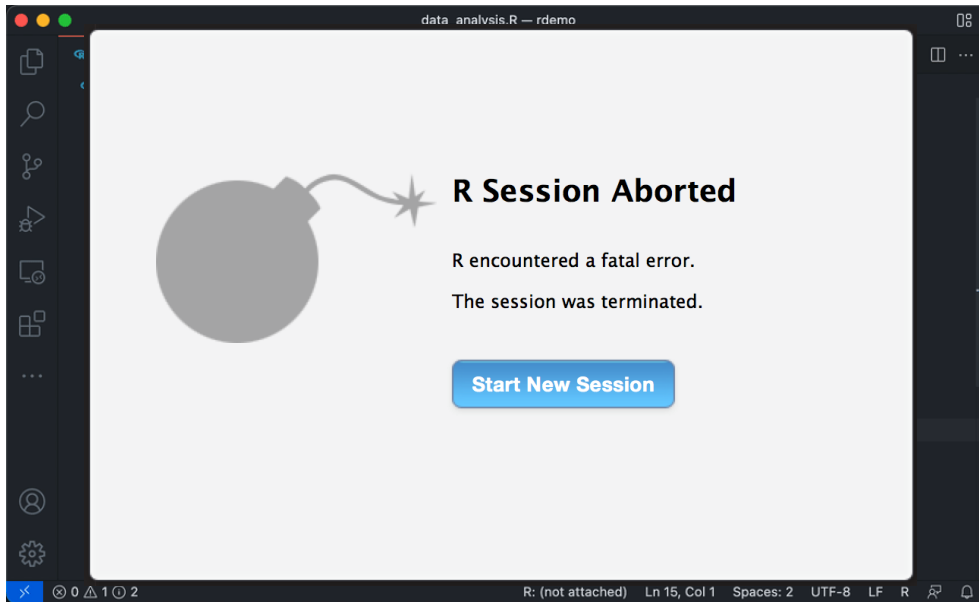
CASE 1: HAS IT HAPPENED TO YOU THAT...?



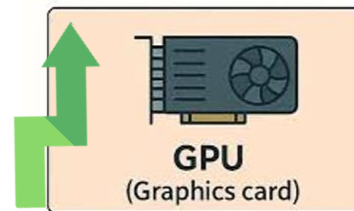
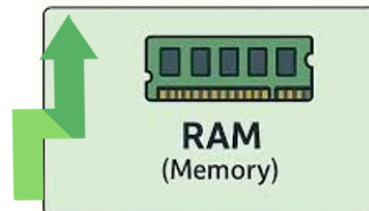
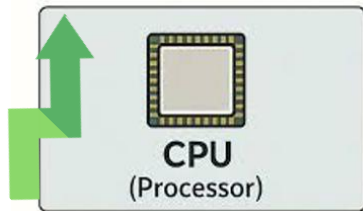
```
data_analysis.R — rdemo
data_analysis.R 1 x
data_analysis.R > ...
1 library(data.table)
2
3 set.seed(123)
4 n = 1000
5 dt <- data.table(id = 1:n)
6 dt[, x1 := rnorm(.N, mean = 0, sd = 2)]
7 dt[, x2 := runif(.N, min = -1, max = 1)]
8 dt[, y := 2 * x1 + x2 + 0.5 * rnorm(.N)]
9
10 model <- lm(y ~ x1 + x2, data = dt)
11 rows <- 1:nrow(dt)
12 summary(model)
13
14 plot(model)
15
```



CASE 1: HAS IT HAPPENED TO YOU THAT...?

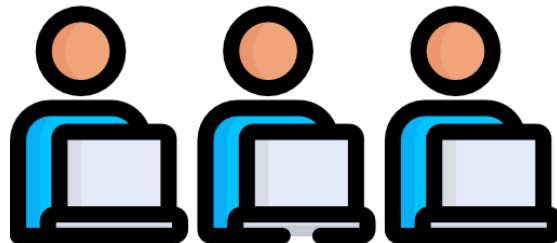
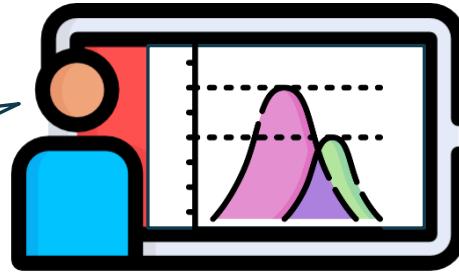


```
data_analysis.R — rdemo
data_analysis.R 1 x
data_analysis.R > ...
1 library(data.table)
2
3 set.seed(123)
4 n = 1000
5 dt <- data.table(id = 1:n)
6 dt[, x1 := rnorm(.N, mean = 0, sd = 2)]
7 dt[, x2 := runif(.N, min = -1, max = 1)]
8 dt[, y := 2 * x1 + x2 + 0.5 * rnorm(.N)]
9
10 model <- lm(y ~ x1 + x2, data = dt)
11 rows <- 1:nrow(dt)
12 summary(model)
13
14 plot(model)
15 |
```

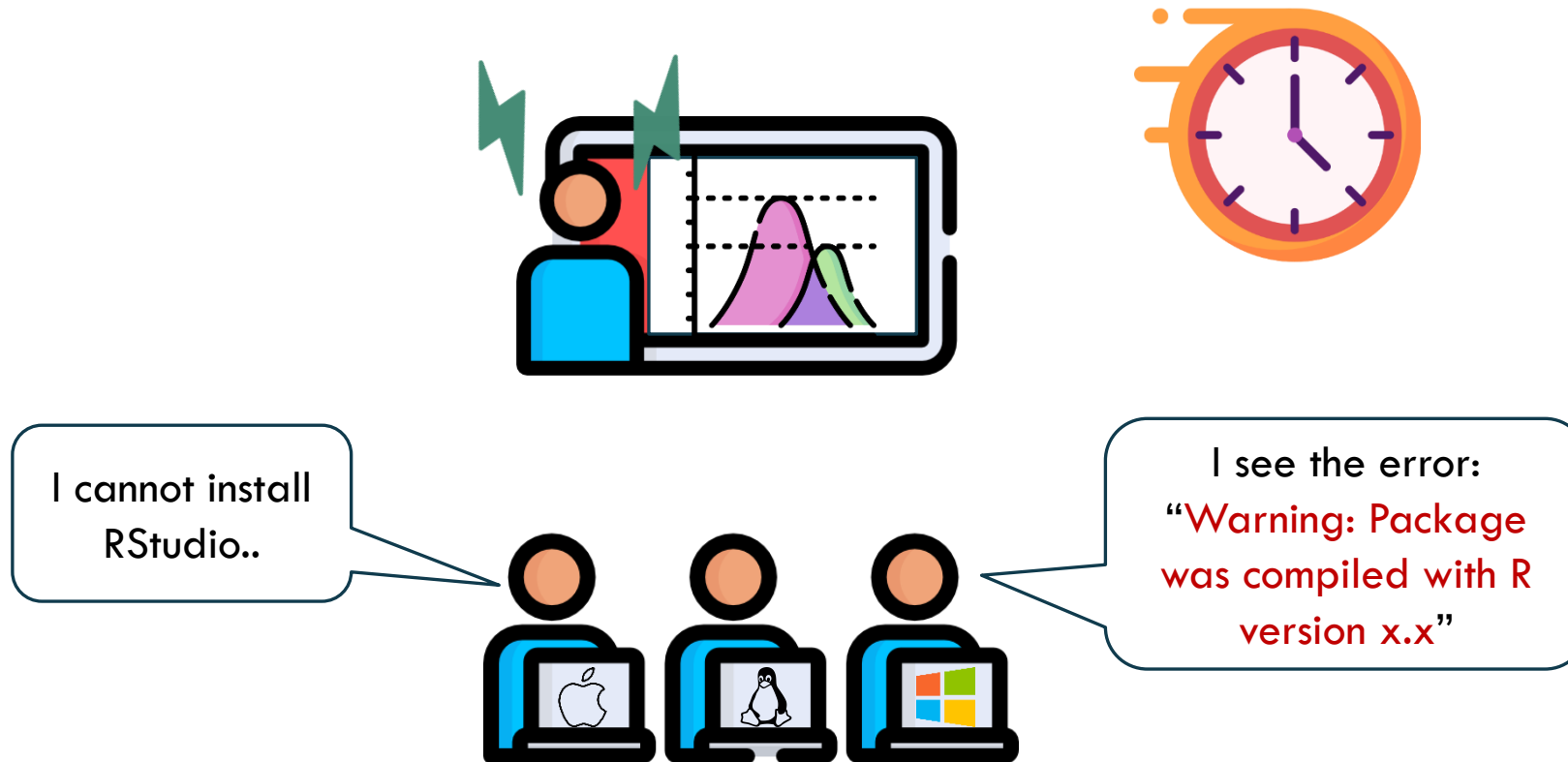


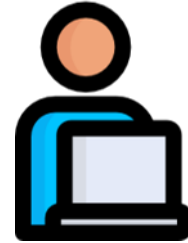
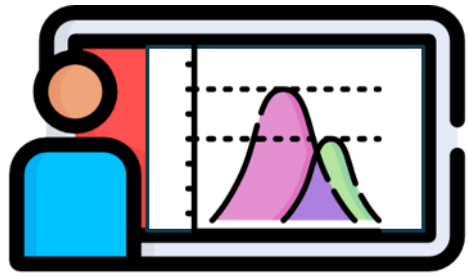
CASE 2: HAS IT HAPPENED TO YOU THAT...?

Today, we will
learn about the
normal distribution!
Open R and follow
the code!



CASE 2: HAS IT HAPPENED TO YOU THAT...?





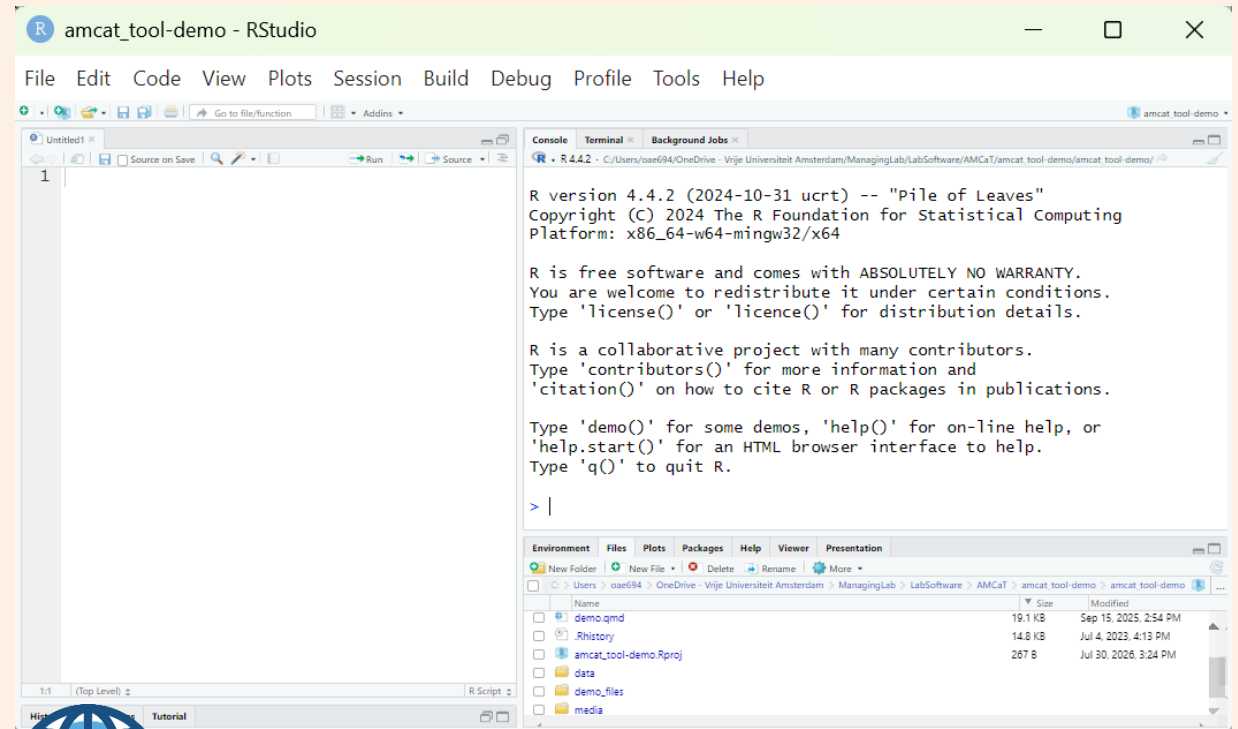
Student 1



Student 2



Student 3

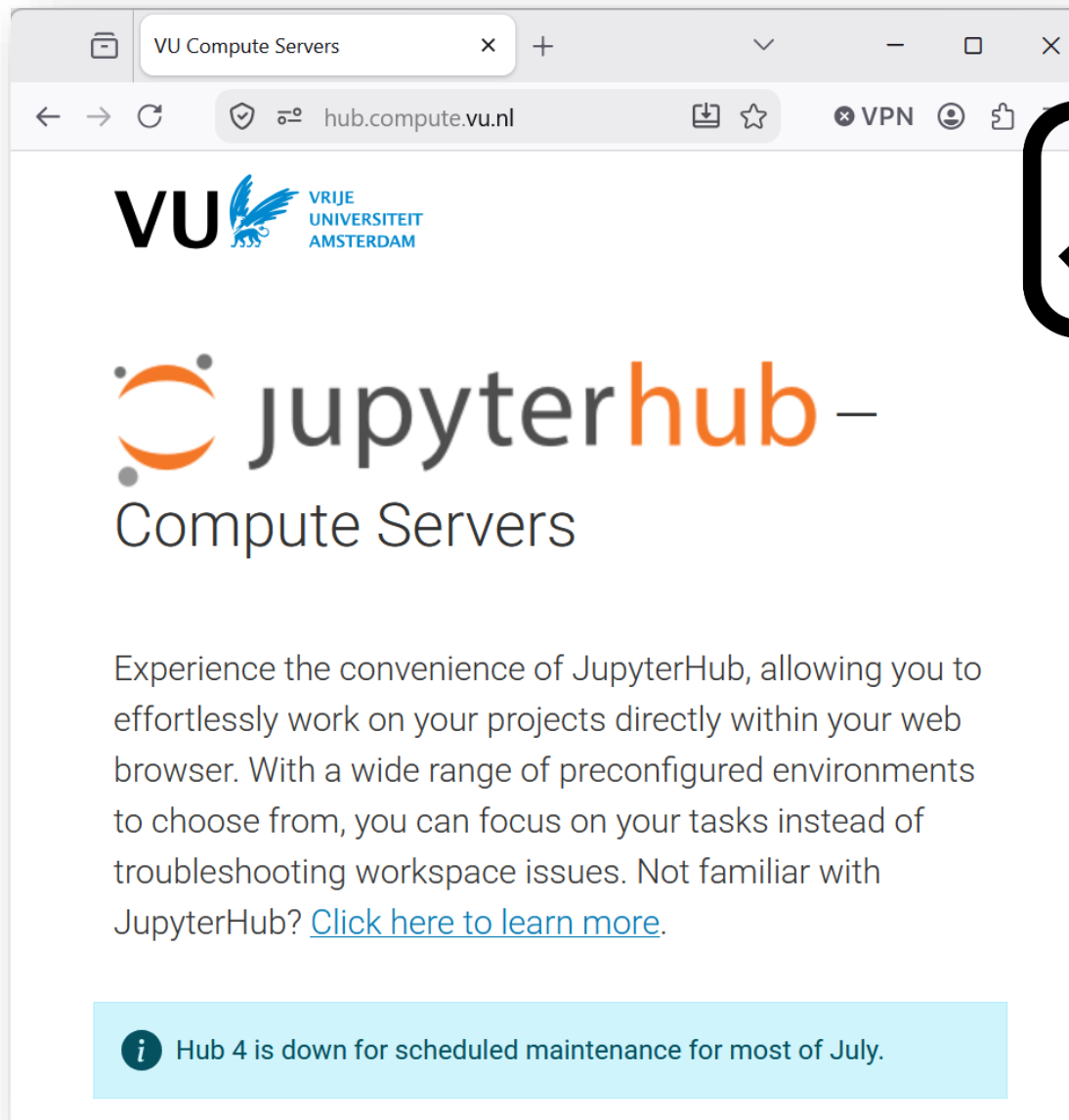


- ❖ All students access the same R and RStudio version and environment independently.
- ❖ You only focus on your intended learning outcomes.



2.2 ACCESS

You can access VU JupyterHub here: <https://hub.compute.vu.nl/>



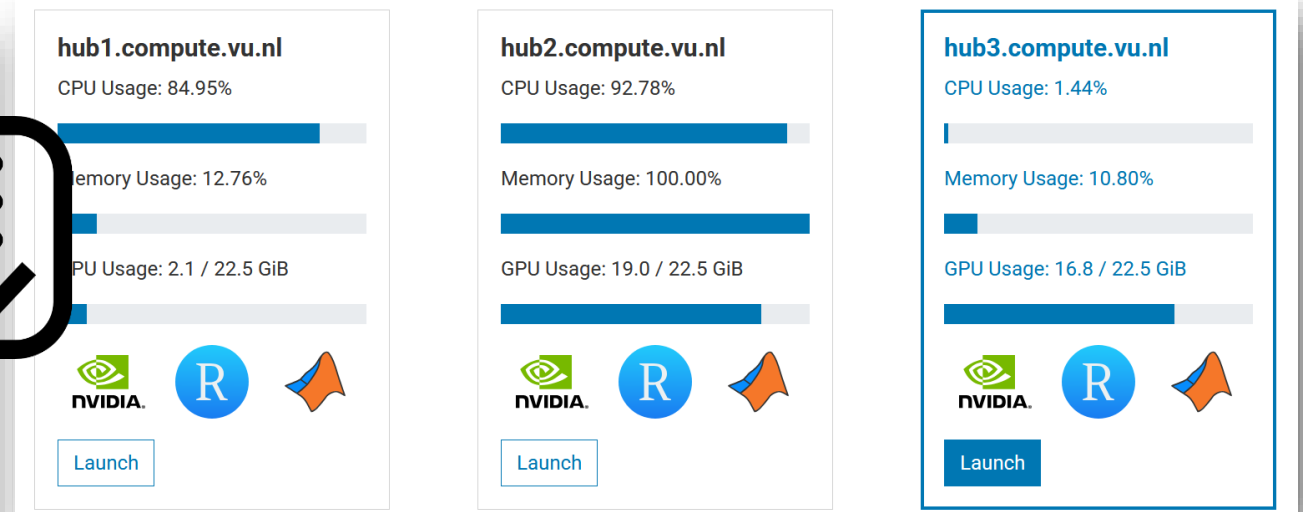
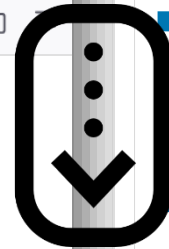
The screenshot shows the VU JupyterHub website in a web browser. The browser's address bar displays 'hub.compute.vu.nl'. The page features the VU logo (Vrije Universiteit Amsterdam) and the JupyterHub logo. Below the logos, there is a paragraph of text describing the service and a link to learn more. At the bottom, a light blue banner contains an information icon and a message about Hub 4 being down for maintenance.

VU VRIJE UNIVERSITEIT AMSTERDAM

jupyterhub – Compute Servers

Experience the convenience of JupyterHub, allowing you to effortlessly work on your projects directly within your web browser. With a wide range of preconfigured environments to choose from, you can focus on your tasks instead of troubleshooting workspace issues. Not familiar with JupyterHub? [Click here to learn more.](#)

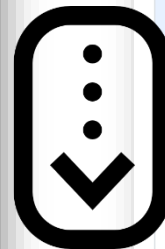
i Hub 4 is down for scheduled maintenance for most of July.



Three server status cards are displayed side-by-side. Each card shows the server name, CPU and Memory usage percentages, GPU usage (in GiB), and icons for NVIDIA, R, and JupyterLab. A 'Launch' button is at the bottom of each card.

Server	CPU Usage	Memory Usage	GPU Usage
hub1.compute.vu.nl	84.95%	12.76%	2.1 / 22.5 GiB
hub2.compute.vu.nl	92.78%	100.00%	19.0 / 22.5 GiB
hub3.compute.vu.nl	1.44%	10.80%	16.8 / 22.5 GiB

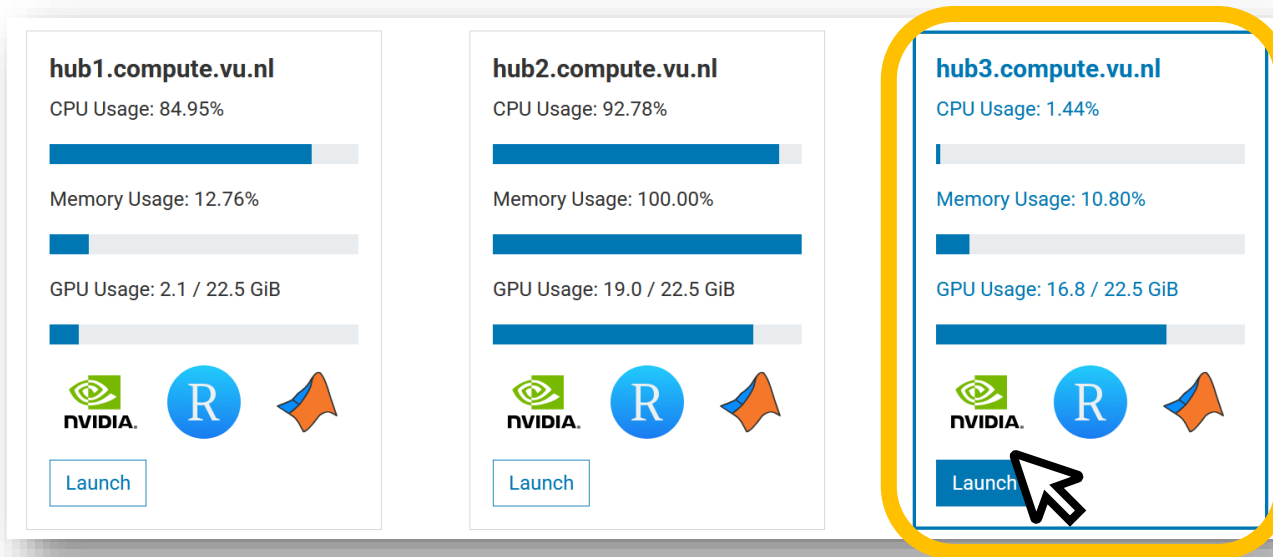
Frequently Asked Questions



Compute server transition: What are the new hardware specs? ^

Notable hardware upgrades:

- **CPU:** AMD EPYC 9375F with 32 physical cores and 64 threads (double the previous amount).
- **MEMORY:** 256GB DDR5 RAM (previously DDR4).
- **GPU:** NVIDIA L4 with 24GB GDDR6 memory (previously 8GB).



Server with lowest CPU load highlighted. Click on an instance to navigate to it. Displayed usage statistics are periodically updated.

Just enter your
VU credentials.



The screenshot shows a web browser window with the JupyterHub logo and the VU logo. Below the logos is a 'Sign in' form with fields for 'Username' (containing 'VUnetID') and 'Password', and a 'Sign in' button.

JupyterHub

hub3.compute.vu.nl/hub/spawn

jupyterhub Home Token oae694 Logout

Environment Selector

Select the environment you want to use. If you are using JupyterHub for a course, select the environment you were instructed to use by your educator, or the one bearing the name of the course. You can also select one of the general-purpose environments.

Select an Image:

Supply Chain Lab

Start

JupyterHub

hub3.compute.vu.nl/hub/spawn

jupyterhub Home Token oae694 Logout

Environment Selector

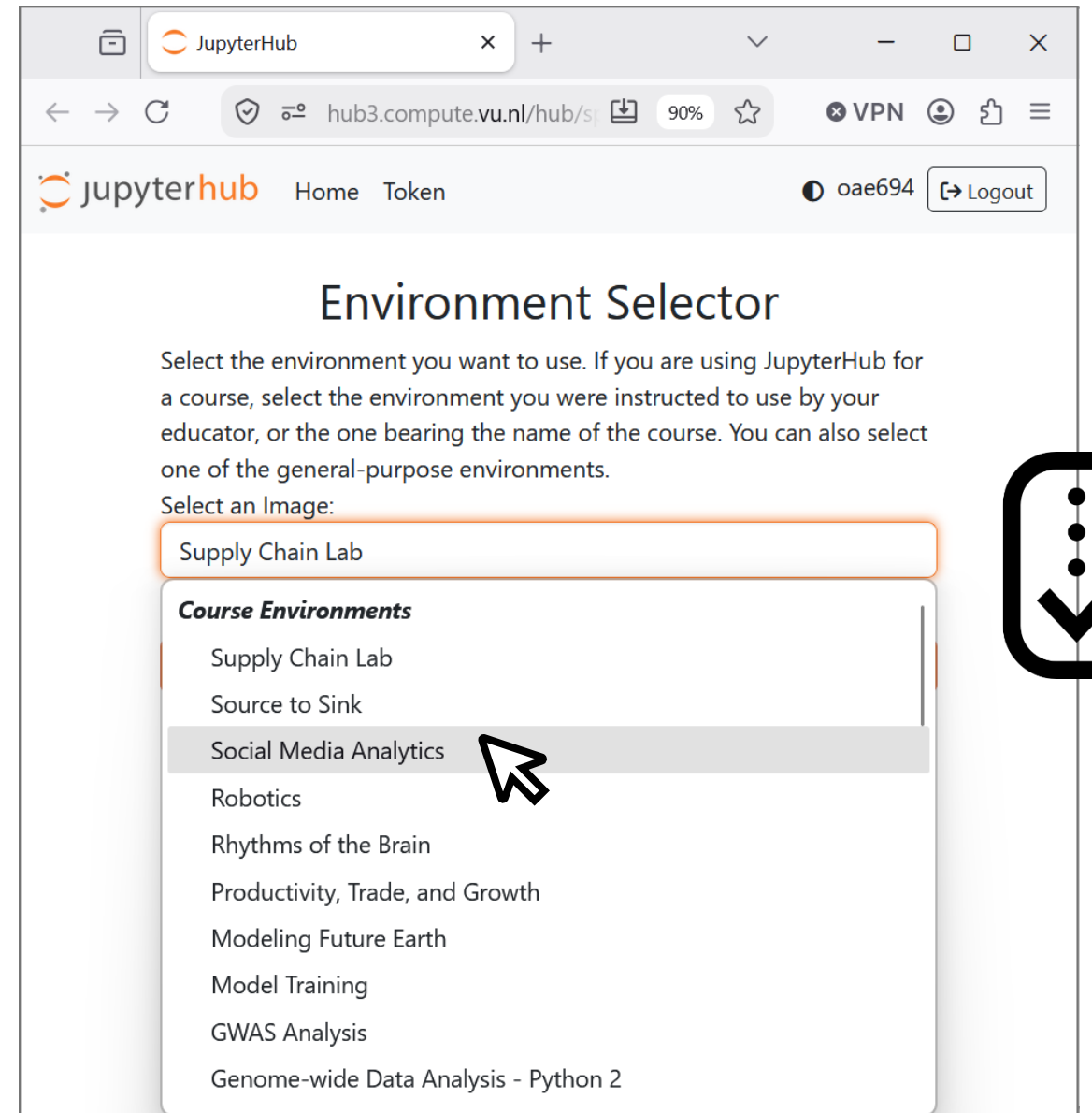
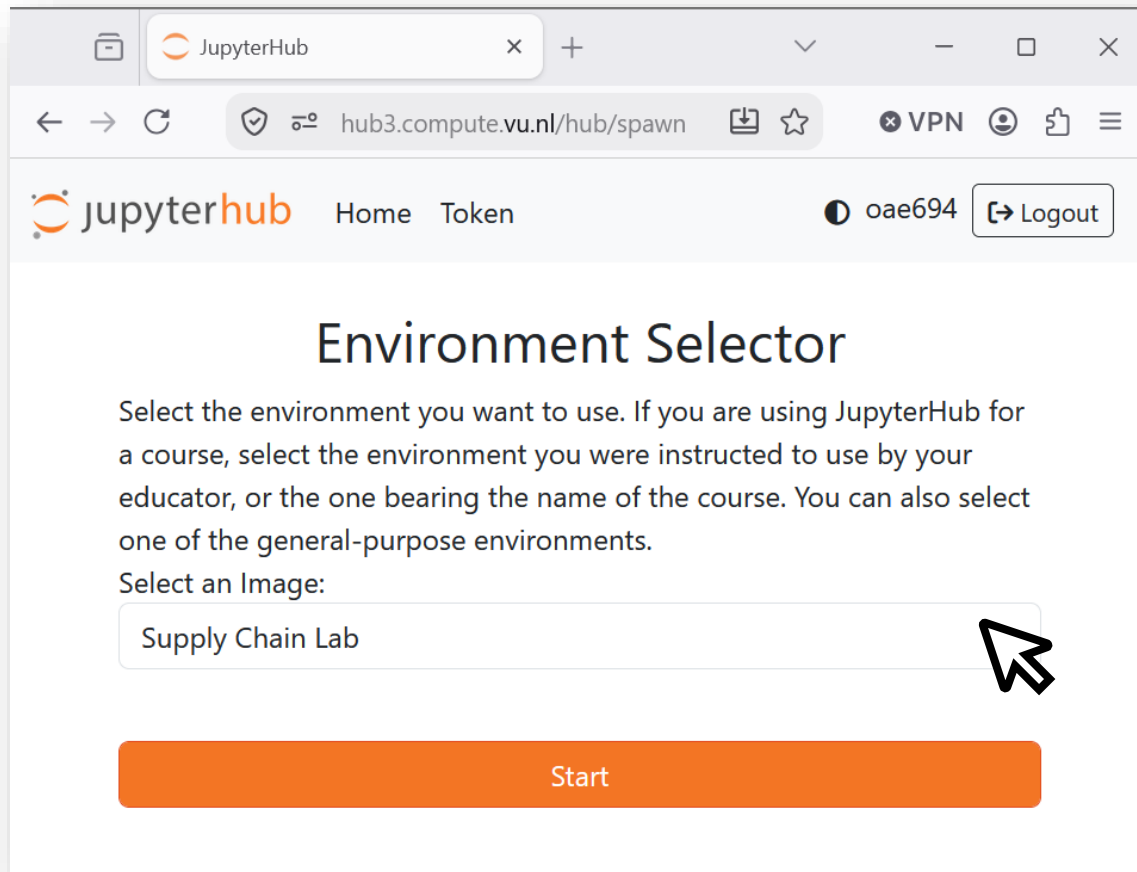
Select the environment you want to use. If you are using JupyterHub for a course, select the environment you were instructed to use by your educator, or the one bearing the name of the course. You can also select one of the general-purpose environments.

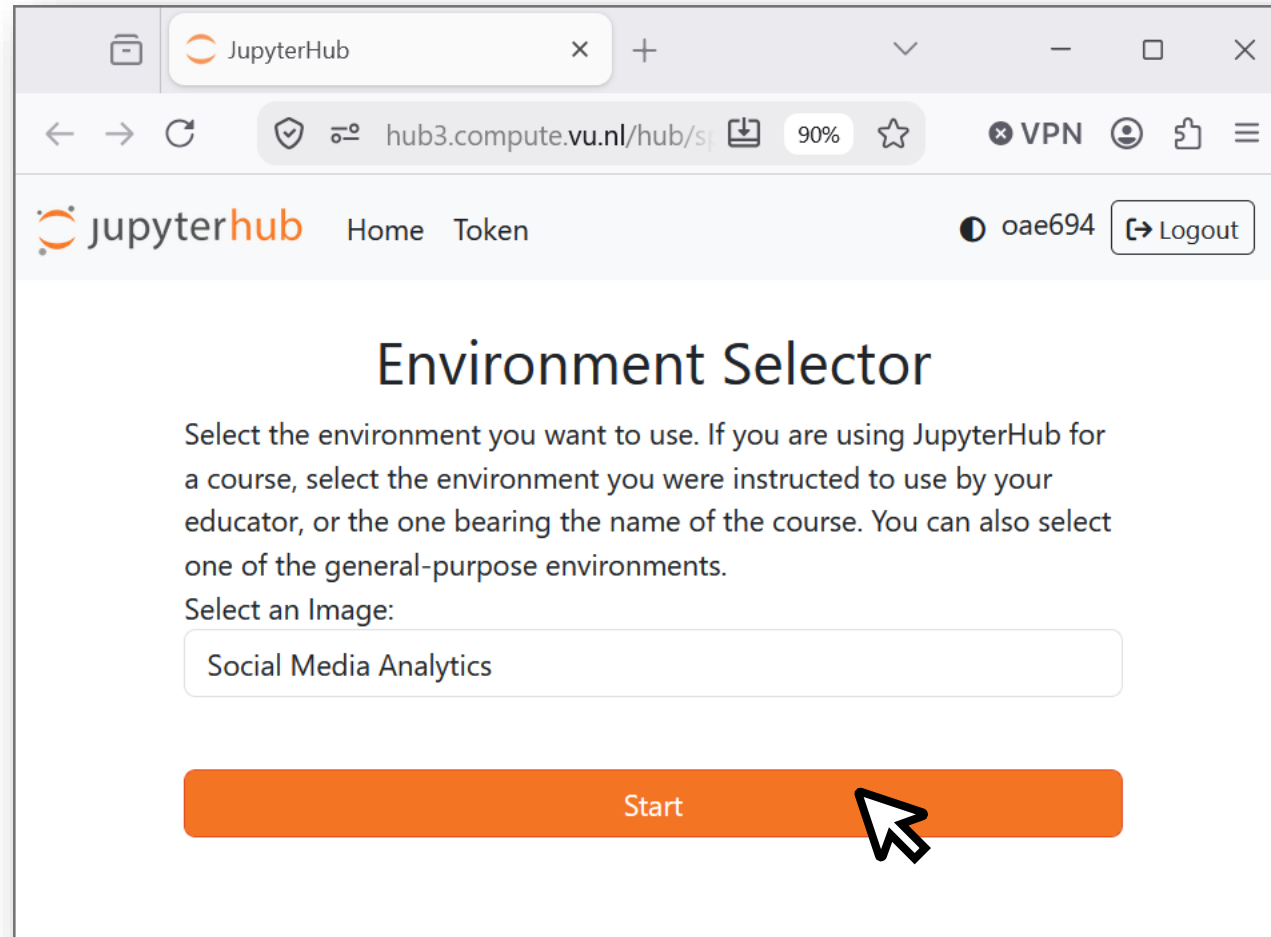
Select an Image:

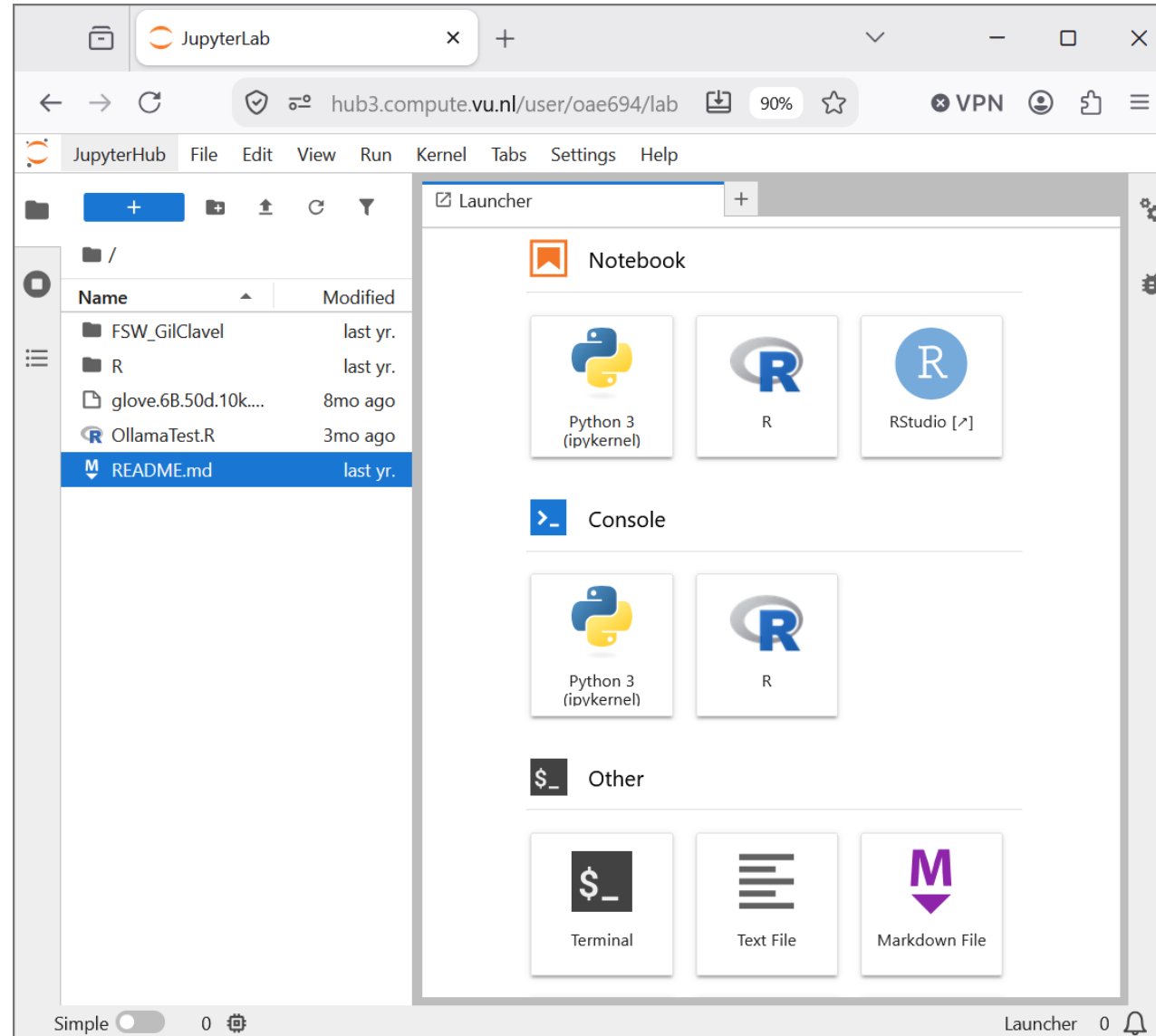
Supply Chain Lab

Course Environments

- Supply Chain Lab
- Source to Sink
- Social Media Analytics
- Robotics
- Rhythms of the Brain
- Productivity, Trade, and Growth
- Modeling Future Earth
- Model Training

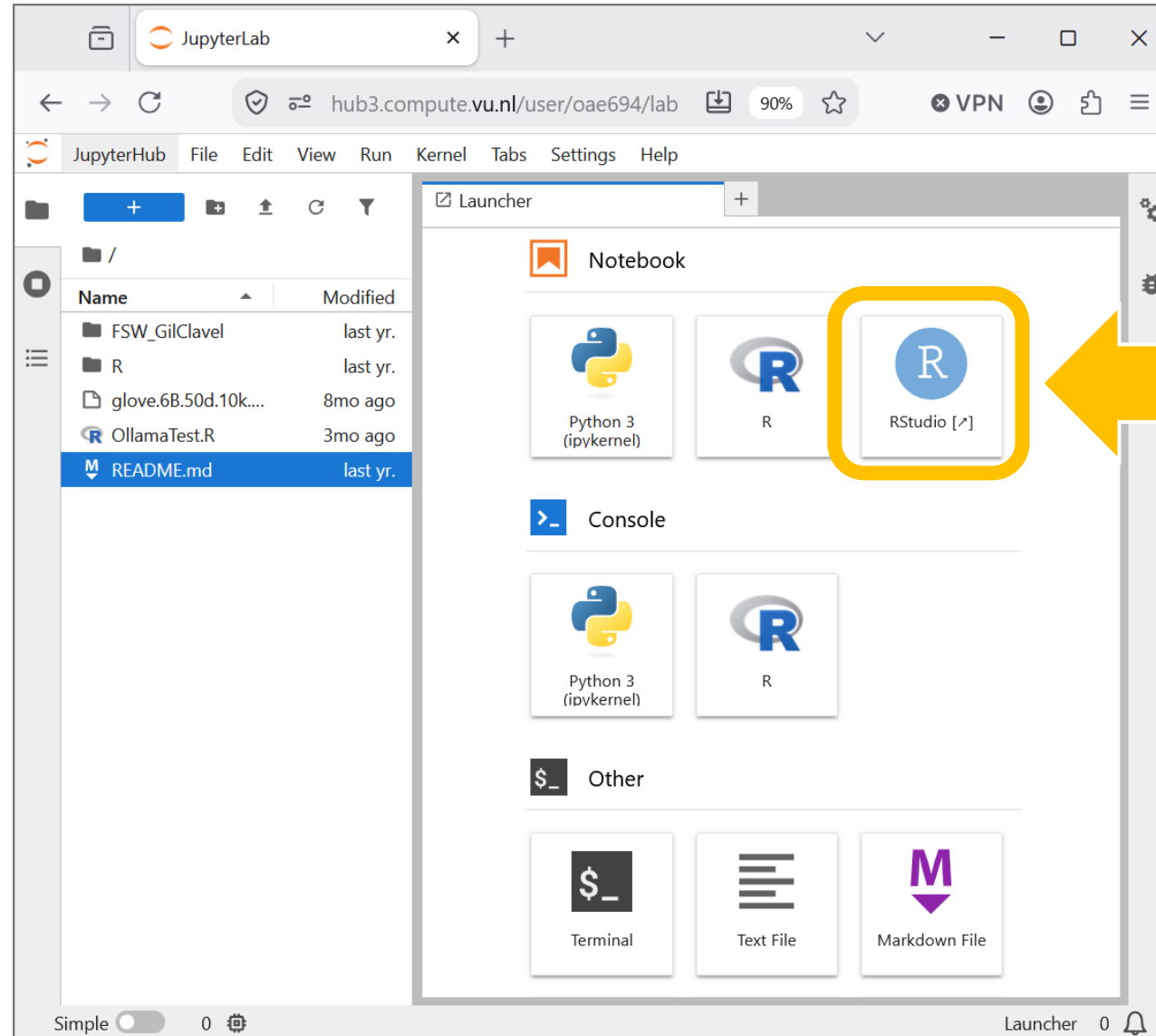








2.3 ENVIRONMENT USAGE



You can access
RStudio from
here.

RStudio Server

hub3.compute.vu.nl/user/oae694/rstudio/

File Edit Code View Plots Session Build Debug Profile Tools Help

oae694 Project: (None)

Untitled1

Source on Save Run

1

Console

R 4.4.3 · ~/

R version 4.4.3 (2025-02-28) -- "Trophy Case"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: x86_64-conda-linux-gnu

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> |

Environment Files Plots Connections Packages Help

Import Dataset 106 MiB

R Global Environment

Environment is empty

1:1 (Top Level) R Script

Viewer Presentation

RStudio Server

hub3.compute.vu.nl/user/oe694/rstudio/

90%

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function

Addins

oae694

Project: (None)

Filter

	Package	LibPath	Version	Priority	Depends
askpass	askpass	/opt/conda/envs/r_sma_env/lib/R/library	1.2.1	A	NA
assertthat	assertthat	/opt/conda/envs/r_sma_env/lib/R/library	0.2.1	A	NA
backports	backports	/opt/conda/envs/r_sma_env/lib/R/library	1.5.0	A	R (>= 3.0.0)
base	base	/opt/conda/envs/r_sma_env/lib/R/library	4.4.3	base	NA
base64enc	base64enc	/opt/conda/envs/r_sma_env/lib/R/library	0.1-6	A	R (>= 2.9.0)
bit	bit	/opt/conda/envs/r_sma_env/lib/R/library	4.6.0	A	R (>= 3.4.0)
bit64	bit64	/opt/conda/envs/r_sma_env/lib/R/library	4.6.0-1	A	R (>= 3.4.0), bit (>= 4.0.0)
bitops	bitops	/opt/conda/envs/r_sma_env/lib/R/library	1.0-9	A	NA
blob	blob	/opt/conda/envs/r_sma_env/lib/R/library	1.3.0	A	NA
boot	boot	/opt/conda/envs/r_sma_env/lib/R/library	1.3-32	recommended	R (>= 3.0.0), graphics, stats
broom	broom	/opt/conda/envs/r_sma_env/lib/R/library	1.0.12	A	R (>= 4.1)
bslib	bslib	/opt/conda/envs/r_sma_env/lib/R/library	0.10.0	A	R (>= 2.10)
cachem	cachem	/opt/conda/envs/r_sma_env/lib/R/library	1.1.0	A	NA
callr	callr	/opt/conda/envs/r_sma_env/lib/R/library	3.7.6	A	R (>= 3.4)
caret	caret	/opt/conda/envs/r_sma_env/lib/R/library	7.0-1	A	ggplot2, lattice (>= 0.20)
caTools	caTools	/opt/conda/envs/r_sma_env/lib/R/library	1.18.3	A	R (>= 3.6.0)
cellranger	cellranger	/opt/conda/envs/r_sma_env/lib/R/library	1.1.0	A	R (>= 3.0.0)
class	class	/opt/conda/envs/r_sma_env/lib/R/library	7.3-23	recommended	R (>= 3.0.0), stats, utils
cli	cli	/opt/conda/envs/r_sma_env/lib/R/library	3.6.5	A	R (>= 3.4)

Showing 2 to 15 of 222 entries, 10 total columns

Console

Terminal

Background Jobs

R 4.4.3 · ~/

R version 4.4.3 (2025-02-28) -- "Trophy Case"

Copyright (C) 2025 The R Foundation for Statistical Computing

Platform: x86_64-conda-linux-gnu

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

Session restored from your saved work on 2026-Jul-31 19:05:52 UTC (1 hour ago)

> View(installed.packages())

Environment

Files

Plots

Connections

Packages

Help

New Folder

New File

Upload

Delete

Rename

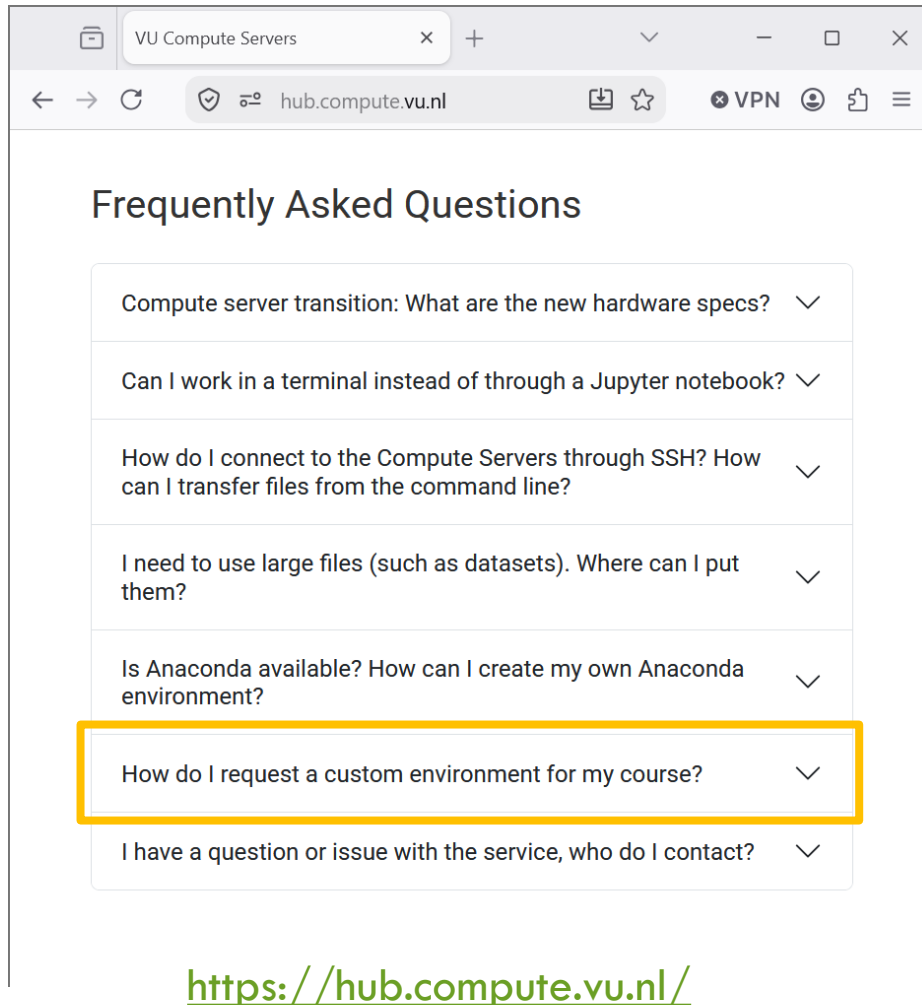
More

Home

	Name	Size	Modified
	.Rhistory	16.5 KB	Jul 31, 2026, 2:58 PM
	OllamaTest.R	3.3 KB	Apr 2, 2026, 2:31 PM
	README.md	200 B	Dec 4, 2025, 2:20 PM

VU TOOLS | DR. SOFIA GIL-CLAVEL | 2026 28

REQUEST AN ENVIRONMENT FOR YOUR COURSE



How do I request a custom environment for my course? ^

Do you want to use JupyterHub for a course you are teaching? You can request a custom environment for your course by filling in this form:

- [JupyterHub – Request an online development environment for your course.](#)

Mention the software you'd like to have included (e.g. libraries for Python, R, Matlab) and their versions, if you need specific ones. For Python libraries, you can, for example, include the contents of your `requirements.txt` file, if you have one. Also mention other specific requirements for your course, such as a shared data directory.

You can also request a shared folder with the data you will be using during the course. Include that information in the email.



2.4 SCRIPTS, FOLDERS, AND STORAGE OF SMALL DATA

 **Be aware of your disk quota!** Typically, users have a 3GB limit on their home folder. If you want to work with large datasets, read below about how to use the local data directory.

In your general folder ("/home/<VUNetID>") you have a disk quota of 3GB.

So, you must use this folder just **for your scripts and to allocate data momentarily**.

Source: <https://hub.compute.vu.nl/>

RStudio Server

hub3.compute.vu.nl/user/oea694/rstudio/

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Project: (None)

Untitled1

1

Console

R 4.4.3 ~/
Type `license()` or `licence()` for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type `contributors()` for more information and
`'citation()'` on how to cite R or R packages in publications.

Type `'demo'` for some demos, `'help()'` for on-line help, or
`'help.start'` for an HTML browser interface to help.
Type `'q()'` to quit R.

>

Environment Files Plots Connections Packages Help

Folder File Upload Delete Rename

Home

	Name	Size	Modified
☐	.Rhistory	16.5 KB	Jul 31, 2026, 2:58 PM
☐	OllamaTest.R	3.3 KB	Apr 2, 2026, 2:31 PM
☐	README.md	399 B	Dec 4, 2025, 2:30 PM
☐	glove.6B.50d.10k.w2v.txt	4.1 MB	Nov 12, 2025, 6:21 PM
☐	FSW_GilClavel		
☐	R		
☐	VUtools		

1

2. Create a new folder called "VUtools".

3. Access it.

1:1 (Top Level) R Script

Viewer Presentation

RStudio Server

hub3.compute.vu.nl/user/oe694/rstudio/

File Edit Code View Plots Session Build Debug Profile Tools Help

Untitled1 x

1

Upload the R-Mardown file from the previous sessions
"TrackingYourLogic.Rmd"

Upload Files

Target directory:
~/VUtools Browse...

File to upload:
Browse... No file selected.

TIP: To upload multiple files or a directory, create a zip file. The zip file will be automatically expanded after upload.

OK Cancel

Environment Files Plots Connections Packages Help

Folder File Upload Delete Rename

Home > VUtools

	Name	Size	Modified
	..		
	TrackingYourLogic.Rmd	501 B	Jul 31, 2026, 3:13 PM

1:1 (Top Level) R Script

Viewer Presentation

EXERCISE 2.4

1. Render the file.
2. Were you able to open the pdf?
3. Where is the pdf stored?

EXERCISE 2.5

1. Add a chunk at the end of the R-Markdown file and save the ChickWeight as a csv.
2. Where is the ChickWeight data stored?
3. What would you have to modify in your script to change the storage folder to “/home/<VUNetID>”.



2.5 INSTALLING & LOADING R-PACKAGES

LIBRARIES LOCATION

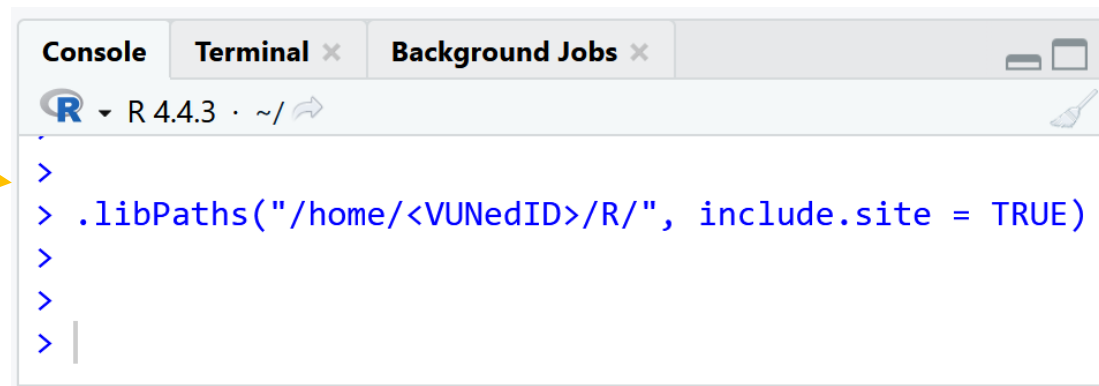
As you saw in the section “Environment usage”, all the libraries are located in:

“/opt/conda/envs/r_sma_env/lib/R/library”

As user, you do not have permissions to add anything to that folder, only VU IT can do so.

If you need some specific libraries for your specific project, you can add them in your general folder (“/home/<VUNetID>”). To do so, in RStudio do the following:

1. Create a folder called “R”.
2. Run the line:
3. Verify that the folder was included by running:
`.libPaths()`



```
R 4.4.3 · ~/
>
> .libPaths("/home/<VUNetID>/R/", include.site = TRUE)
>
>
>
```

INSTALLING R-PACKAGES

To install an R package, it is necessary to do so in our general folder ("/home/<VUNetID>"). For this, we will use the function `install.packages()` with the following parameters:

- `lib = "/home/<VUNetID>/R/"`
- `dependencies = TRUE`
- `destdir = "/home/<VUNetID>/R/"`
- `clean = TRUE`

EXERCISE 2.6

Add a new section to the R-Markdown. In it, do the following:

1. Add a new chunk and install the package ``janeaustenr``.
2. Load the package ``janeaustenr``.

RStudio Server

hub1.compute.vu.nl/user/oea694/rstudio/

VPN

R

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function

oae694

Project: (None)

TrackingYourLogic_answers.Rmd installed.packages()

Package	LibPath	Version	Priority
desc	/home/oea694/R	1.4.3	NA
janeaustenr	/home/oea694/R	1.0.0	NA
pkgbuild	/home/oea694/R	1.4.8	NA
pkgload	/home/oea694/R	1.5.3	NA
praise	/home/oea694/R	1.0.0	NA
rprojroot	/home/oea694/R	2.1.1	NA
askpass	/opt/conda/envs/r_sma_env/lib/R/library	1.2.1	NA
assertthat	/opt/conda/envs/r_sma_env/lib/R/library	0.2.1	NA
backports	/opt/conda/envs/r_sma_env/lib/R/library	1.5.0	NA
base	/opt/conda/envs/r_sma_env/lib/R/library	4.4.3	base
base64enc	/opt/conda/envs/r_sma_env/lib/R/library	0.1-6	NA
bit	/opt/conda/envs/r_sma_env/lib/R/library	4.6.0	NA
bit64	/opt/conda/envs/r_sma_env/lib/R/library	4.6.0-1	NA
bitops	/opt/conda/envs/r_sma_env/lib/R/library	1.0-9	NA
blob	/opt/conda/envs/r_sma_env/lib/R/library	1.3.0	NA
boot	/opt/conda/envs/r_sma_env/lib/R/library	1.3-32	recomm

Showing 1 to 16 of 227 entries, 16 total columns

Console Terminal Background Jobs

R 4.4.3

> View(installed.packages())

Environment Files Plots Connections Packages Help

R: Search Paths for Packages

Find in Topic

.libPaths gets/sets the library trees within which packages are looked for.

Usage

.libPaths(new, include.site = TRUE)

.Library



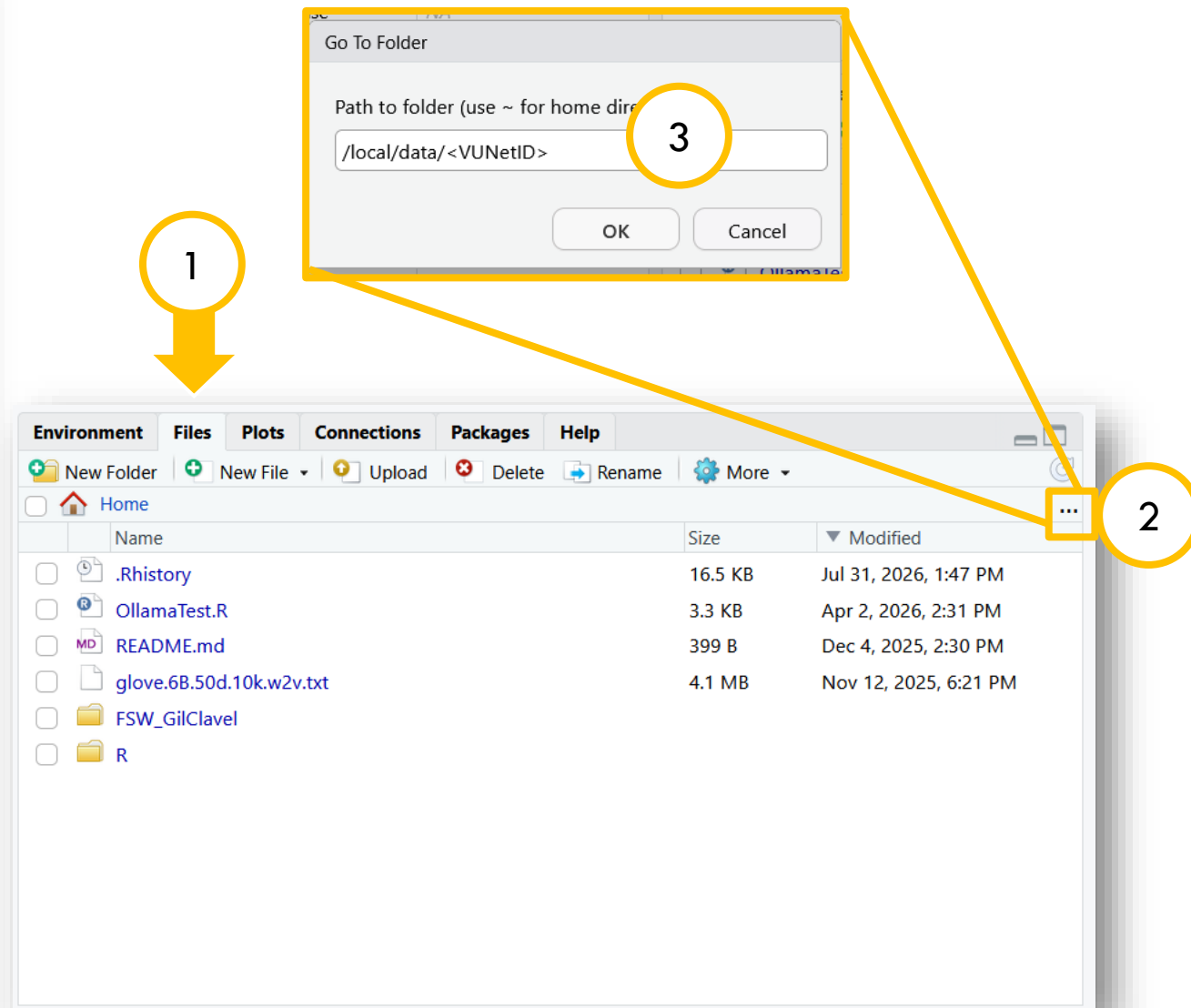
2.6 STORAGE OF BIGGER DATA

I need to use large files (such as datasets). Where can I put them? ^

You can create a directory in `/local/data/<VUNetID>/` (such as `/local/data/abc123/my_dataset/`), and place your dataset there using SSH/SFTP. Usually, this directory is automatically created for you on login. Inside you can find a `README.txt` file containing details about the `/local` folder. Be aware that these folders are local to each compute instance, and are not synchronized. Therefore, they are using the instance's own storage space.

Make sure that there is enough space for your dataset before placing it (you can verify by running `df -h /local/data` in the Terminal).

The compute instances make up a shared service. Please remove your folders from `/local/data/<VUNetID>/` when you don't need them anymore. This way, you can leave storage space for others. The directories in `/local/data` are removed if the directory name is not a valid VUNetID. Every first day of the month, the directories are moved to `/local/.old/`. If your data is missing, check this directory first and move your relevant files back. You are responsible of keeping your data backed up.

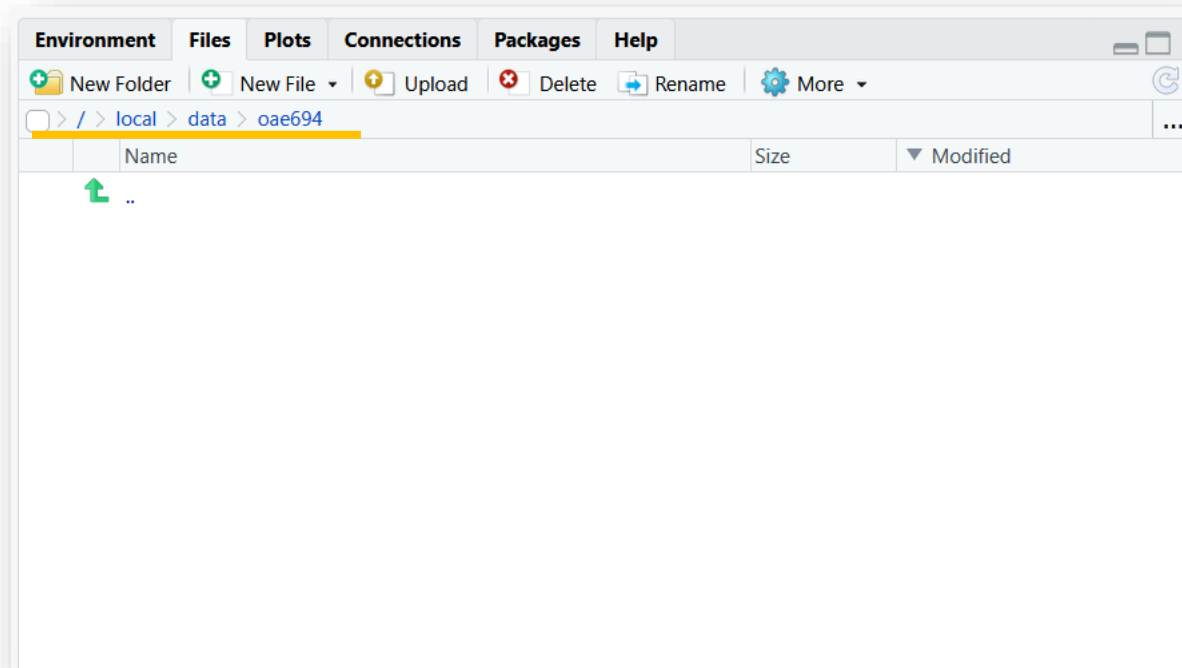


I need to use large files (such as datasets). Where can I put them? 

You can create a directory in `/local/data/<VUNetID>/` (such as `/local/data/abc123/my_dataset/`), and place your dataset there using SSH/SFTP. Usually, this directory is automatically created for you on login. Inside you can find a `README.txt` file containing details about the `/local` folder. Be aware that these folders are local to each compute instance, and are not synchronized. Therefore, they are using the instance's own storage space.

Make sure that there is enough space for your dataset before placing it (you can verify by running `df -h /local/data` in the Terminal).

The compute instances make up a shared service. Please remove your folders from `/local/data/<VUNetID>/` when you don't need them anymore. This way, you can leave storage space for others. The directories in `/local/data` are removed if the directory name is not a valid VUNetID. Every first day of the month, the directories are moved to `/local/.old/`. If your data is missing, check this directory first and move your relevant files back. You are responsible of keeping your data backed up.



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Make sure that there is enough space for your dataset before placing it (you can verify by running `df -h /local/data` in the Terminal).

The compute instances make up a shared service. Please remove your folders from `/local/data/<VUNetID>/` when you don't need them anymore. This way, you can leave storage space for others. The directories in `/local/data` are removed if the directory name is not a valid VUNetID. Every first day of the month, the directories are moved to `/local/.old/`. If your data is missing, check this directory first and move your relevant files back. You are responsible of keeping your data backed up.

The screenshot shows the JupyterLab web interface. At the top, a yellow box contains the text: "Before uploading any data there, we are going to check whether there is enough space." Below this, on the left sidebar, a file browser shows a directory listing with files like 'FSW_GilClavel', 'R', 'glove.6B.50d.10k..', 'OllamaTest.R', and 'README.md'. A yellow box with the text "1. Go back to main screen." is positioned over the file browser. In the center, a yellow box with the text "2. Enter the 'Terminal'." has a large yellow arrow pointing down to a 'Terminal' icon (a black square with a white '\$_') in the bottom panel. Other icons in the bottom panel include 'Text File' and 'Markdown File'. The bottom status bar shows 'Simple' mode, a toggle switch, and a 'Launcher' button.

Before uploading any data there, we are going to check whether there is enough space.

1. Go back to main screen.

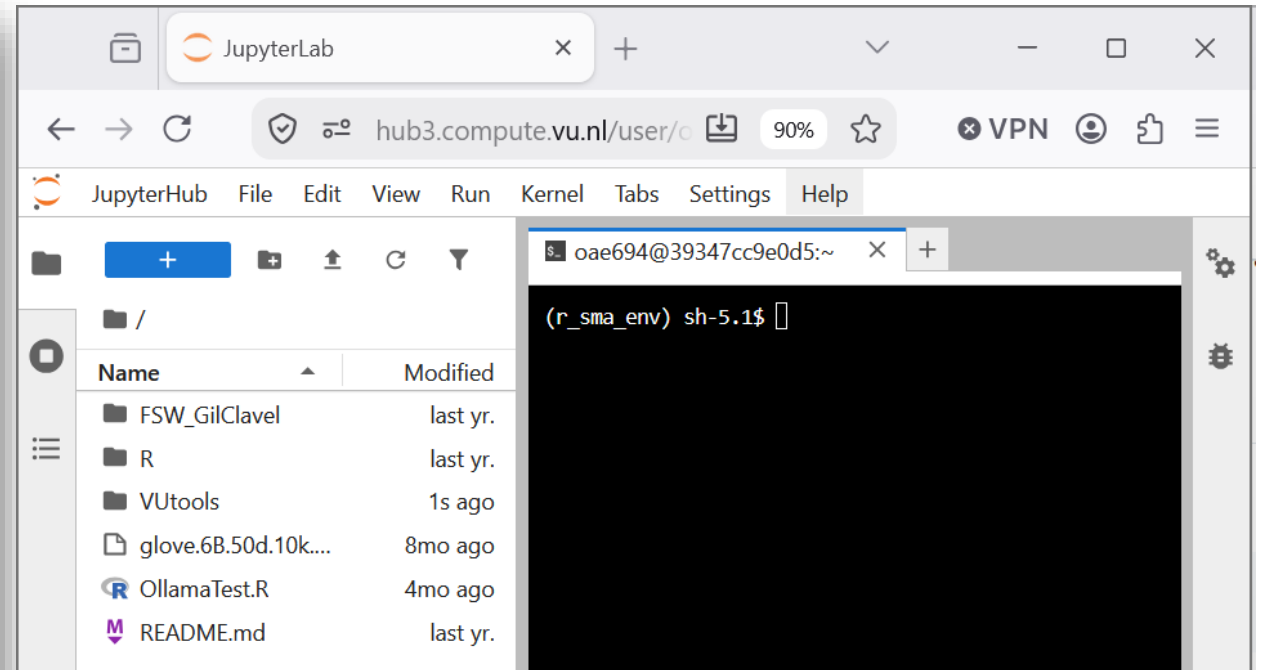
2. Enter the "Terminal".

I need to use large files (such as datasets). Where can I put them?

You can create a directory in `/local/data/<VUNetID>/` (such as `/local/data/abc123/my_dataset/`), and place your dataset there using SSH/SFTP. Usually, this directory is automatically created for you on login. Inside you can find a `README.txt` file containing details about the `/local` folder. Be aware that these folders are local to each compute instance, and are not synchronized. Therefore, they are using the instance's own storage space.

Make sure that there is enough space for your dataset before placing it (you can verify by running `df -h /local/data` in the Terminal).

The compute instances make up a shared service. Please remove your folders from `/local/data/<VUNetID>/` when you don't need them anymore. This way, you can leave storage space for others. The directories in `/local/data` are removed if the directory name is not a valid VUNetID. Every first day of the month, the directories are moved to `/local/.old/`. If your data is missing, check this directory first and move your relevant files back. You are responsible of keeping your data backed up.



Now, you just need to type `df -h /local/data` in the terminal:

These are spaces.

```
(r_sma_env) sh-5.1$ df -h /local/data
Filesystem              Size  Used Avail Use% Mounted on
/dev/mapper/vg_local-lv_local 1.5T 332G 1.2T  23% /local
(r_sma_env) sh-5.1$
```

EXERCISE 2.7

1. Go back to RStudio.
2. Add a chunk to the R-Markdown “TrackingYourLogic.Rmd”.
3. Run the lines:

```
JA<-janeaustenr::austen_books()
```

4. Save the data frame JA in the folder “/local/data/<VUNetID>/”.

I need to use large files (such as datasets). Where can I put them? 

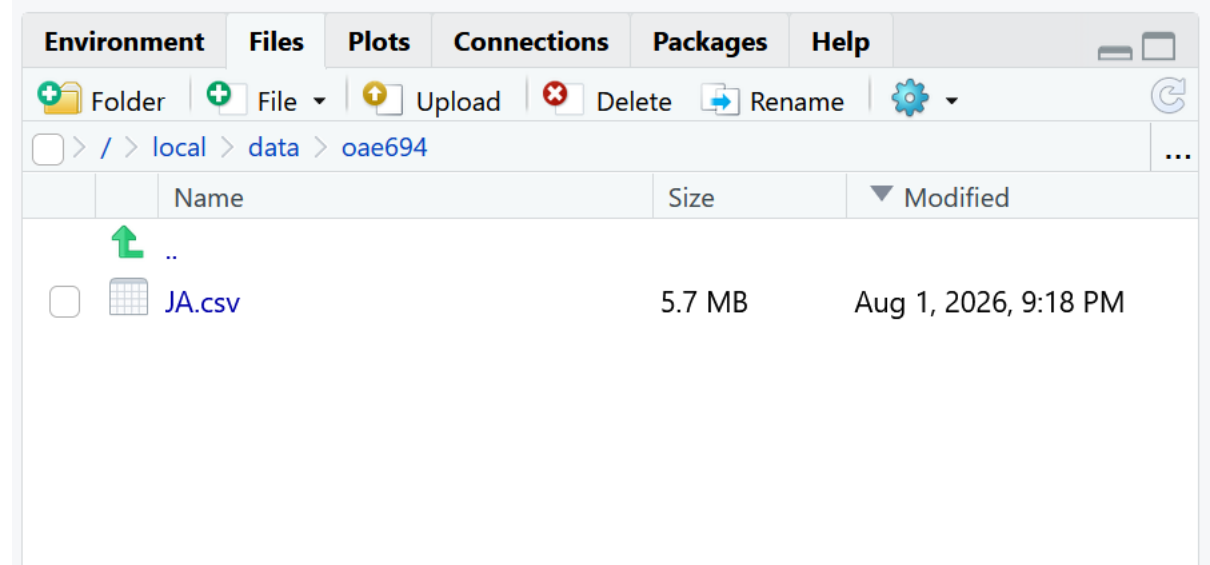
You can create a directory in `/local/data/<VUNetID>/` (such as `/local/data/abc123/my_dataset/`), and place your dataset there using SSH/SFTP. Usually, this directory is automatically created for you on login. Inside you can find a `README.txt` file containing details about the `/local` folder. Be aware that these folders are local to each compute instance, and are not synchronized. Therefore, they are using the instance's own storage space.

Make sure that there is enough space for your dataset before placing it (you can verify by running `df -h /local/data` in the Terminal).

The compute instances make up a shared service. Please remove your folders from `/local/data/<VUNetID>/` when you don't need them anymore. This way, you can leave storage space for others. The directories in `/local/data` are removed if the directory name is not a valid VUNetID. Every first day of the month, the directories are moved to `/local/.old/`. If your data is missing, check this directory first and move your relevant files back. You are responsible of keeping your data backed up.

Let's delete your data!

1. Go to the folder “/local/data/<VUNetID>”.



2. Select the item and press “Delete”.



2.7 DOWNLOADING FILES

JupyterLab

hub3.compute.vu.nl/user/oea694/lab

VPN

JupyterHubFileEditViewRunKernelTabsSettingsHelp

+

+

↑

↺

⏏

/

Name	Modified
FSW_GilClavel	last yr.
R	32m ago
VUtools	34m ago
glove.6B.50d.10k....	8mo ago
OllamaTest.R	4mo ago
README.md	last yr.

Launcher

Notebook



Python 3
(ipykernel)



R



RStudio [↗]

Console



Python 3
(ipykernel)



R

Other













Simple 0 

Launcher 0 

VU TOOLS | DR. SOFIA GIL-CLAVEL | 2026

49

JupyterLab interface showing the file browser on the left and the main workspace on the right.

File Browser (Left Panel):

- Root directory: /
- Files and folders:
 - FSW_GilClavel (last yr.)
 - R (32m ago)
 - VUtools (34m ago)
 - glove.6B.50d.1
 - OllamaTest.R
 - README.md

Main Workspace (Right Panel):

Launcher

Notebook

- Python 3 (ipykernel)
- R
- RStudio [↗]

Console

- Python 3 (ipykernel)
- R

Other

Simple ☐ 0

Launcher 0

JupyterLab

hub3.compute.vu.nl/user/oe694/lab/tree/VUtools/ChickWeight.csv

JupyterHub File Edit View Run Kernel Tabs Settings Help

/ VUtools /

Name	Modified
ChickWeight.csv	2d ago
TrackingYourLogic...	2d ago
TrackingYourLogic...	yesterday

Launcher ChickWeight.csv

	A	B	C	D	E
1		weight	Time	Chick	Diet
2	1	42	0	1	1
3	2	51	2	1	1
4	3	59	4	1	1
5	4	64	6	1	1
6	5	76	8	1	1
7	6	93	10	1	1
8	7	106	12	1	1
9	8	125	14	1	1
10	9	149	16	1	1
11	10	171	18	1	1
12	11	199	20	1	1
13	12	205	21	1	1
14	13	40	0	2	1
15	14	49	2	2	1
16	15	58	4	2	1
17	16	72	6	2	1
18	17	84	8	2	1
19	18	103	10	2	1
20	19	122	12	2	1

Simple 0

ChickWeight.csv 0

JupyterLab interface showing the File menu open. The browser address bar displays `hub3.compute.vu.nl/user/oeae694/lab/tree/VUtools/ChickWeight.csv`.

The File menu options are:

- New Launcher (Ctrl+Shift+L)
- Open from Path...
- Open from URL...
- Open Recent
- New View for CSV File
- New Console for Activity
- Close Tab (Alt+W)
- Close and Shut Down (Ctrl+Shift+Q)
- Close All Tabs
- Save CSV File (Ctrl+S)
- Save CSV File As... (Ctrl+Shift+S)
- Save All
- Reload CSV File from Disk
- Revert CSV File to Checkpoint...
- Rename CSV File...
- Duplicate CSV File
- Download
- Save and Export Notebook As
- Workspaces

The main area displays a CSV file named `ChickWeight.csv` with the following data:

	C	D	E
Time	Chick	Diet	
0	1	1	
2	1	1	
4	1	1	
6	1	1	
8	1	1	
10	1	1	
12	1	1	
14	1	1	
16	1	1	
18	1	1	
20	1	1	
21	1	1	
0	2	1	
2	2	1	
4	2	1	
6	2	1	
8	2	1	
10	2	1	
12	2	1	

JupyterLab interface showing the File menu open. The browser address bar displays `hub3.compute.vu.nl/user/oe694/lab/tree/VUtools/ChickWeight.csv`. The File menu options include:

- New
- New Launcher (Ctrl+Shift+L)
- Open from Path...
- Open from URL...
- Open Recent
- New View for CSV File
- New Console for Activity
- Close Tab (Alt+W)
- Close and Shut Down (Ctrl+Shift+Q)
- Close All Tabs
- Save CSV File (Ctrl+S)
- Save CSV File As... (Ctrl+Shift+S)
- Save All
- Reload CSV File from Disk
- Revert CSV File to Checkpoint...
- Rename CSV File...
- Duplicate CSV File
- Download** (highlighted by a mouse cursor)
- Save and Export Notebook As...
- Workspaces

The background shows a CSV file named `ChickWeight.csv` with the following data:

	C	D	E
Time	Chick	Diet	
0	1	1	
2	1	1	
4	1	1	
6	1	1	
8	1	1	
10	1	1	
12	1	1	
14	1	1	
16	1	1	
18	1	1	
20	1	1	
21	1	1	
0	2	1	
2	2	1	
4	2	1	
6	2	1	
8	2	1	
10	2	1	
12	2	1	

JupyterLab

hub3.compute.vu.nl/user/oe694/lab/tree/VUtools/ChickWeight.csv

JupyterHub File Edit View Run Kernel Tabs Settings Help

/ VUtools /

Name	Modified
ChickWeight.csv	2d ago
TrackingYourLogic...	2d ago
TrackingYourLogic...	yesterday

Launcher

ChickWeight.csv

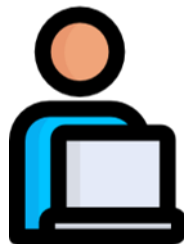
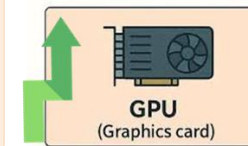
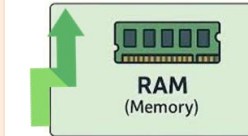
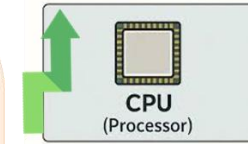
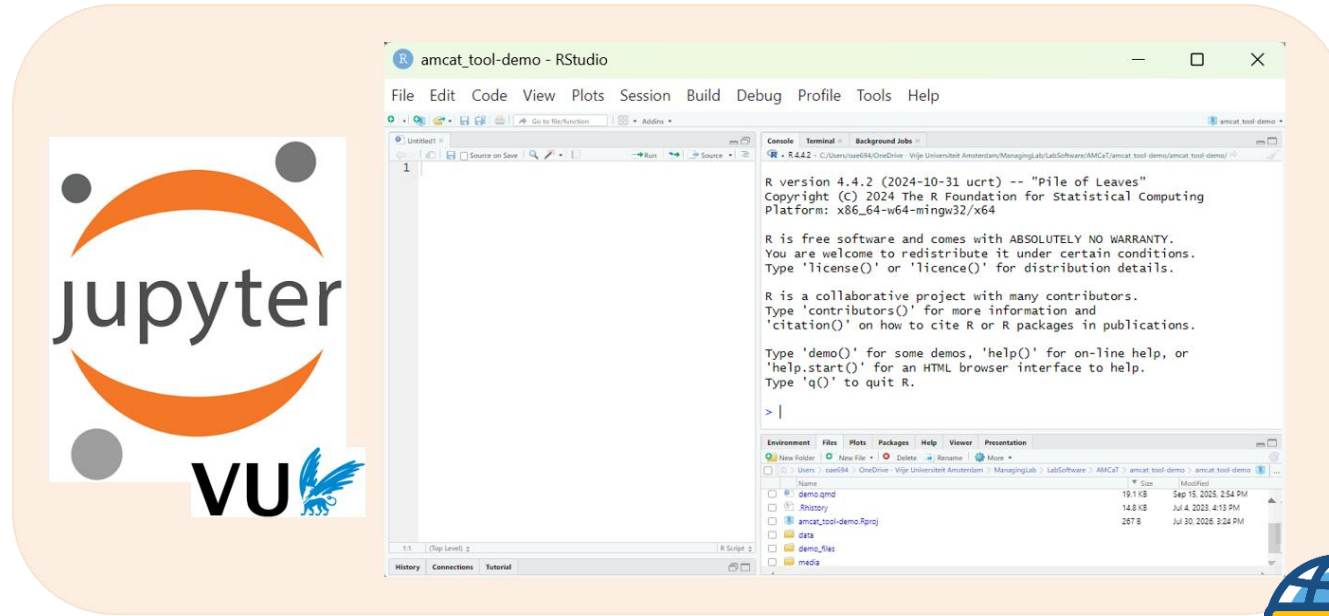
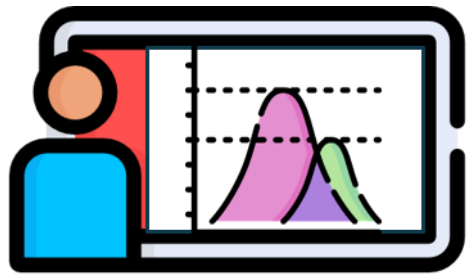
	A	B	C	D	E
1		weight	Time	Chick	Diet
2	1	42	0	1	1
3	2	51	2	1	1
4	3	59	4	1	1
5	4	64	6	1	1
6	5	76	8	1	1
7	6	93	10	1	1
8	7	106	12	1	1
9	8	125	14	1	1
10	9	149	16	1	1
11	10	171	18	1	1
12	11	199	20	1	1
13	12	205	21	1	1
14	13	40	0	2	1
15	14	49	2	2	1
16	15	58	4	2	1
17	16	72	6	2	1
18	17	84	8	2	1
19	18	103	10	2	1
20	19	122	12	2	1

ChickWeight.csv
Completed — 11,7 KB

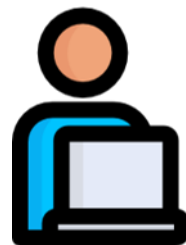
Show all downloads

Simple 0

ChickWeight.csv 0



Student 1

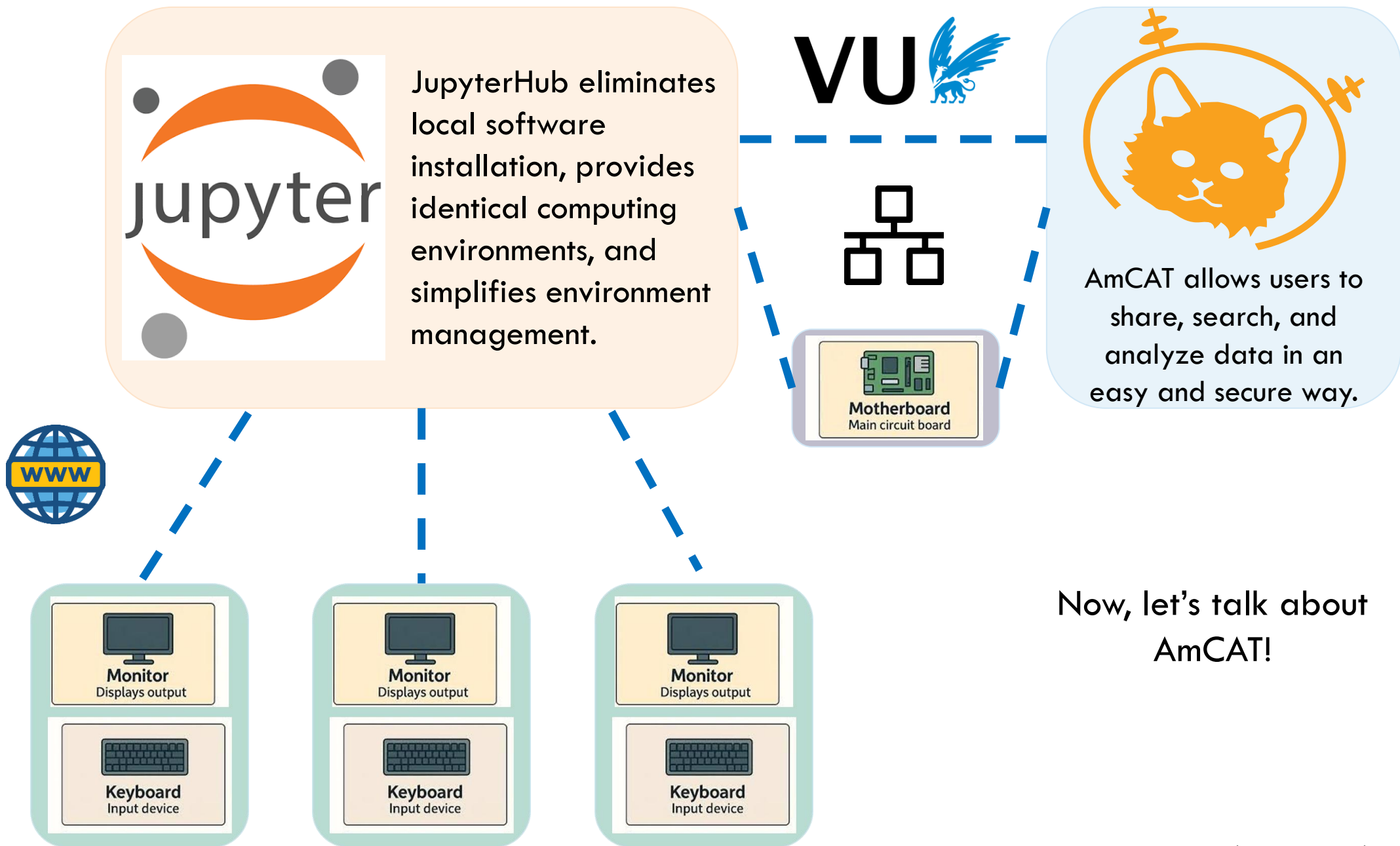


Student 2



Student 3





Now, let's talk about AmCAT!



Let's 1st sign
in to AmCAT

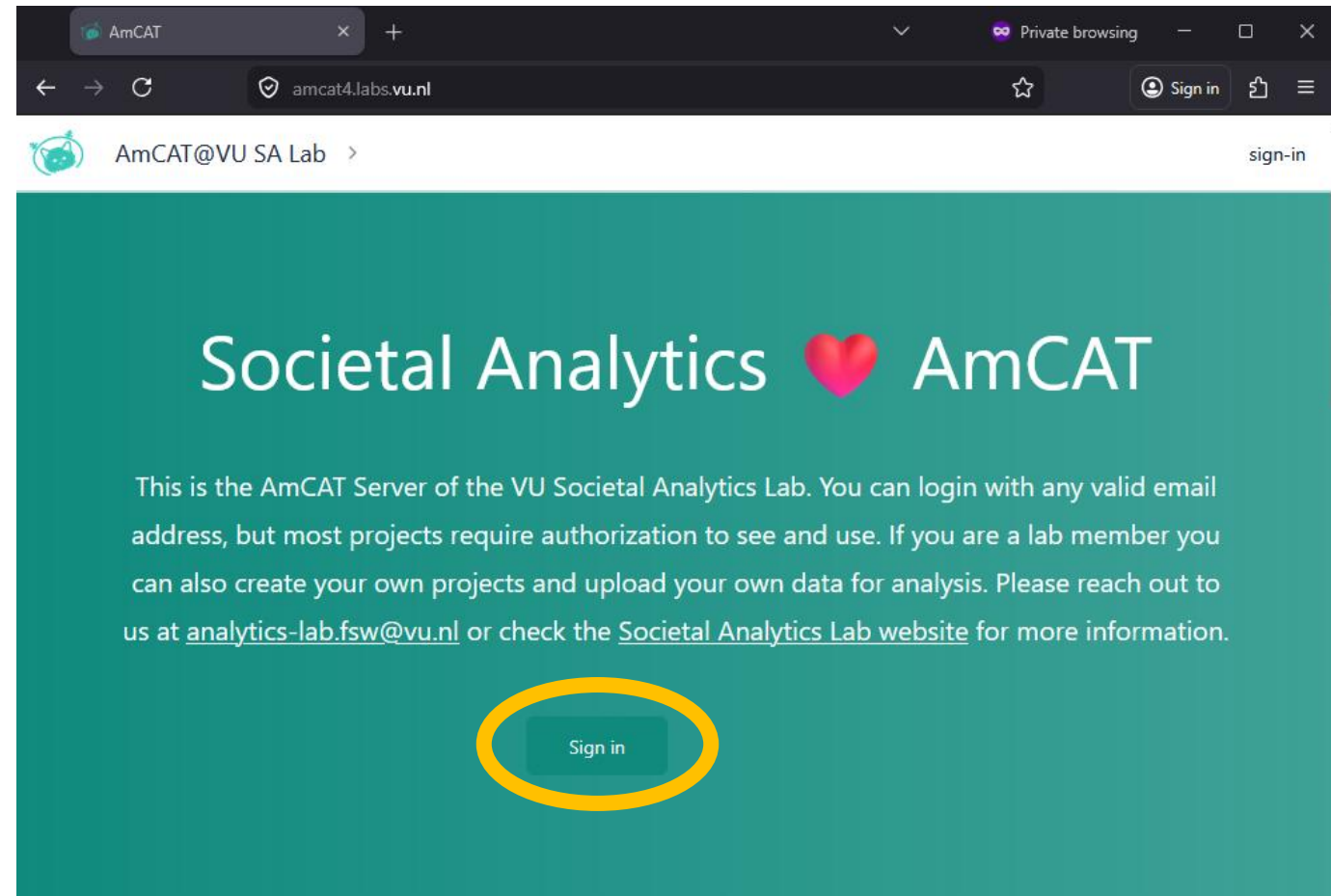


Signing In

To sign in, you just need to access the following link:

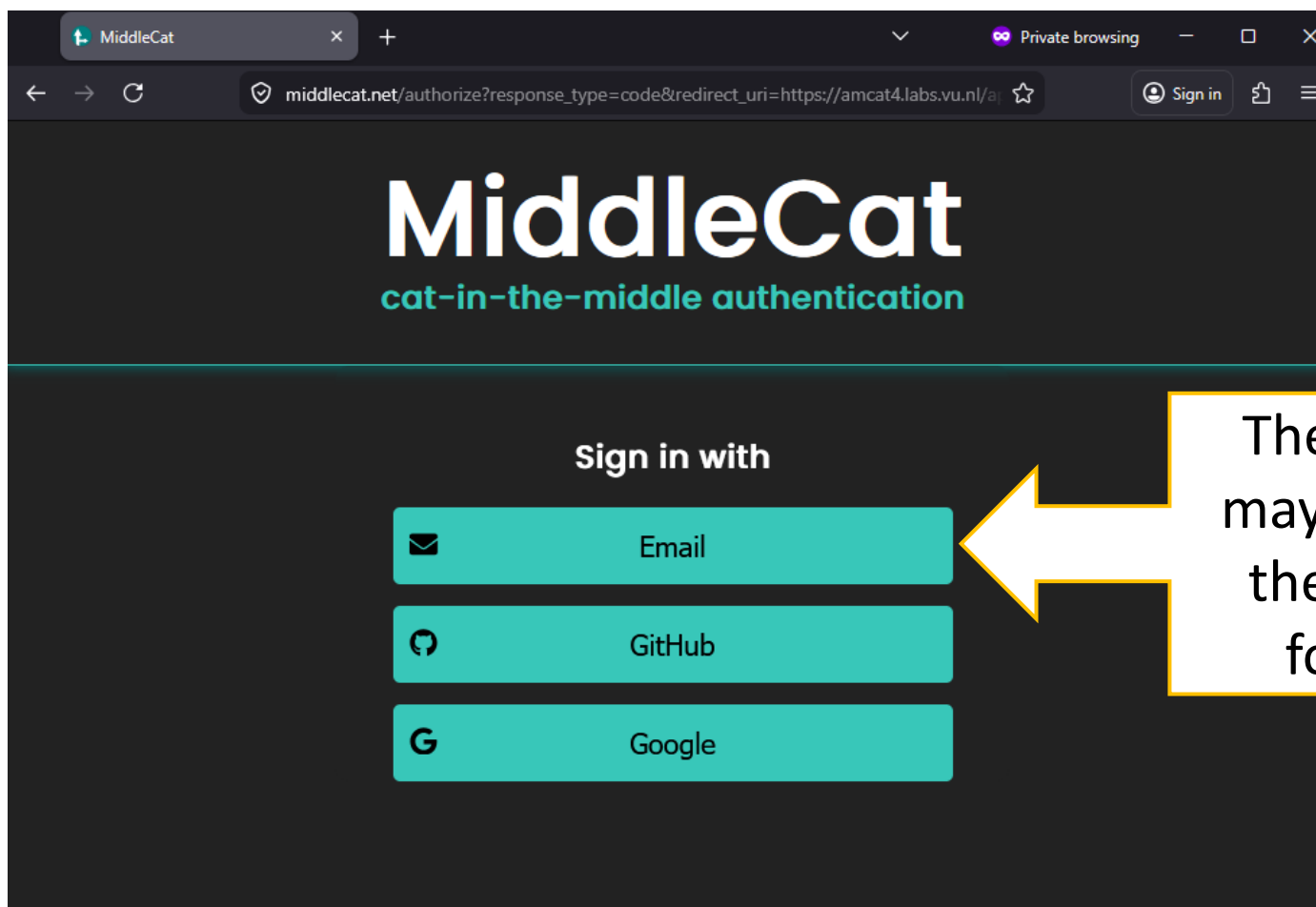
<https://amcat4.labs.vu.nl/>

From there, you will see this webpage:





Authentication



The email
may land in
the spam
folder.

5 MINS BREAK



3.1 Introduction

3.2 Access

3.3 Using the GUI

- Uploading data
- Visualizing
- Giving & requesting access

3.4 Using the API

- Access through R
- Uploading data
- Analyzing data

3. AMCAT



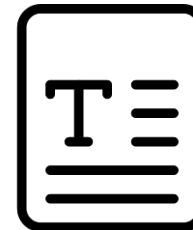
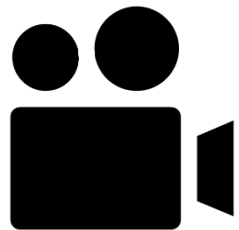
3.1 INTRODUCTION



AMCAT GENERAL GOAL

**To standardize
access and analysis
of multimedia data
with easy-to-use software.**

But... what is multimedia data?
Why is AmCAT necessary?

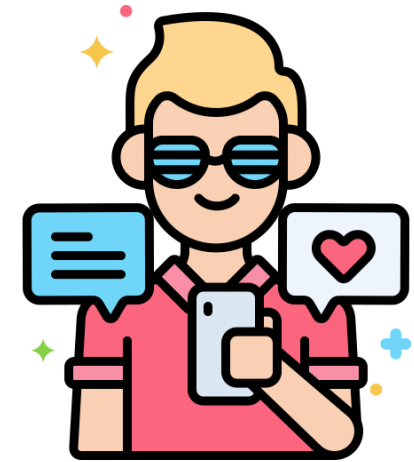


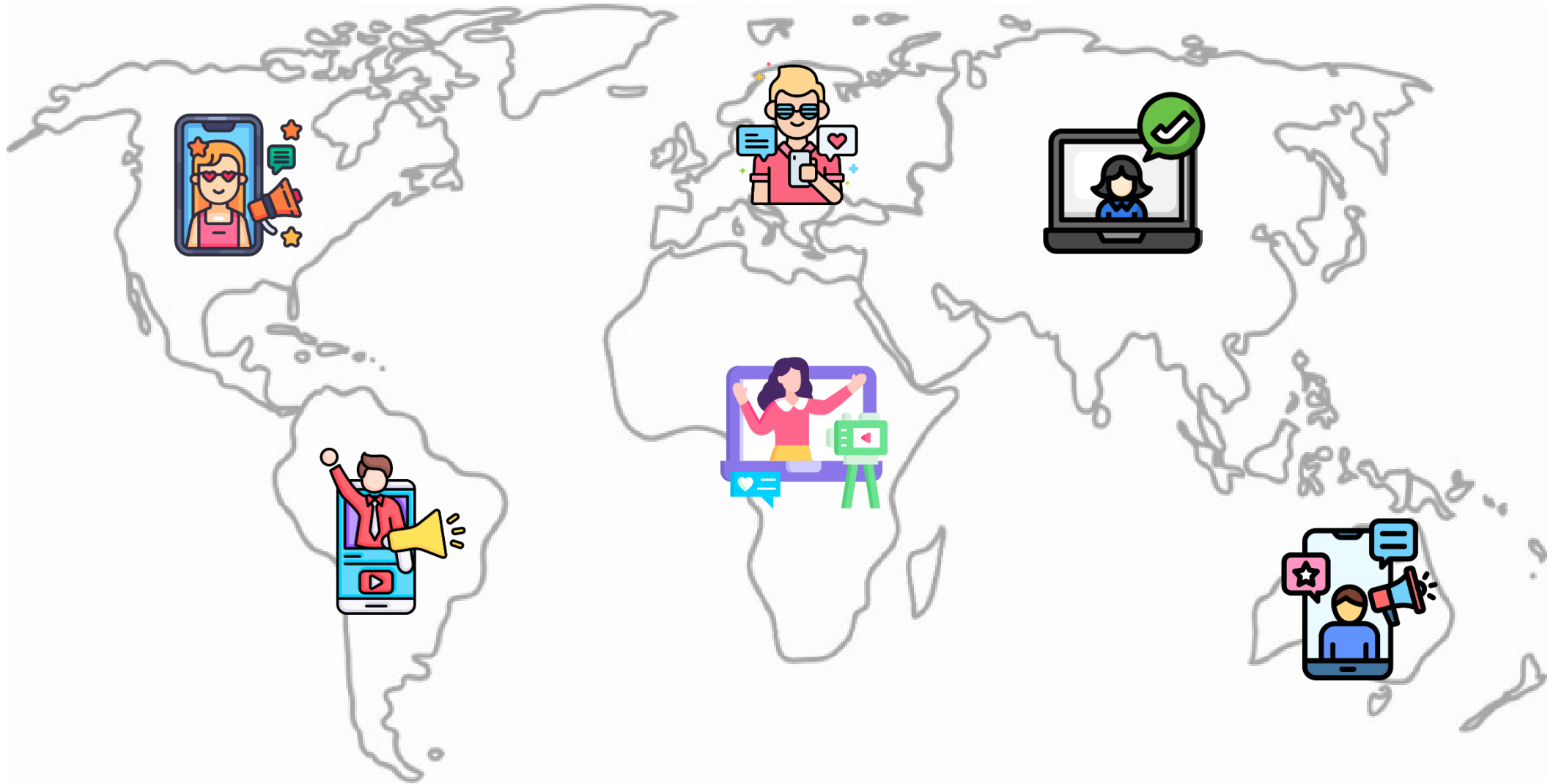


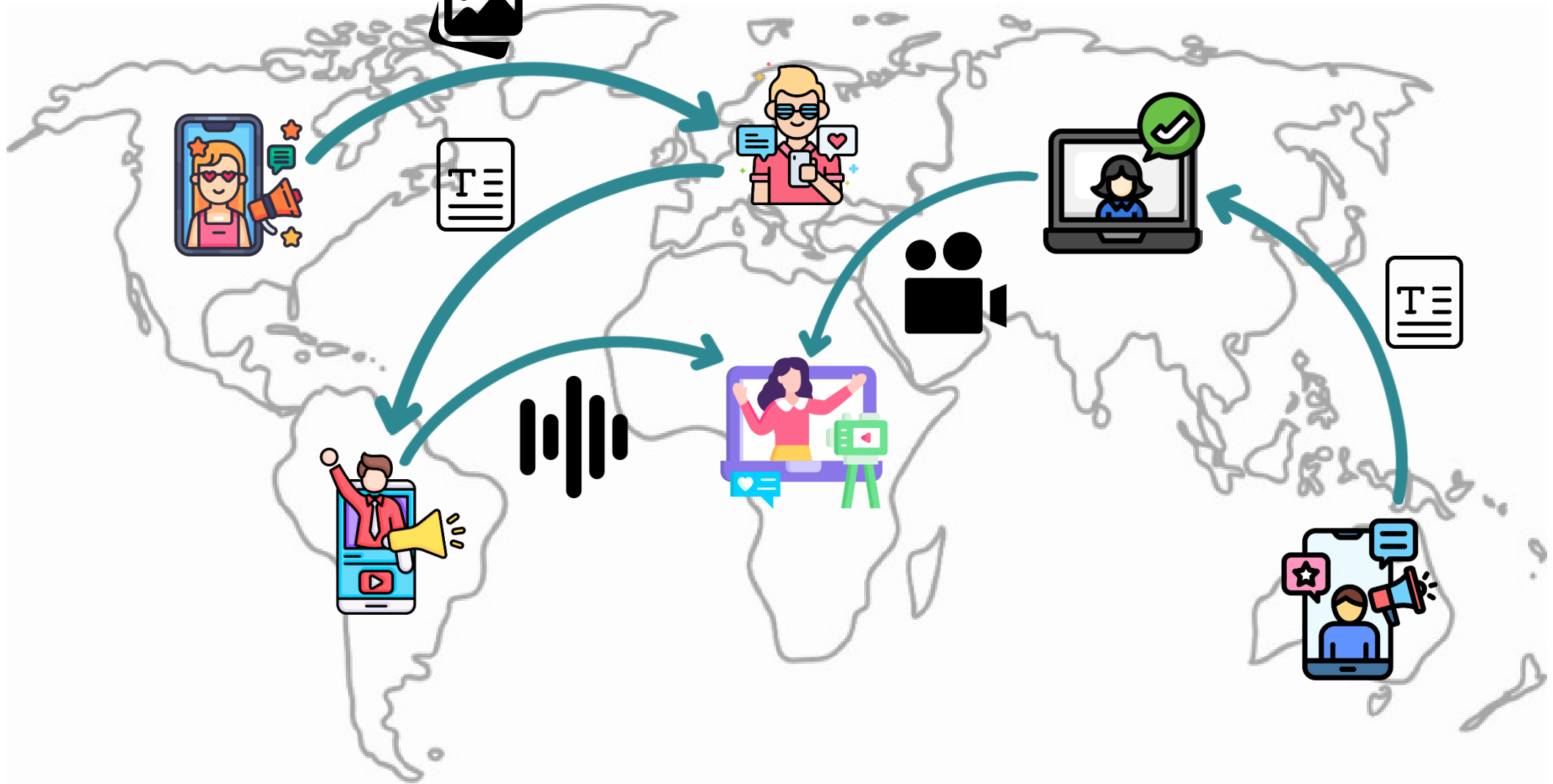




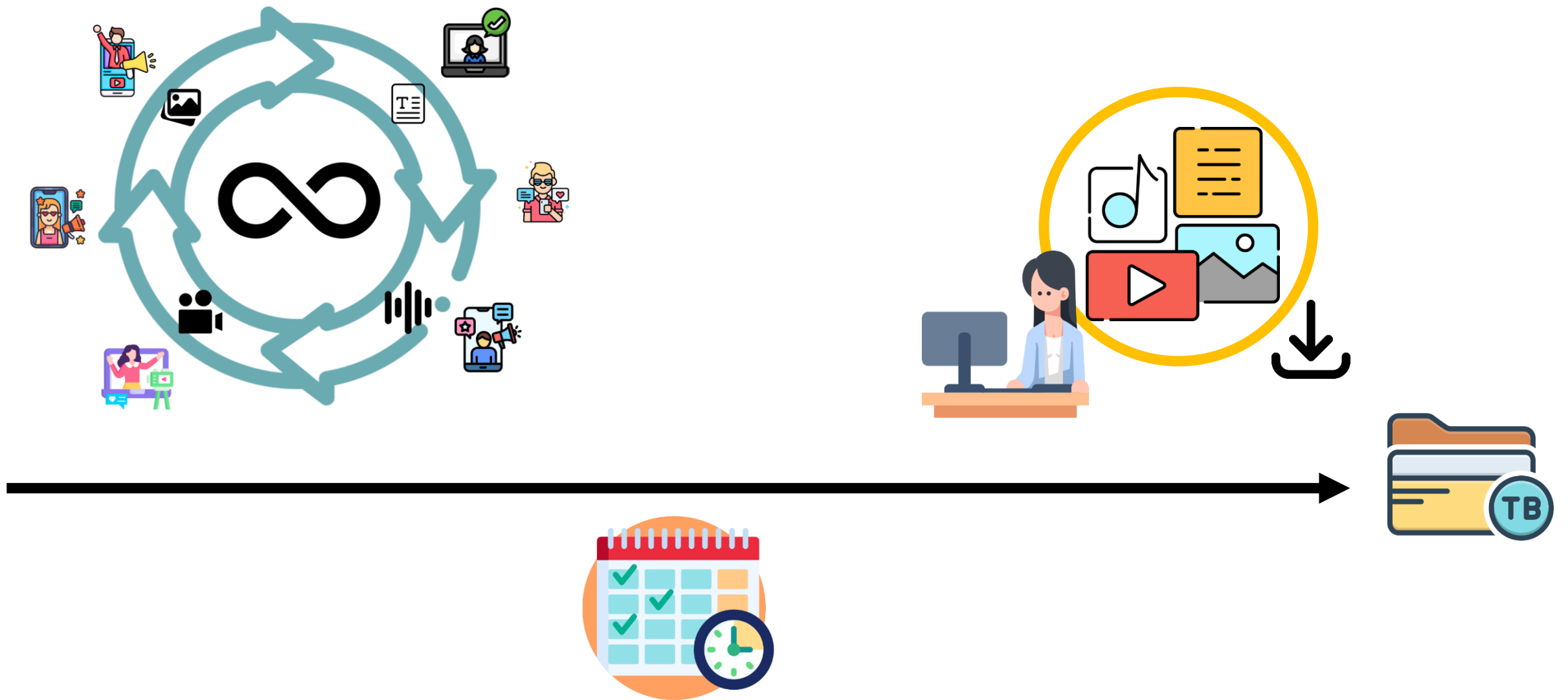




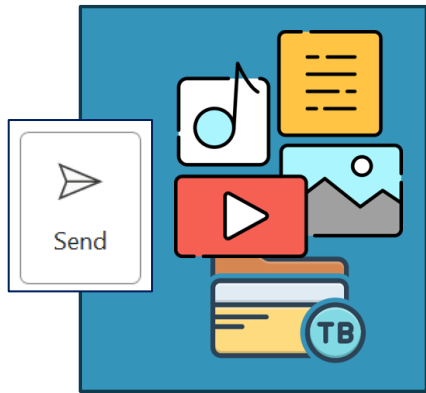


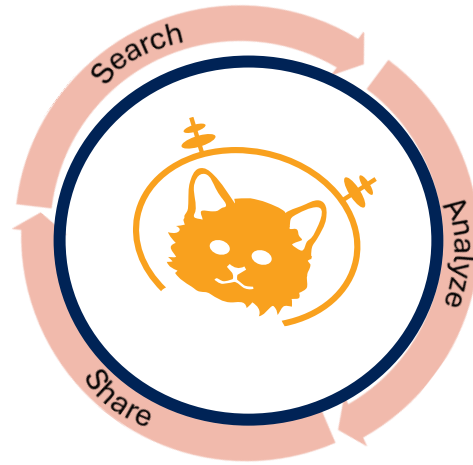


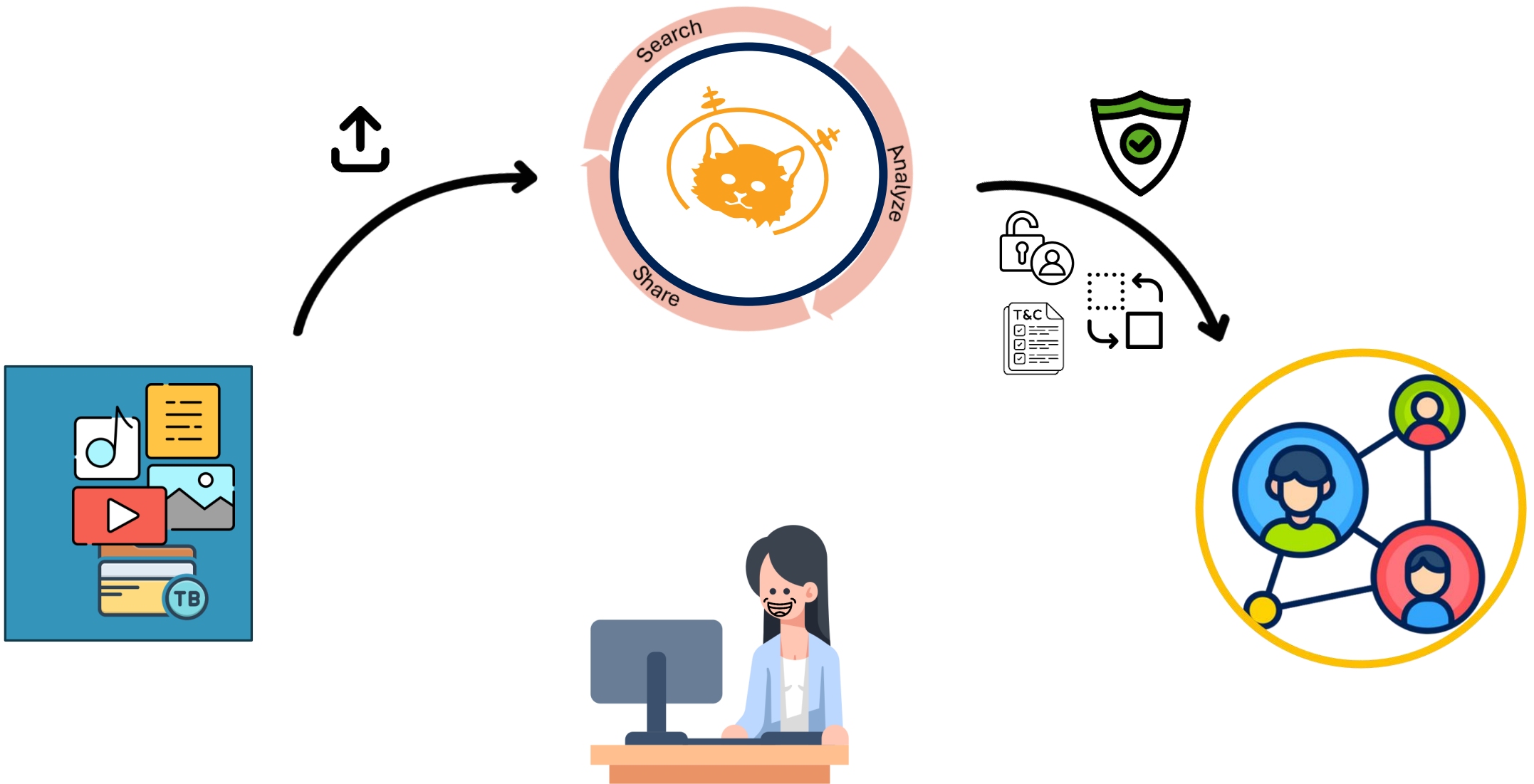




CASE 1: HAS IT HAPPENED TO YOU THAT...?

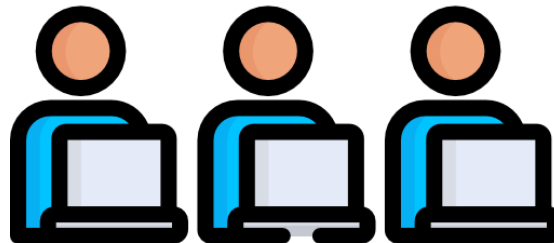
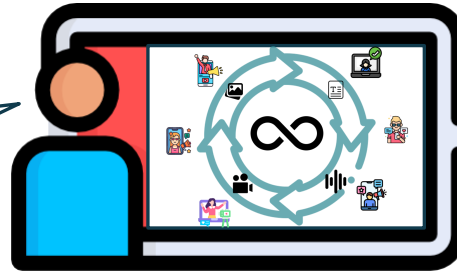




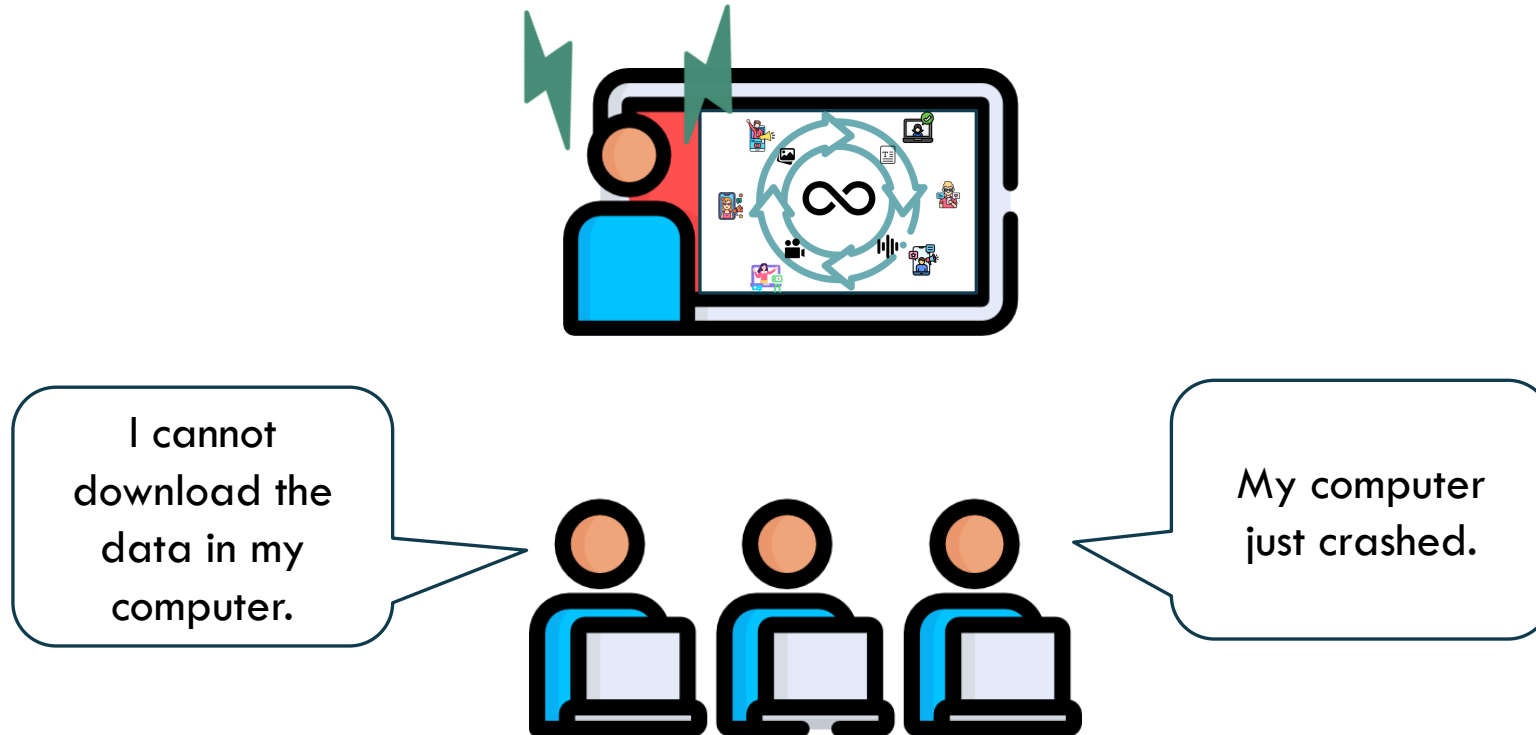


CASE 2: HAS IT HAPPENED TO YOU THAT...?

Today, we will learn how to perform content analysis of social media data.



CASE 2: HAS IT HAPPENED TO YOU THAT...?





Developed by



Kasper Welbers



Wouter van Atteveldt



Johannes Gruber



Sofia Gil-Clavel

by researchers for researchers, **society**, and data savvy users.



3.2 Let's 1st sign in to AmCAT

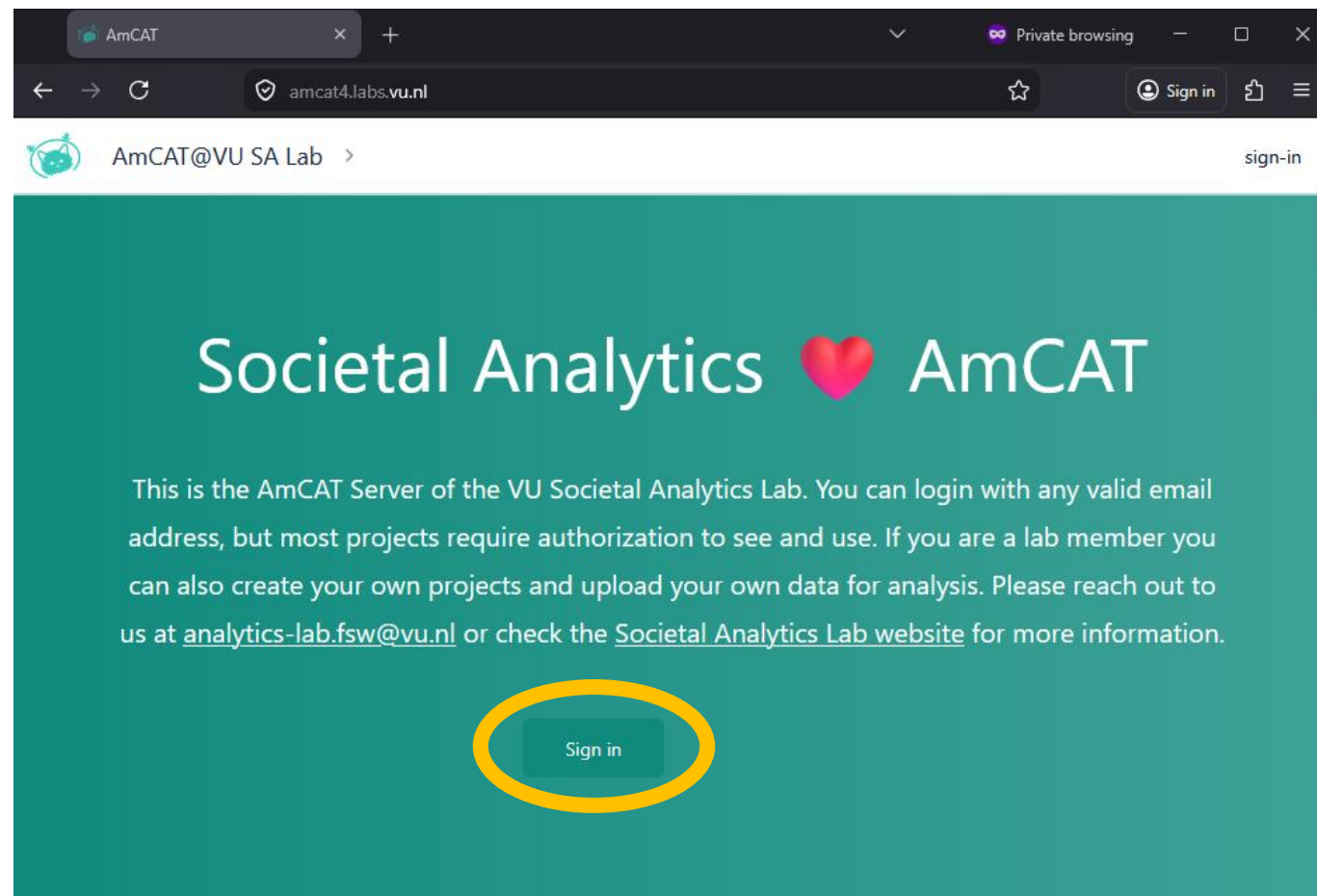


Signing In

To sign in, you just need to access the following link:

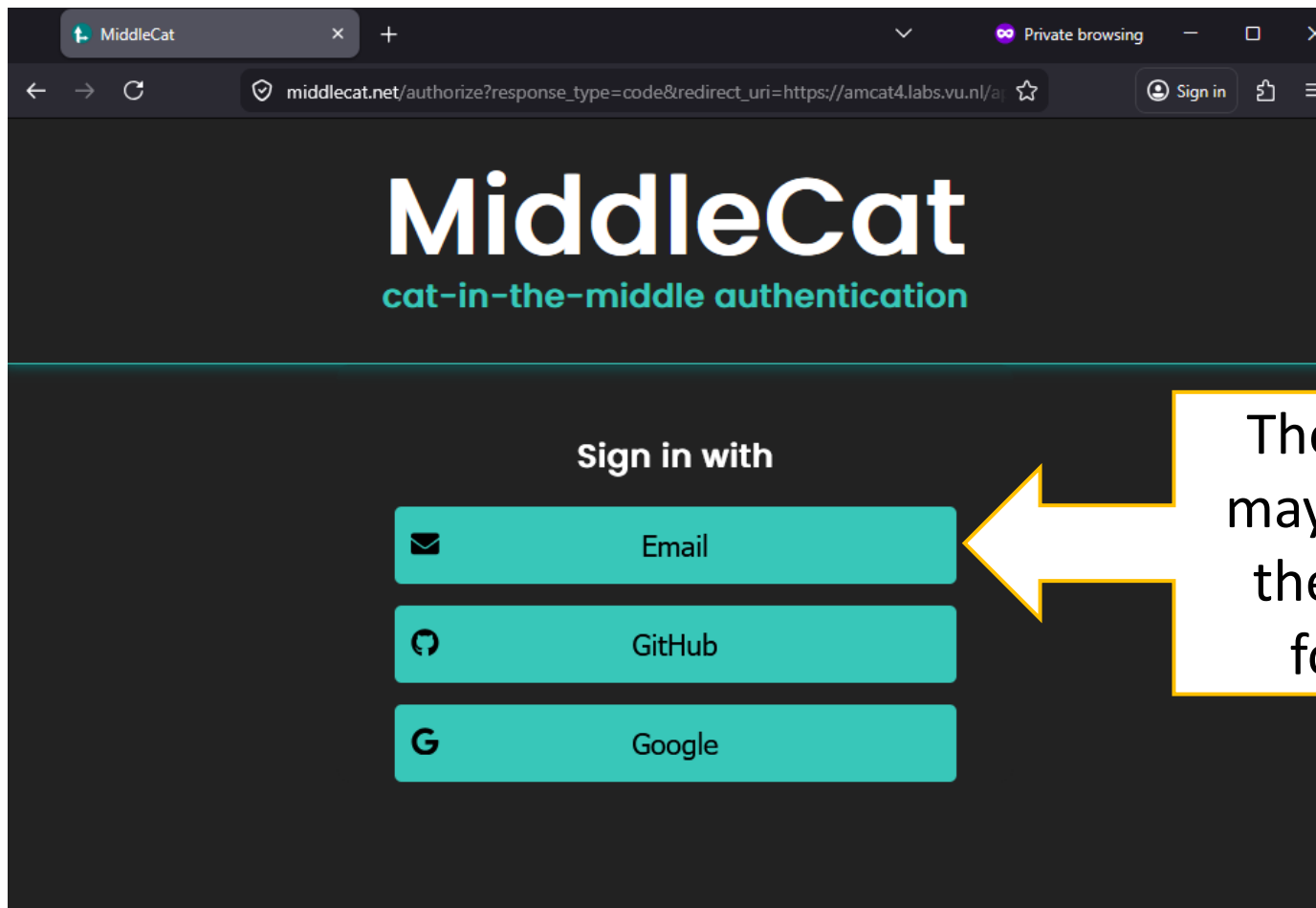
<https://amcat4.labs.vu.nl/>

From there, you will see this webpage:

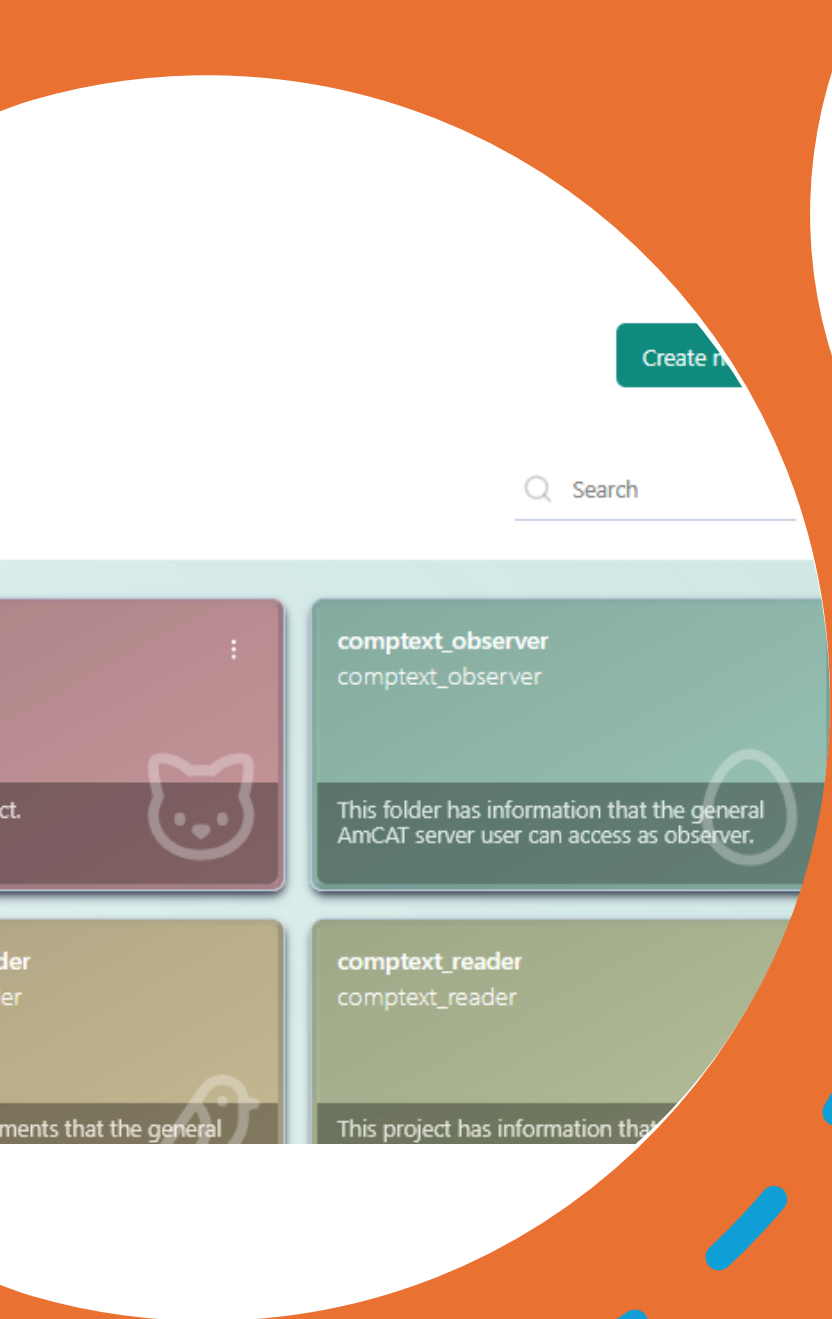




Authentication



The email
may land in
the spam
folder.



3.3 Using the GUI



We will cover

Uploading



Visualizing



Giving & Requesting
Access





Uploading

The screenshot shows a web browser window with the address bar displaying `amcat4.labs.vu.nl/projects`. The page title is "AmCAT@VU SA Lab" and the breadcrumb is "select project". The main content area is titled "Projects" and contains a button labeled "Create new project" which is circled in yellow. Below the button, there is a search bar and a list of projects. The list is currently empty, showing "No folders found".



Uploading

Use the following:
`comtext_<your initials>_<a number>`

request. If approved, the project will be created for you and you will be granted admin access to it.

Name

Project ID

Description

Folder

Project request Message

Submit request

Give me a
minute while I
accept your
requests.





AmCAT

amcat4.labs.vu.nl/projects

Sign in

AmCAT@VU SA Lab > select project

Projects Server role

Projects

Create new project

Search

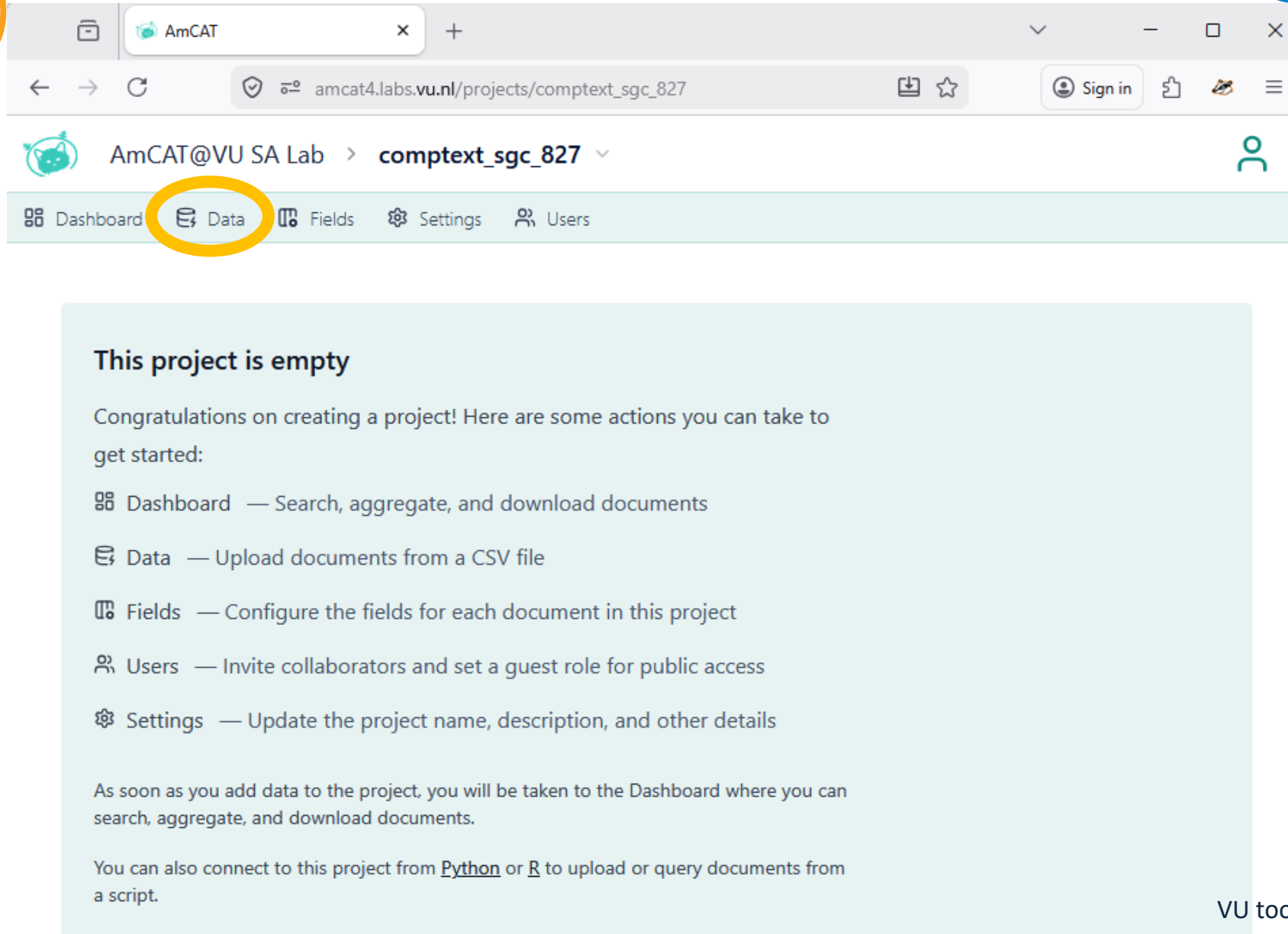
Projects

No folders found

complex_sgc_827
complex_sgc_827

My first AmCAT project.

Uploading



AmCAT

amcat4.labs.vu.nl/projects/comptext_sgc_827

Sign in

AmCAT@VU SA Lab > comptext_sgc_827

Dashboard Data Fields Settings Users

This project is empty

Congratulations on creating a project! Here are some actions you can take to get started:

- Dashboard — Search, aggregate, and download documents
- Data — Upload documents from a CSV file
- Fields — Configure the fields for each document in this project
- Users — Invite collaborators and set a guest role for public access
- Settings — Update the project name, description, and other details

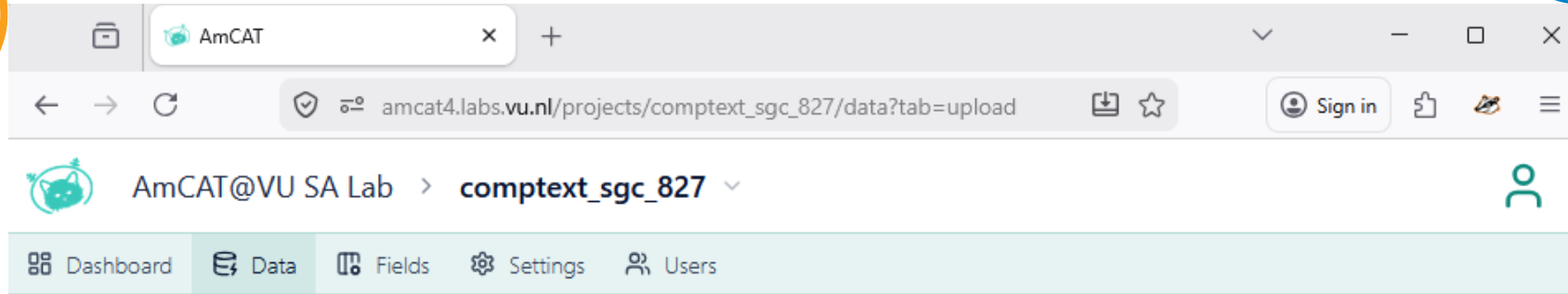
As soon as you add data to the project, you will be taken to the Dashboard where you can search, aggregate, and download documents.

You can also connect to this project from [Python](#) or [R](#) to upload or query documents from a script.

Uploading



Uploading



Upload

Multimedia

Upload the
“Example_Index_2.csv”

Click to upload or drag and drop
CSV, TSV, XLSX, ODS — or a ZIP of .txt / .docx / .pdf documents

Or select a folder of documents

Information on Uploading Documents

Upload documents to this project. You can upload a spreadsheet (CSV, TSV, XLSX) or a folder or zip with with text, PDF, or DOCX documents. After uploading the file, you can choose what to do with each field or column in the data before uploading.

Column actions



Uploading

The screenshot shows a web browser window with the address bar displaying `amcat4.labs.vu.nl/projects/comptext_sgc_827/data?tab=upload`. The page title is "AmCAT@VU SA Lab" and the project name is "comptext_sgc_827". The navigation menu includes "Dashboard", "Data", "Fields", "Settings", and "Users". The "Upload" tab is selected, and the "Multimedia" sub-tab is active. The main content area has a large light blue box with the text "Click to upload or drag and drop" and "CSV, TSV, XLSX, ODS — or a ZIP of .txt / .docx / .pdf documents". Below this is a link "Or select a folder of documents" with a folder icon. At the bottom, there is a section titled "Information on Uploading Documents" with a collapse icon. The text in this section reads: "Upload documents to this project. You can upload a spreadsheet (CSV, TSV, XLSX) or a folder or zip with with text, PDF, or DOCX documents. After uploading the file, you can choose what to do with each field or column in the data before uploading." Below this text is a heading "Column actions".



Fields format

AmCAT

amcat4.labs.vu.nl/projects/compte

Upload

Multimedia

CSV column	Action	AmCAT field	Type
id	Exclude	—	
author	Exclude	—	
year	Exclude	—	
title	Exclude	—	
chapter	Exclude	—	
text	Exclude	—	

Field types

keyword

Short labels or categories (e.g. country, language). Searched as exact values.

tag

Like keyword, but a document can have multiple tags.

text

Longer free text (e.g. article body). Analysed word-by-word for full-text search.

date

Date or date/time values.

integer

Whole numbers without decimals.

number

Numeric values with decimals.

boolean

True or false values.

url / image / video / audio

Links to web pages or media files. Displayed as a clickable link or inline media.

object

Structured JSON objects. Not analysed or parsed by Elasticsearch.



Fields format

AmCAT

amcat4.labs.vu.nl/projects/comptext_sgc_827

AmCAT@VU SA Lab > comptext_sgc_827

Dashboard Data Fields Settings Users

Upload Multimedia

Field types

keyword

Short labels or categories (e.g. country, language). Searched as exact values.

tag

Like keyword, but a document can have multiple tags.

text

Longer free text (e.g. article body). Analysed word-by-word for full-text search.

date

Date or date/time values.

integer

Whole numbers without decimals.

number

Numeric values with decimals.

boolean

True or false values.

url / image / video / audio

Links to web pages or media files. Displayed as a clickable link or inline media.

object

Structured JSON objects. Not analysed or parsed by Elasticsearch.

CSV column	Action	AmCAT field	Type	
id	New field	id	keyword	
author	New field	author	keyword	
year	New field	year	integer	
title	New field	title	text	
chapter	New field	chapter	keyword	
text	New field	text	text	



Choosing identifier

AmCAT

amcat4.labs.vu.nl/projects/comptext_sgc_827/data?tab=upload

Sign in

AmCAT@VU SA Lab > comptext_sgc_827

Dashboard Data Fields Settings Users

Upload Multimedia

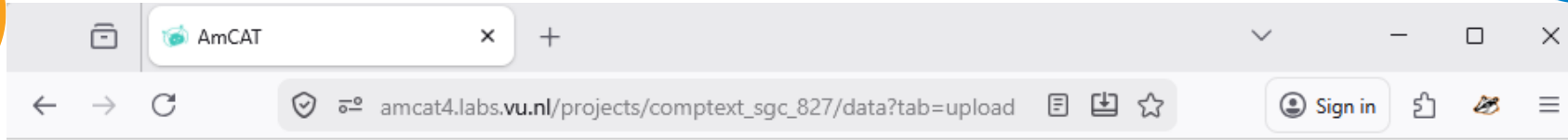
Change file

CSV column	Action	AmCAT field	Type	
id	New field	id	keyword	
author	New field	author	keyword	
year	New field	year	integer	
title	New field	title	text	
chapter	New field	chapter	keyword	
text	New field	text	text	

Use as document identifier



Uploading



CSV column	Action	AmCAT field	Type	
id	New field ▾	id	keyword ▾	🔑
author	New field ▾	author	keyword ▾	🔑
year	New field ▾	year	integer ▾	🔑
title	New field ▾	title	text ▾	🔑
chapter	New field ▾	chapter	keyword ▾	🔑
text	New field ▾	text	text ▾	🔑

Confirm upload

Upload 15 documents

<> Show code

Create ▾

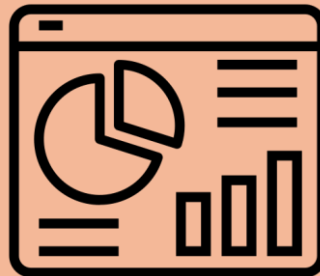


We will cover

Uploading



Visualizing



Giving & Requesting
Access





Visualizing

AmCAT

amcat4.labs.vu.nl/projects/comptext_sgc_827/data?tab=upload

Sign in

Change file

CSV column	Action	AmCAT field	Type	
id	New field	id	keyword	
author	New field	author	keyword	
year	New field	year	integer	
title	New field	title	text	
chapter	New field	chapter	keyword	
text	New field	text	text	

Upload complete

15 documents uploaded successfully.

View uploaded data Upload more data

Confirm upload

Upload 15 documents <> Show code Create

Information on Uploading Documents



Visualizing

The screenshot shows the AmCAT web interface in a browser window. The address bar displays the URL `amcat4.labs.vu.nl/projects/comptext_sgc_827/dashboard?tab=s`. The page header includes the AmCAT logo, the text "AmCAT@VU SA Lab", and a dropdown menu for the project "comptext_sgc_827". A navigation bar contains links for "Dashboard", "Data", "Fields", "Settings", and "Users". Below the navigation bar is a search bar with the placeholder text "search" and a filter icon. The main content area features a section titled "Information on Search syntax & filters" with a collapse arrow. The text below the title explains that queries can be entered to search through the project and that results will be displayed below. A table titled "Query syntax" lists various query patterns and their meanings. At the bottom, a section titled "Filters" is partially visible.

Query syntax	
<code>climate AND policy</code>	Both terms must appear
<code>climate OR weather</code>	Either term
<code>climate NOT gas</code>	Exclude a term
<code>(climate OR weather) AND policy</code>	Group with parentheses
<code>"climate change"</code>	Exact phrase
<code>"climate policy"~5</code>	Words within 5 positions of each other
<code>climat*</code>	Wildcard (prefix match)



AmCAT

amcat4.labs.vu.nl/projects/comptext_sgc_827/fields

Sign in

AmCAT@VU SA Lab > comptext_sgc_827

Dashboard Data Fields Settings Users

Visualizing

Fields [Add field](#)

ID	Field	Type	Dashboard	METAREADER access
	author	keyword keyword		invisible
	chapter	keyword keyword		invisible
	id	keyword keyword		invisible
	text	text text		invisible
	title	text text		invisible
	url	url url		invisible



Visualizing

AmCAT

amcat4.labs.vu.nl/projects/comptext_sgc_827/fields

AmCAT@VU SA Lab > comptext_sgc_827

Dashboard Data Fields Settings Users

Fields Add field

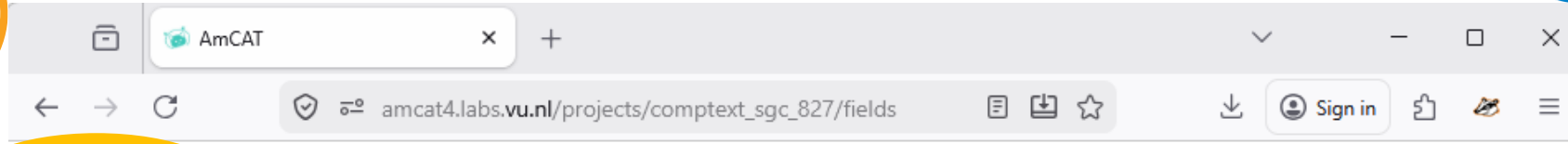
ID	Field	Type	Dashboard	METAREADER access
	author	keyword keyword		invisible
	chapter	keyword keyword		invisible
	id	keyword keyword		invisible
	text	text text		invisible
	title	text text		invisible

☒ show in document
 ☒ show in list
 ☐ show as heading (title)
 ☐ show graph

You can choose where to display them.

1

2



Visualizing

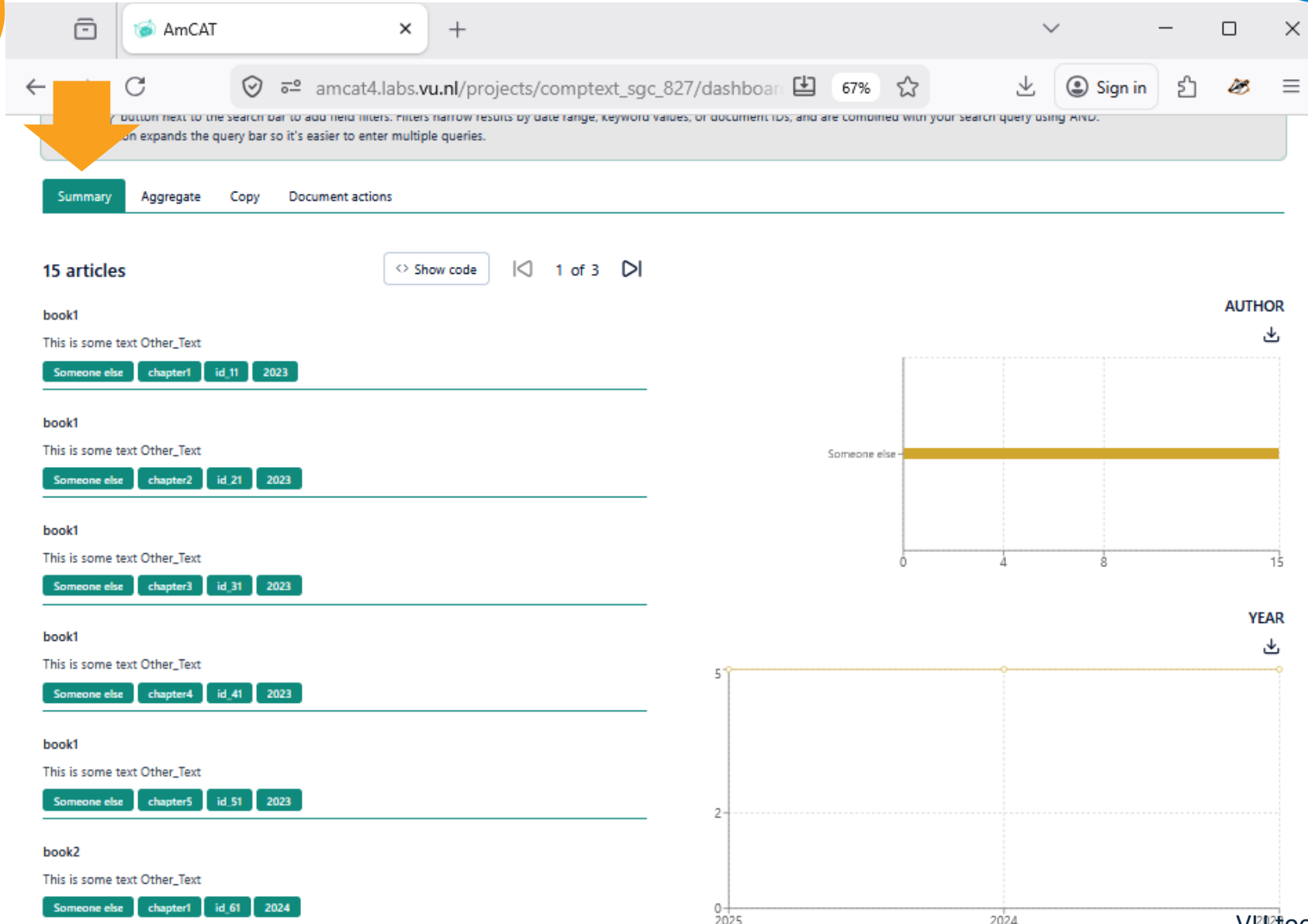
Fields + Add field



ID	Field	Type	Dashboard	METAREADER access
	author	keyword keyword		invisible
	chapter	keyword keyword		invisible
	id	keyword keyword		invisible
	text	text text		invisible
	title	text text		invisible
	year	integer long		invisible



Visualizing





Visualizing

AmCAT

amcat4.labs.vu.nl/projects/comptext_sgc_827/dashboard

67%

Sign in

Wildcard (prefix match)

climat*

Filters

Use the button to the search bar to add field filters. Filters narrow results by date range, keyword values, or document IDs, and are combined with your search query using AND. The button in the query bar so it's easier to enter multiple queries.

Summary **Aggregate** Copy Document actions

Display

Line Graph

Aggregate

Count

Horizontal (X) axis

Select field

Multiple lines

Select field

<> Show code Submit

Information on aggregation

The aggregate tab groups documents by one or two fields and visualizes the result. Configure the options on the left and click **Submit** to run.

Display Choose how to visualize the result: a line graph or bar chart for trends over time, a table for cross-tabulations, or a list for ranked counts.

Aggregate What to measure. Defaults to Count (number of documents), but you can also compute the sum, average, minimum, or maximum of a numeric field.

Axis The field to group by, such as a date (with an interval like week or month), a keyword, or a tag. A second axis is optional and adds a further grouping dimension (e.g. multiple lines in a line graph).



Visualizing

AmCAT

amcat4.labs.vu.nl/projects/comptext_sgc_827/dashboard

67%

Sign in

Wildcard (prefix match)

climat*

Filters

Use the button next to the search bar to add field filters. Filters narrow results by date range, keyword values, or document IDs, and are combined with your search query using AND. The button expands the query bar so it's easier to enter multiple queries.

Summary **Aggregate** Copy Document actions

Display

Line Graph

Aggregate

Count

Horizontal (X) axis

Select field

Multiple lines

Select field

Show code Submit

Information on aggregation

The aggregate tab groups documents by one or two fields and visualizes the result. Configure the options on the left and click **Submit** to run.

Display	Choose how to visualize the result: a line graph or bar chart for trends over time, a table for cross-tabulations, or a list for ranked counts.
Aggregate	What to measure. Defaults to Count (number of documents), but you can also compute the sum, average, minimum, or maximum of a numeric field.
Axis	The field to group by, such as a date (with an interval like week or month), a keyword, or a tag. A second axis is optional and adds a further grouping dimension (e.g. multiple lines in a line graph).



We will cover

Uploading



Visualizing



Giving & Requesting
Access





Giving Access

AmCAT

amcat4.labs.vu.nl/projects/comptext_sgc_827/dashboard?tab=s

Sign in

AmCAT@VU SA Lab > comptext_sgc_827

Dashboard Data **Fields** Settings Users

search

Information on Search syntax & filters

You can enter queries here to search through the project. Results will be displayed below this box. You can minimize this box by clicking on the header above.

Query syntax

climate AND policy	Both terms must appear
climate OR weather	Either term
climate NOT gas	Exclude a term
(climate OR weather) AND policy	Group with parentheses
"climate change"	Exact phrase
"climate policy"~5	Words within 5 positions of each other
climat*	Wildcard (prefix match)

Filters





AmCAT

amcat4.labs.vu.nl/projects/comptext_sgc_827/fields

Sign in

AmCAT@VU SA Lab > comptext_sgc_827

Dashboard Data Fields Settings Users

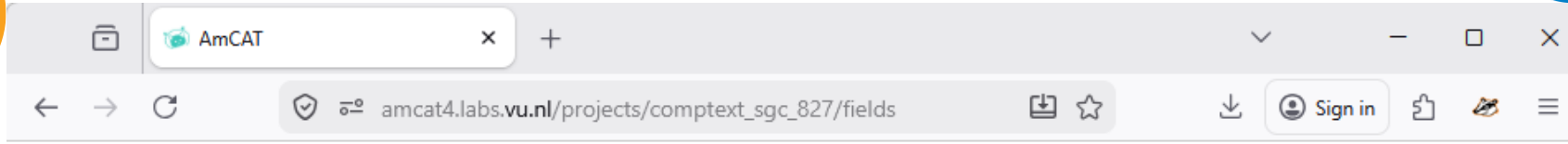
Fields [Add field](#)

ID	Field	Type	Dashboard	METAREADER access
	author	keyword keyword		invisible
	chapter	keyword keyword		invisible
	id	keyword keyword		invisible
	text	text text		invisible
	title	text text		invisible
	url	url		invisible

Giving Access



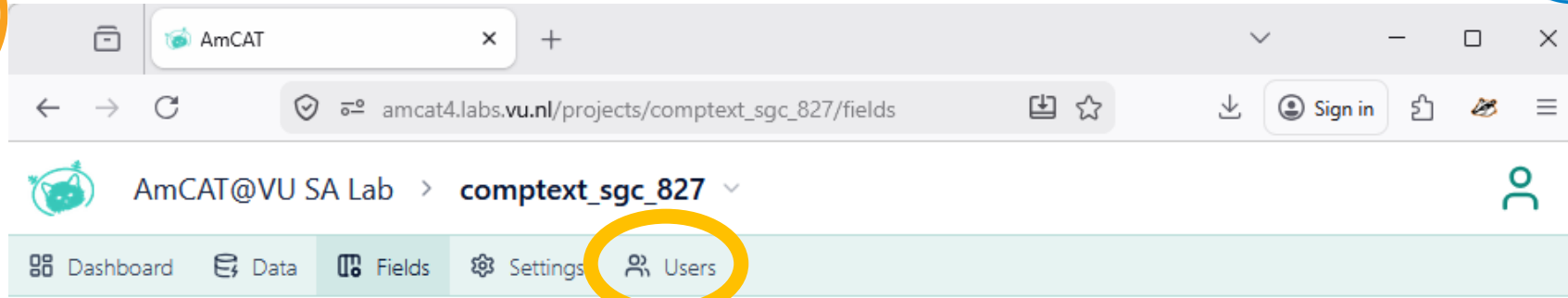
Giving Access



Fields + Add field Q

ID	Field	Type	Dashboard	METAREADER access
	author	keyword keyword		invisible 1
	chapter	keyword keyword		invisible visible 2
	id	keyword keyword		invisible
	text	text text		invisible
	title	text text		invisible
	year	integer long		invisible

You can make visible, as many variable as you want.



Giving Access

Fields + Add field 🔍

ID	Field	Type	Dashboard	METAREADER access
	author	keyword keyword		invisible ▼
	chapter	keyword keyword		invisible ▼
	id	keyword keyword		invisible ▼
	text	text text		invisible ▼
	title	text text		invisible ▼
	url	url url		invisible ▼



Let's make groups of 5 people.

You and your team will explore the
access control features.





Giving Access

AmCAT

amcat4.labs.vu.nl/projects/comptext_sgc_827/users

AmCAT@VU SA Lab > comptext_sgc_827

Dashboard Data Fields Settings Users

Guest role NONE

+ Add user

Email ↑↓	Role ↑↓	
b.s.gilclavel@vu.nl	ADMIN	

Show code

Information on user management

Project roles

Every project has its own users table with the following roles and incremental permissions.

OBSERVER	Can find the project and see project metadata, but cannot search documents.
METAREADER	Can also search documents, but only view document metadata. Project ADMINs can determine which fields are



Giving Access

AmCAT

amcat4.labs.vu.nl/projects/comptext_sgc_827/users

AmCAT@VU SA Lab > comptext_sgc_827

Dashboard Data Fields Settings Users

Guest role NONE

+ Add user

Email ↑↓

b.s.gilclavel@vu.nl

Add Users

user1@userland.com
user2@userland.com
*@my-university.com

1 → OBSERVER

2 →

- OBSERVER
- METAREADER
- READER
- WRITER
- ADMIN

Create

Information on management

Project roles

Every project has its own users table with the following permissions.

OBSERVER Can find the project and see project metadata, but cannot search documents.

METAREADER Can search documents but not view document metadata. Project ADMINs can determine which fields are



Giving Access

Add Users

user1@userland.com
user2@userland.com
*@my-university.com

OBSERVER

?

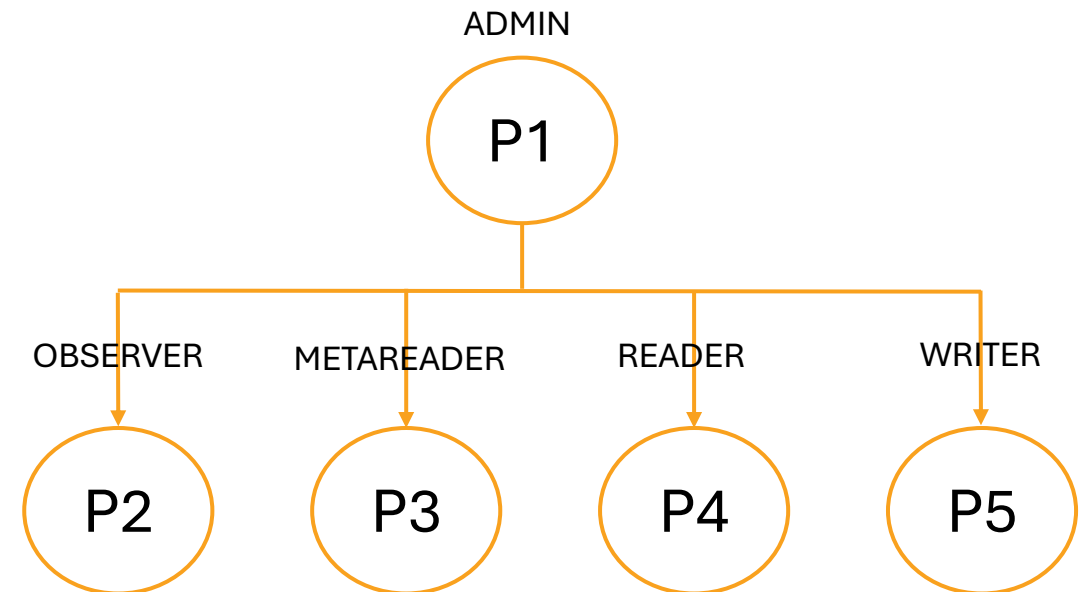
- OBSERVER
- METAREADER
- READER
- WRITER
- ADMIN

Create

By default, you are ADMIN of your project.

This allows you to:

- Upload documents.
- Delete documents and your project.
- Give other people access to your project.

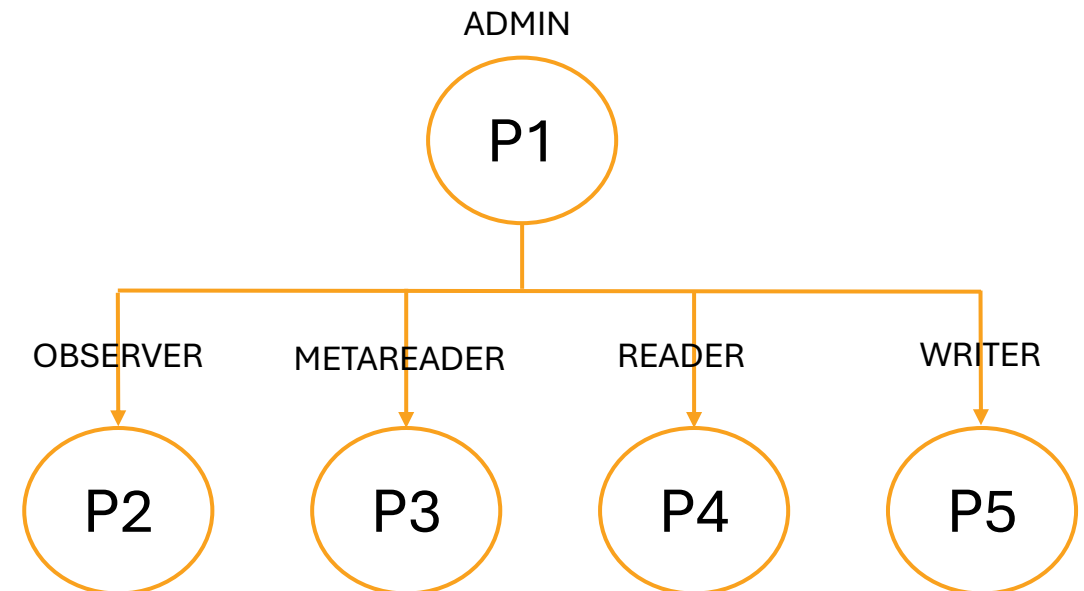




Giving Access

Exercise:

1. To each member in your team give access to your project using the different types of access.
2. Based on the access that the others gave you: What can you see of each project?





Giving Access

AmCAT

amcat4.labs.vu.nl/projects/comptext_sgc_827/users

AmCAT@VU SA Lab > comptext_sgc_827

Dashboard Data Fields Settings Users

Guest role NONE

Add user

Email ↑↓	Role ↑↓	
b.s.gilclavel@vu.nl	ADMIN	
wouter.van.atteveldt@vu.nl	WRITER	
k.welbers@vu.nl		
a.g.bollicimoreno@vu.nl		

Are you sure you want to delete this user? Type "yes"

type "yes" to confirm

Delete Cancel

Observer
Metareader
Reader
• Writer
Admin

You can always modify.

You can always delete.



Giving General Access

AmCAT

amcat4.labs.vu.nl/projects/comptext_sgc_827/users

AmCAT@VU SA Lab > comptext_sgc_827

Dashboard Data Fields Settings Users

Guest role NONE

Add NONE

Observer

Metareader

Reader

Writer

General users of the AmCAT server can check your data based on the variables that you made visible and what you allow them to see.

METAREADER access

invisible

WRITER

METAREADER

READER

Show code



Giving General Access

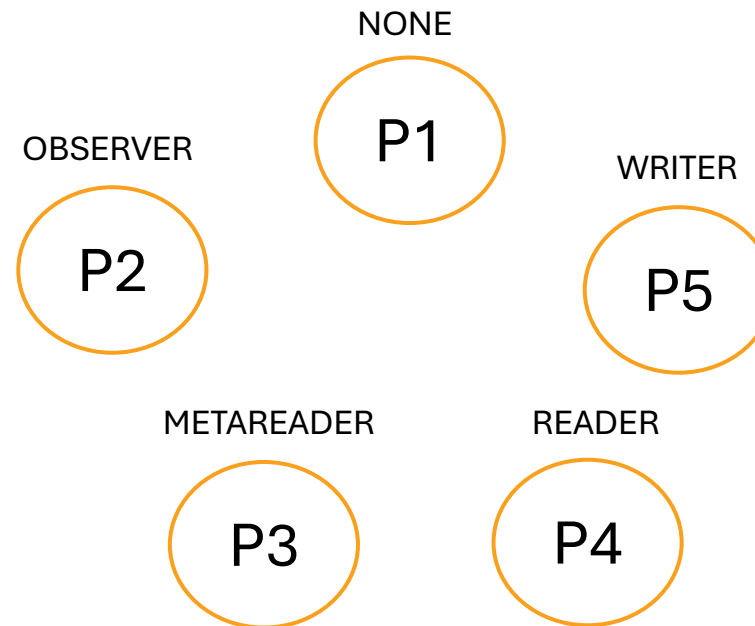
Guest role

NONE 

- NONE
- OBSERVER
- METAREADER
- READER
- WRITER

Exercise:

1. Each member of your team should choose one different type of “Guest role” for their project.
2. What can you see of each project?
3. What can you do in each project?





Requesting Access

The screenshot shows a web browser window with the URL `amcat4.labs.vu.nl/projects`. The page displays a list of projects under the heading "Projects". On the left, it says "No folders found". The projects are listed in a grid:

- comptext_sgc_827**
comptext_sgc_827
My first AmCAT project. (with a cat icon)
- comptext_observer**
comptext_observer
This folder has information that the general user can access as observer.
- comptext_metareader**
comptext_metareader
This project has documents that the general AmCAT server user can access as metareader. (with a cat icon)
- comptext_reader**
comptext_reader
This project has information that the general AmCAT user can access as reader. (with a cat icon)
- comptext_writer**
comptext_writer
This project has information that the general AmCAT server user can access as writer. (with a cat icon)

What if I see an interesting project?
Is it possible to request access?

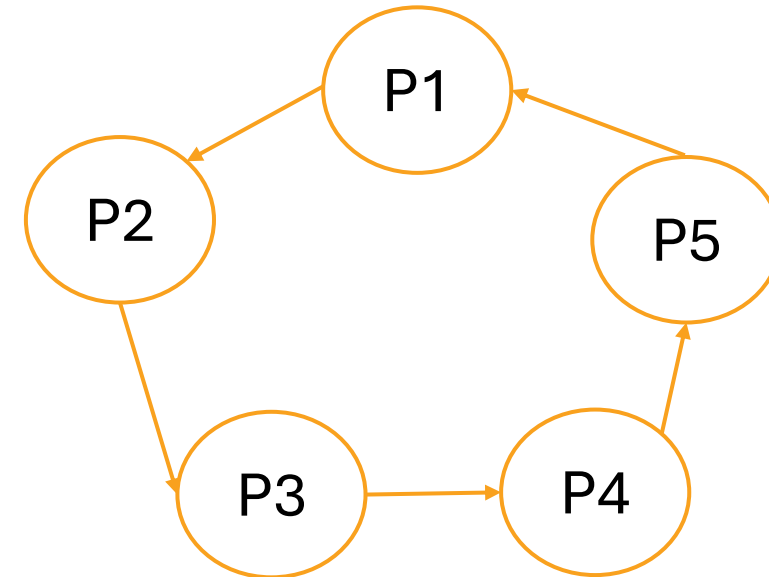
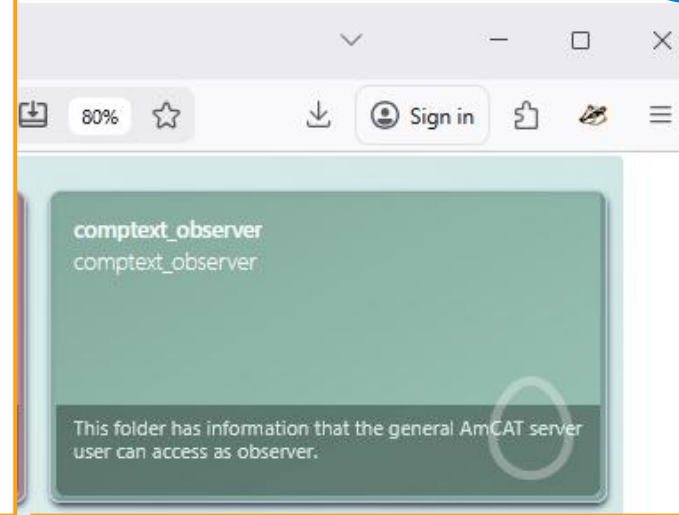
Yes, it is possible!



Requesting Access

Exercise:

1. In your team, look for the **project of the person on your right**.
2. **Access it.**
3. **Wait for the next instructions.**





In another person's project

Requesting Access

The screenshot shows a web browser window with the URL `amcat4.labs.vu.nl/projects/comptext_observer/settings`. The page title is "AmCAT@VU SA Lab > comptext_observer". The "Access" tab is highlighted with a yellow circle. Below the tabs, the project name "comptext_observer" is displayed, followed by a description: "This folder has information that the general AmCAT server user can access as observer." Under the "Folder" section, it says "(none)". A light blue box contains the text "Contact information" and "This project does not have public contact information." Below this, the project details are listed: "Project ID: comptext_observer" and "Size: 498 Bytes". At the bottom, there is a button labeled "Information on project settings" with a dropdown arrow.

AmCAT

amcat4.labs.vu.nl/projects/comptext_observer/settings

AmCAT@VU SA Lab > comptext_observer

Settings Access

comptext_observer

This folder has information that the general AmCAT server user can access as observer.

Folder (none)

Contact information

This project does not have public contact information.

Project ID: comptext_observer
Size: 498 Bytes

Information on project settings



Requesting Access

In another person's project

AmCAT

amcat4.labs.vu.nl/projects/comptext_observer/access

Sign in

AmCAT@VU SA Lab > comptext_observer

Settings Access

You have the **OBSERVER** role on this project. ?

Request a role change

Select role ▼

- delete role
- METAREADER
- READER
- WRITER
- ADMIN

Message to administrator

You can add a message here to explain why you need this role.



In another person's project

Requesting Access

AmCAT

amcat4.labs.vu.nl/projects/comptext_observer/access

Sign in

AmCAT@VU SA Lab > comptext_observer

Settings Access

You have the **OBSERVER** role on this project. ?

Request a role change **READER** ▾

Message to administrator

1 → Dear Admin,
Could you please let me explore your data? Perhaps, we can collaborate.
- Sofia

→ 2 Send request



Requesting Access

In another person's project

AmCAT

amcat4.labs.vu.nl/projects/comptext_observer/access

Sign in

AmCAT@VU SA Lab > comptext_observer

Settings Access

You have the **OBSERVER** role on this project. ?

⌚ Role change request pending

READER ▾

Message to administrator

Dear Admin,
Could you please let me explore your data? Perhaps, we can collaborate.
- Sofia

Update request



Accepting Requests

AmCAT

amcat4.labs.vu.nl/projects

AmCAT@VU SA Lab > select project

Projects Branding Server users Snapshots

Projects

Create new project Import project

Search

Projects

Dutch Media

ParIEE plenary speeches data set: AT
parlee_at

See <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/ZY3RV7>

ParIEE plenary speeches data set: PT



Accepting Requests

AmCAT

amcat4.labs.vu.nl/projects

AmCAT@VU SA Lab > select project

Notifications

Server role Project role 1 Project 1

user requests for project **b.s.gilclavel@vu.nl**
READER role
comptext_observer

1

✓ ✕

Dear Admin, Could you please let me explore your data? Perhaps, we can collaborate. - Sofia

0 / 2 requests evaluated

Submit decisions Cancel

2



Accepting Requests

AmCAT

amcat4.labs.vu.nl/projects

AmCAT@VU SA Lab > select project

Notifications

Server role Project role ✓ Project ①

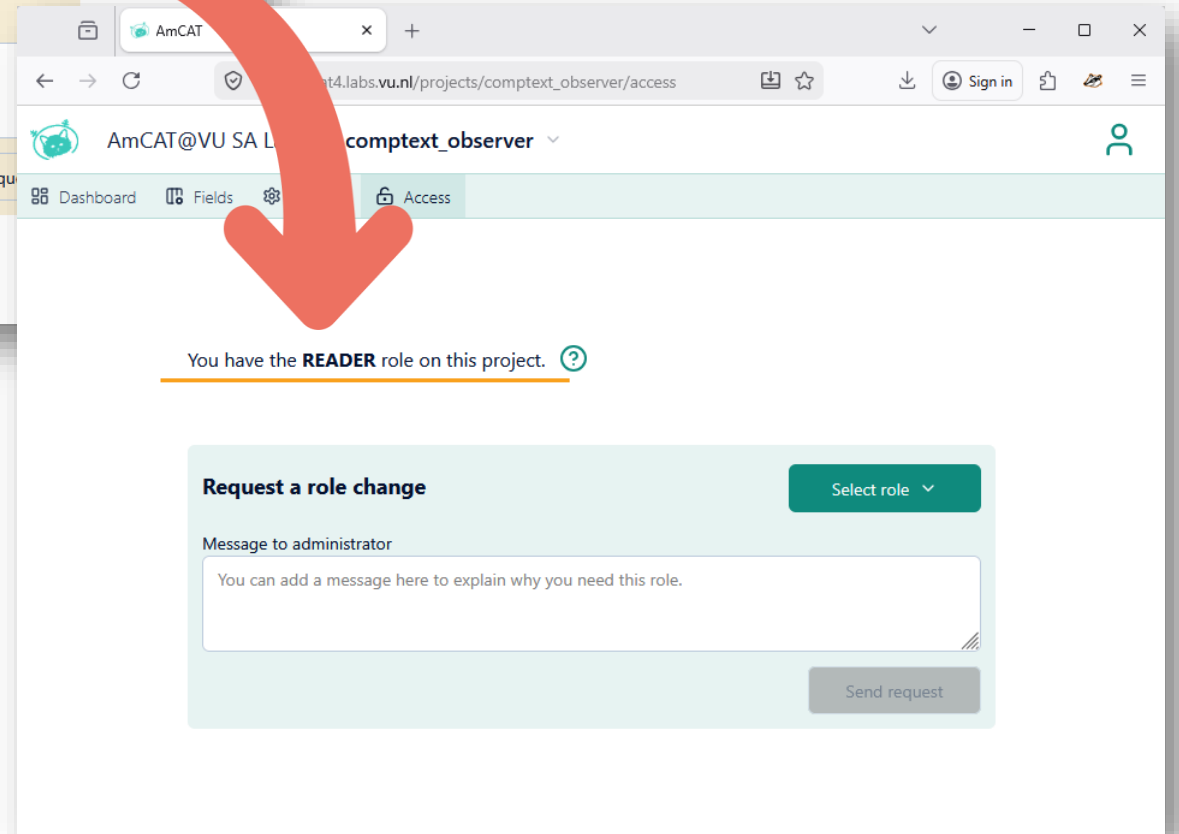
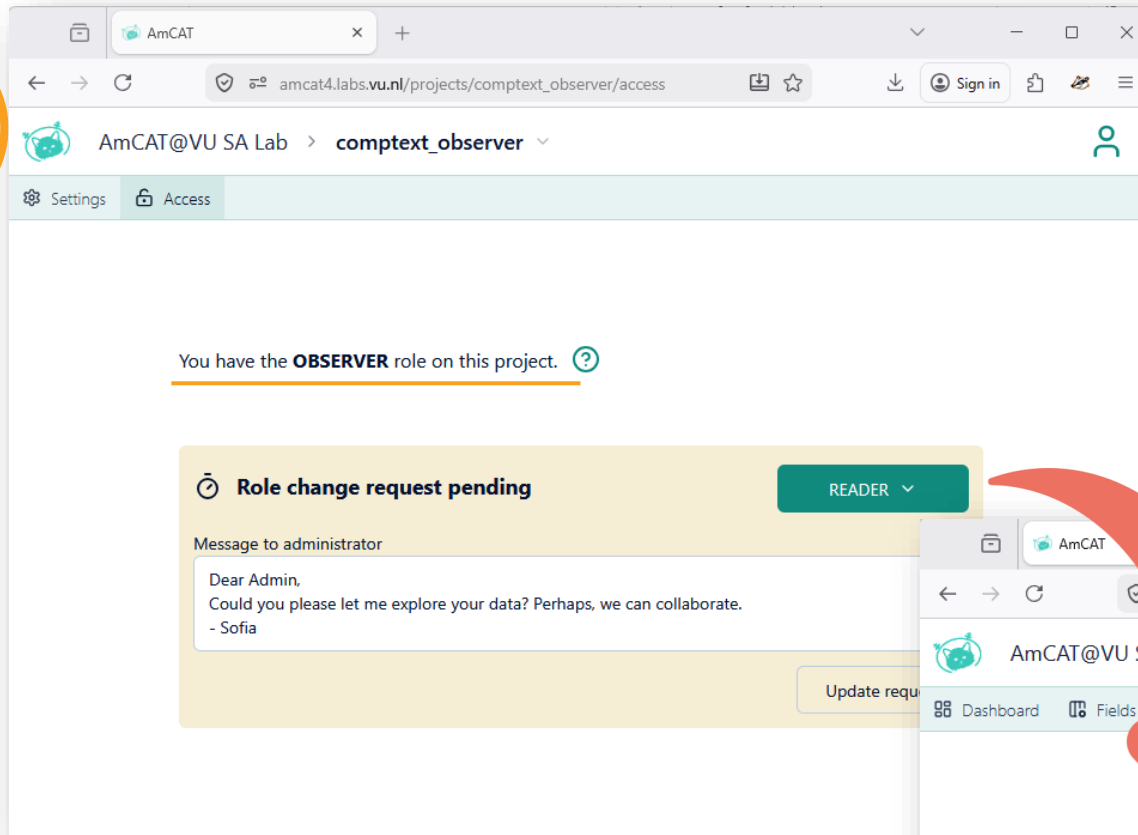
user **b.s.gilclavel@vu.nl** ✓ ✕
requests READER role
for project **comptext_observer**

Dear Admin, Could you please let me explore your data? Perhaps, we can collaborate. - Sofia

1 / 2 requests evaluated **Submit decisions** Cancel



Accepting Requests





3.4 Using the API



We will cover

Access AmCAT in R



Uploading Data



Analyzing Data





Getting an API key

AmCAT

amcat4.labs.vu.nl/projects

AmCAT@VU SA Lab > select project

Projects Server role

Projects

No folders found

comptext_sgc_827
comptext_sgc_827

My first AmCAT project.

comptext_observer

b.s.gilclavel@vu.nl

- Sign-out
- API Keys
- Take Tour
- Light mode
- About



Getting an API key

The screenshot shows a web browser window with the URL `amcat4.labs.vu.nl/api_keys`. The page header includes the AmCAT logo and the text "AmCAT@VU SA Lab > select project". Below the header, there are two tabs: "Projects" and "Server role". The main content area features a prominent green button labeled "Create API Key", which is highlighted with a yellow oval. Below this button, there is an information box titled "Information on API keys" containing the following text:

An **API key** lets you access AmCAT programmatically — from a script or analysis pipeline — without having to log in interactively. Once created, pass the key when connecting to AmCAT from Python or R.

Most screens in the interface have a **Show code** button that generates ready-to-use Python and R code for the current query or dataset. This is a good starting point for your own scripts.

[amcat4py \(Python\)](#) [amcat4r \(R\)](#)



Getting an API key

AmCAT

amcat4.labs.vu.nl/api_keys

AmCAT@VU SA Lab > select project

Projects Server role

API Key Name* Expiration date*

comtext_project 16 / 04 / 2027 , 12 : 38

restrict server role restrict project role

NONE Select role

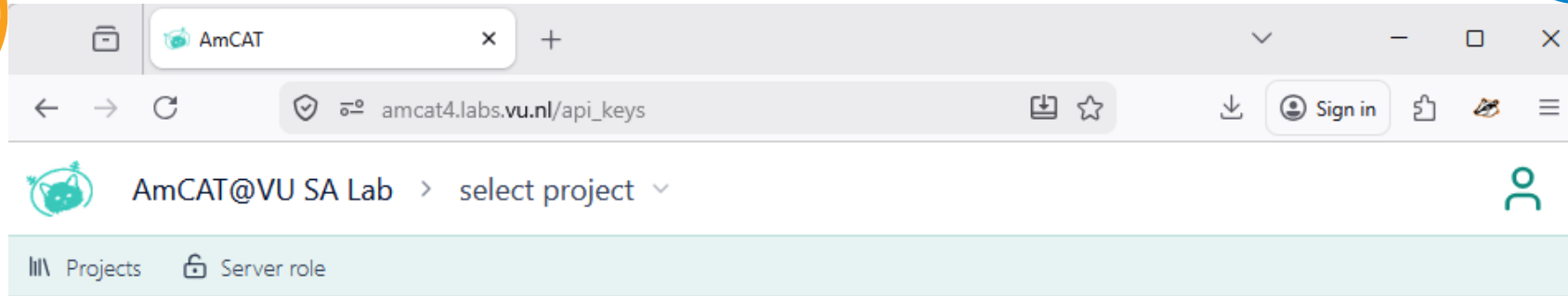
Custom project role restrictions (overrides restrict project role)

Add project role

Create API Key Cancel

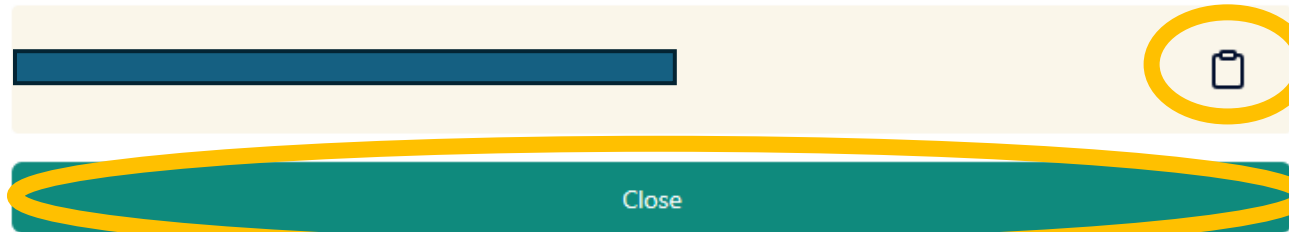


Getting an API key



API Key Created

Please copy and store your API key securely. You won't be able to see it again!



Copy the key and
save it in your
computer.



R/RStudio

1. Open the Social Media Analytics environment of VU JupyterHub.
2. Open and save the R-script where you will be working.
3. Create the variable “key”, in it save your AmCAT API key as a string.
4. Let me know when you are ready to continue.



Access AmCAT in R

```
1 library(amcat4r)
```



Access AmCAT in R

```
library(amcat4r)
```

2

```
amcat_login(server="https://amcat4.labs.vu.nl/api",  
            api_key = "")
```



Access AmCAT in R

```
library(amcat4r)
```

```
amcat_login(server="https://amcat4.labs.vu.nl/api",  
            api_key = "[redacted]")
```

```
Console Terminal Background Jobs x  
R 4.4.2 · C:/Users/oea694/OneDrive - Vrije Universiteit Amsterdam/ManagingLab/LabSoftware/LabSoftware/  
> amcat_login(server="https://amcat4.labs.vu.nl/api",  
+             api_key = [redacted])  
Discarding pre-4.1 authentication tokens  
Should the amcat token be stored only in memory (more secure) or encrypted  
on disk (more convenient as authentication does not need to be repeated  
in a new R session)?  
  
1: store on disk (less secure)  
2: store only in memory (less convenient)  
3: Authenticate each time  
  
selection: 2
```



We will cover

Access AmCAT in R



Uploading Data



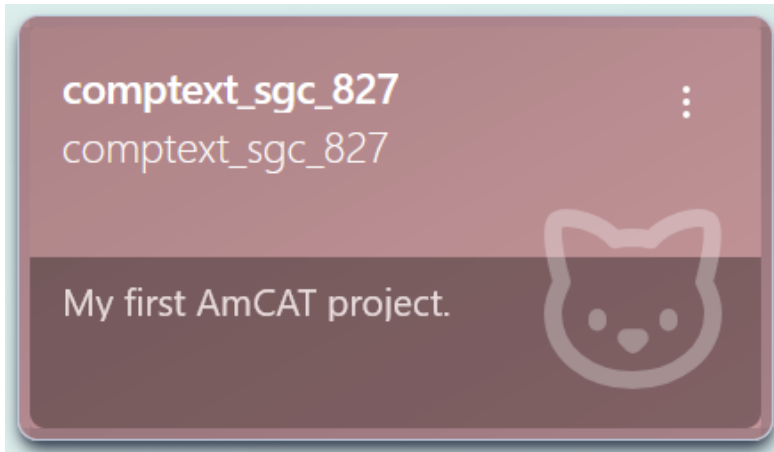
Analyzing Data





Uploading Data

For this part, I will upload the Jane Austen's books. For this, I will use the project that I already created:



```
get_fields("comptext_sgc_827")$name
```

```
> get_fields("comptext_sgc_827")$name  
[1] "author" "chapter" "id"      "text"  
[5] "title"  "year"
```

As long as I upload documents that have these fields, AmCAT will be able to process them.



Uploading Data

1. Upload the R-script AmCAT.R to JupyterHub.
2. Open the document.
3. Let me know when you are done.



Uploading Data

For this part, we will upload the Jane Austen's books. For this, you will use the project that you already created:

```
> get_fields("comptext_sgc_827")$name  
[1] "author" "chapter" "id" "text"  
[5] "title" "year"
```

```
> names(JA_BOOKS)  
[1] "text" "title" "author" "chapter"  
[5] "year" "id"
```

upload_documents(index = "comptext_sgc_827",
documents=JA_BOOKS)



We will cover

Access AmCAT in R



Uploading Data



Analyzing Data





Analyzing Data

`list_indexes()`

`list_indexes()$id`

Yes, index is the same
as project.

```
> list_indexes()$id
[1] "comptext_metareader" "comptext_observer"
[3] "comptext_writer"    "comptext_reader"
[5] "comptext_sgc_827"
```



Analyzing Data

```
query_documents(index="comtext_sgc_827",  
  queries = "text:countenance OR text:felicity",  
  fields=c("year", "title"))
```

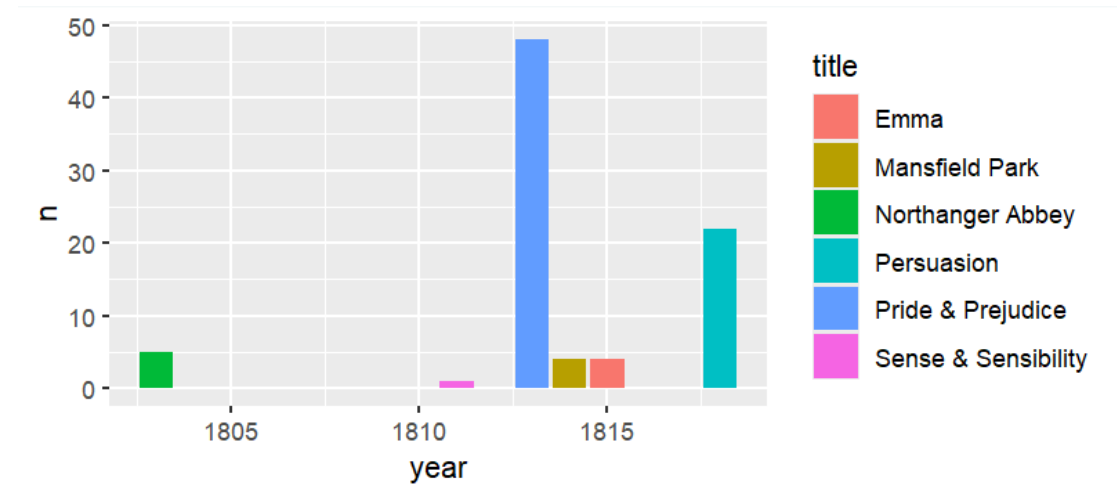
You can use
REGEX.

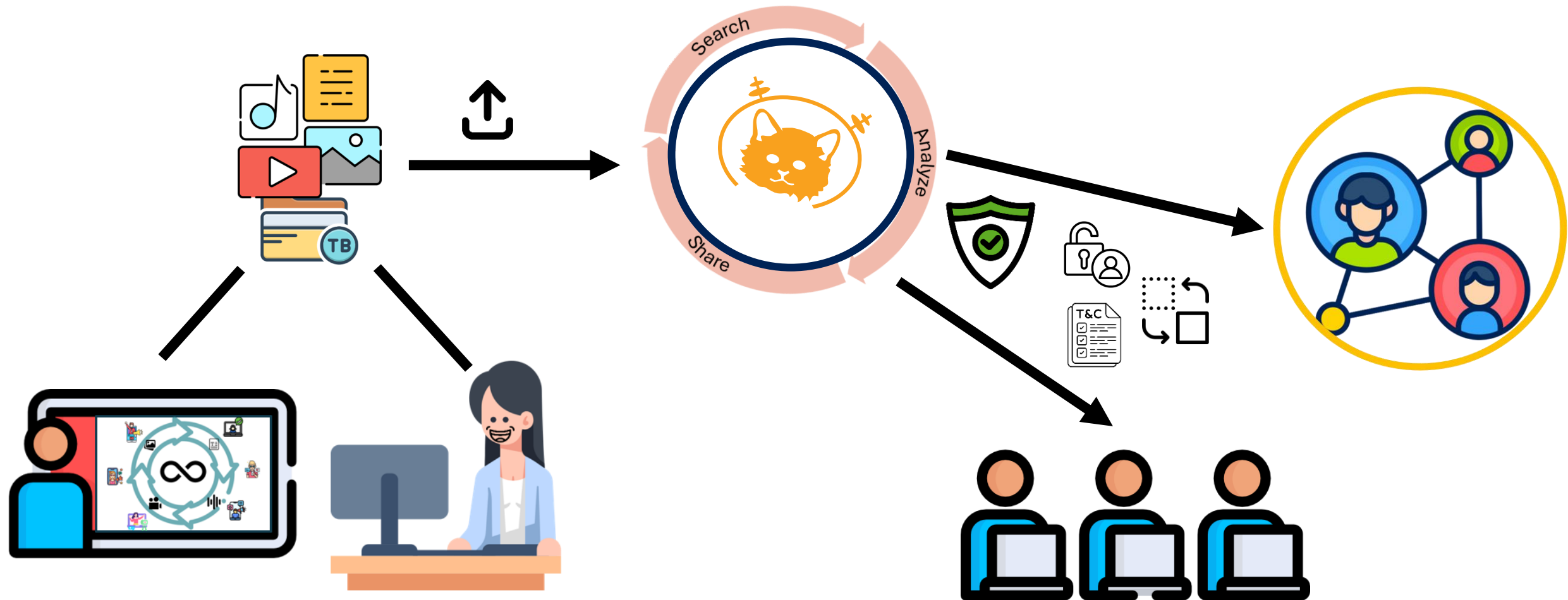
You can
retrieve any of
the fields.

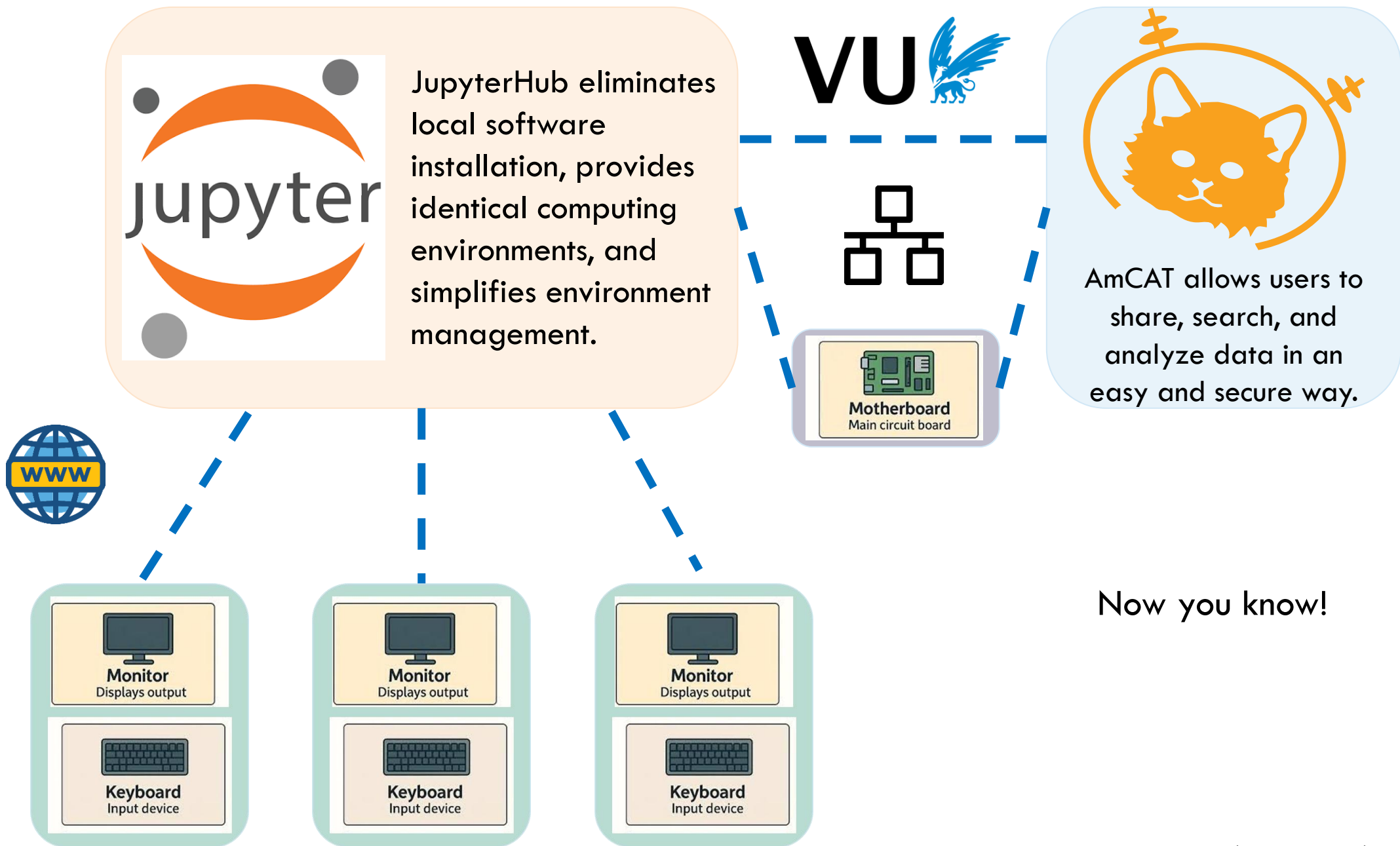


Analyzing Data

```
query_documents(index="comptext_sgc_827",  
                queries = "text:countenance OR text:felicity",  
                fields=c("year", "title"))%>%  
  group_by(title,year)%>%  
  count()%>%  
  ggplot(aes(x=year,y=n,fill=title))+  
  geom_col()
```







Now you know!

REFERENCES

- ❑ Gil-Clavel, Sofia. Constrained by Computing Power? We Can Help You! January 2025. https://societal-analytics.nl/blogs/20250201_computing-power/.
- ❑ Gil-Clavel, Sofia. “AmCAT Workshop - COMPTExT 2026.” Societal Analytics Lab - Blogs, April 2026. https://societal-analytics.nl/blogs/20260422_AmCAT_Workshop/.
- ❑ Gil-Clavel, Sofia. *SofiaG11/CAnD3_2026*. R. May 17, 2026, released May 27, 2026. https://github.com/SofiaG11/CAnD3_2026.
- ❑ Icons from <https://www.flaticon.com/>.



Join us!



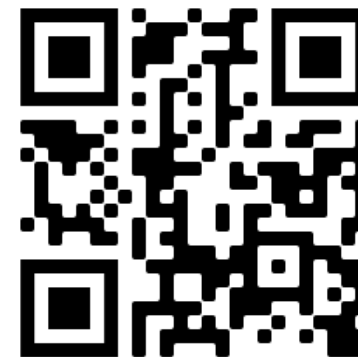
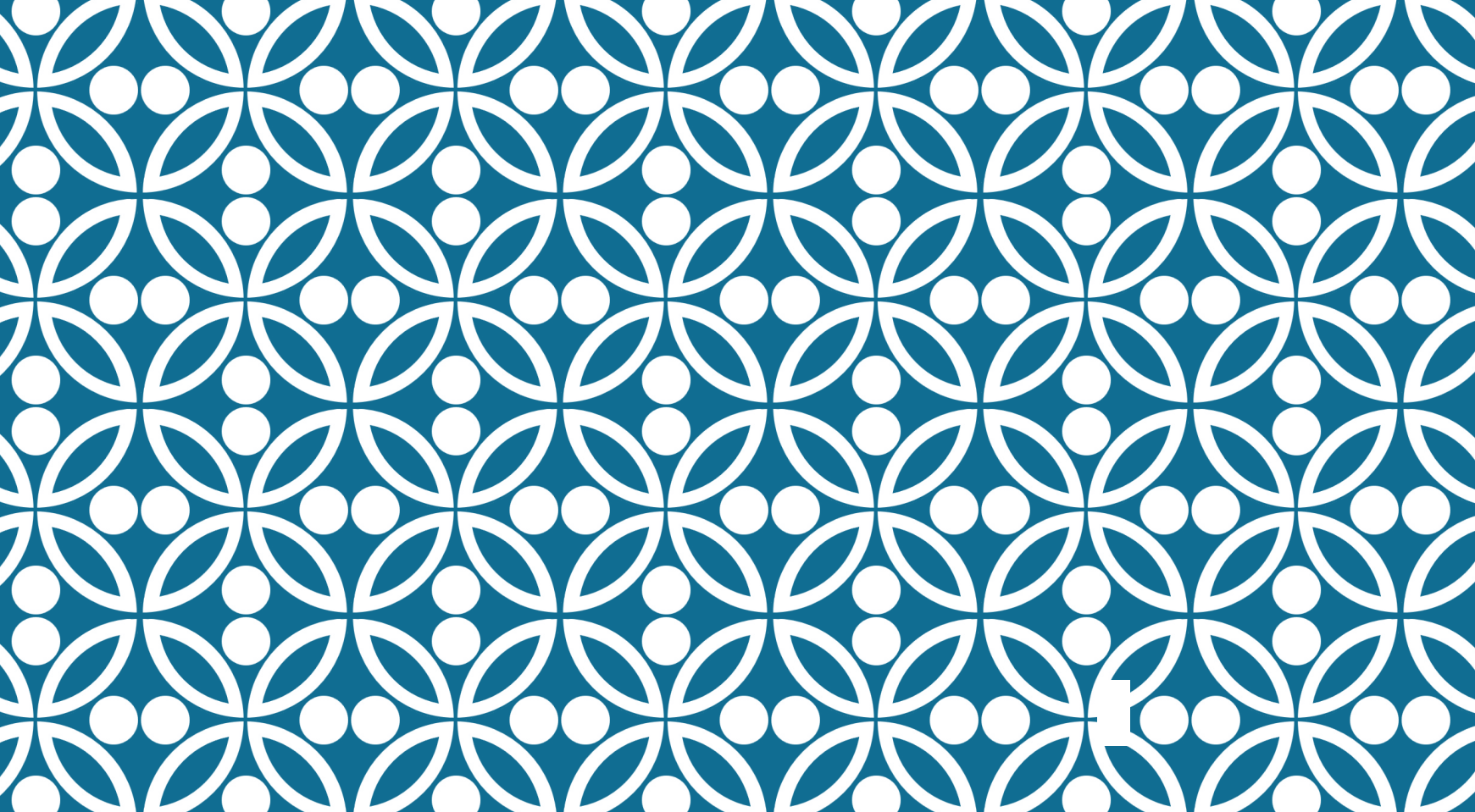
<https://forms.office.com/e/8Bgd2YsasJ>

All about the lab:

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<https://sofiag11.github.io/>

THANKS!

Dr. Sofia Gil-Clavel

—

Let's have a 5
mins break
while I accept
your requests.

